MIRAGE: Supplementary Information, with code

Measuring the Impact and Reality of Rainwater Harvesting Mandates. A meta-analysis of rooftop rainwater harvesting performance in Indian cities

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Contents

1 Data and code supplied with this report **1**

2 Screening and inclusion **1**

3 Reconciliation with the midway check-in **3**

4 Performance dataset **3**

5 Cost dataset and benchmarks **6**

6 Functionality, installation and failure modes **9**

7 Mandate instruments and reference inputs **10**

8 Sources **12**

9 Statement of contributions **14**

10 Limitations in full **14**

11 Code used in this work **16**

11.1 What each script does and where its output is used . . . 16
11.2 How to rerun everything . . . 17
11.3 Where the key numbers are computed . . . 17
11.4 build_dataset.py . . . 18
11.5 analysis.py . . . 22
11.6 make_supplementary_tables.py . . . 27
11.7 style.py . . . 30
11.8 build_deck.js . . . 31

1 Data and code supplied with this report

The submission folder contains, besides this document and the main report: the harmonised dataset (performance observations with computed potential and ratios, cost rows with deflation and levelised cost, functionality estimates, installation figures, failure-mode counts, the mandate table, the screening log and the price-index series used for deflation); the source extraction tables, one row per source record, each carrying the URL and the sentence or field values taken from the page for every number used, together with verbatim extracts from the two CGWB documents and a codebook for every column; the Python scripts that rebuild the dataset, the statistics, the figures and the tables in this document from the source extraction tables (run in the order dataset, analysis, tables); and a results summary listing every number quoted in the report.

2 Screening and inclusion

The work was organised as four parallel searches, called strands below: performance (yield against potential), cost, functionality and failure modes, and mandates. The flow diagram and the log give the counts for each.

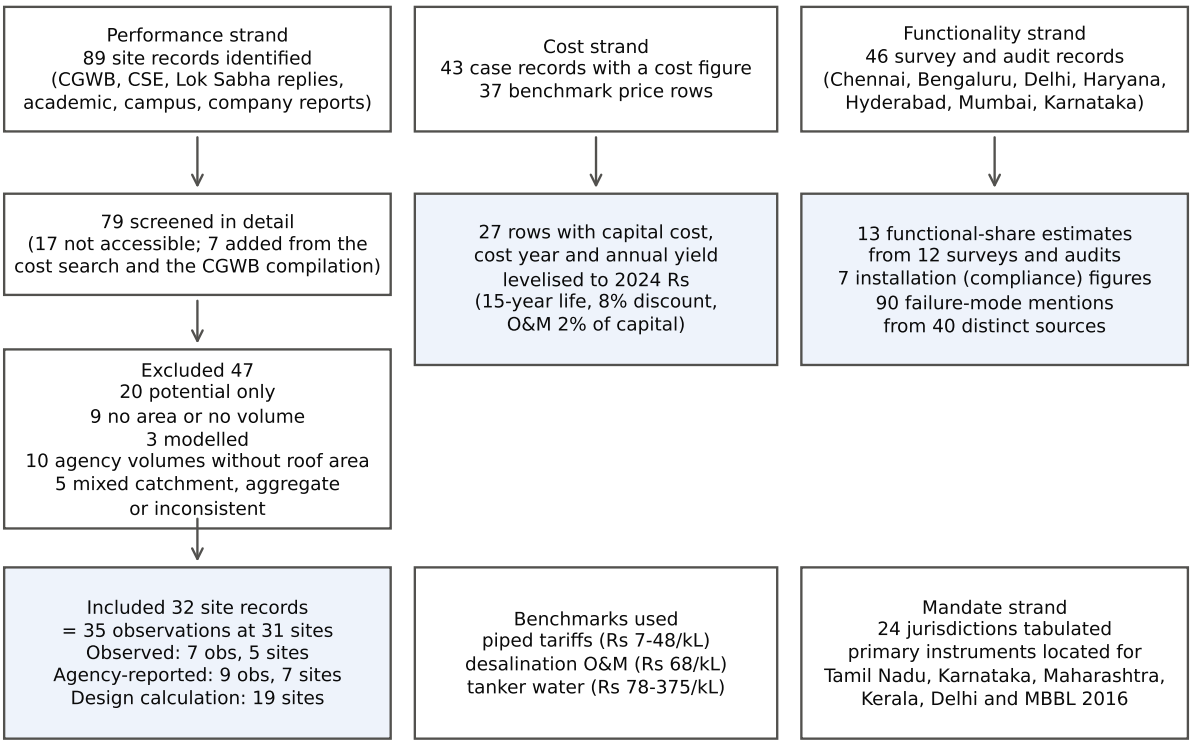


Figure 1: Screening and inclusion flow for the four research strands.

Table 1: Screening and inclusion log (counts behind the flow diagram above).

Step	Count
Performance strand: site records identified (CGWB, CSE, Lok Sabha replies, academic, campus and company reports)	89
Performance strand: records not accessible (documents that could not be opened or were paywalled)	17
Performance strand: records screened in detail (72 from the performance search, 6 CSE model projects from the cost search, 1 case read directly from the CGWB 2011 compilation)	79
Excluded: potential only (no actual volume reported)	20
Excluded: no catchment area or no volume	9
Excluded: modelled volumes (JNTU Hyderabad, 3 structures)	3
Excluded: agency volumes with no roof area (Lok Sabha 2004 list)	10
Excluded: mixed or non-roof catchment, aggregate, or internally inconsistent	5
Included site records	32
Included observations (some sites have several years)	35
Included as observed volumes (instrument or tank record named)	7
Included as agency-reported volumes	9
Included as design calculations (CSE model projects and CGWB design note)	19
Cost strand: case records screened	43
Cost strand: rows with both capital cost and annual yield and a cost year, used for levelised cost	27
Cost strand: benchmark price rows	37
Functionality strand: survey records screened	46
Functionality strand: surveys with a measured functional / defunct split	13
Functionality strand: city compliance (installation) figures	7
Failure-mode mentions extracted	90
Failure-mode distinct sources	40
Failure-mode categories after collapsing	11
Mandate jurisdictions tabulated	24

Inclusion rules for the performance strand. A record entered the ratio analysis if it reported (i) a roof or whole-site catchment area, (ii) an actual harvested, collected or recharged volume for a stated period, and (iii) rainfall for the site or a city for which an IMD normal exists. Records reporting only design potential, only a storage capacity, only a structure count, or volumes for aggregated multi-site portfolios were excluded. Rows were graded by one mechanical rule into three evidence tiers: observed, if the source names a flow meter, a measuring device at the recharge well or a tank-filling record for that site; agency-reported, if an agency states an annual volume with no instrument named, or states it as a share of estimated runoff; design-calculated, if the source itself labels the figure a design calculation. The rule puts both CGWB Maharashtra demonstrations (Ramtek and Warud) in the observed tier, because CGWB describes an identical pipe train with a flow meter for both and uses the same sentence (“As per estimation, a total of X cu. m of rain was received on the roof of the building, out of which Y cu. m was actually recharged”) for both. Warud’s Y is a round 90% of X, which may be an assumed share; the

main report therefore gives the result both ways (Table 5). Shram Shakti Bhawan, whose CGWB figure is labelled "expected recharge", is design-calculated. The JNTU Hyderabad rows (modelled monthly volumes equal to rainfall × area × 0.9) were excluded as modelled rather than reported.

Harmonisation choices. Theoretical potential $P = A \times R \times C/1000$ with $C = 0.80$. The CGWB (2007) manual gives 0.7 for concrete, 0.75 tiled, 0.8 asbestos and 0.9 GI sheet (Table 7.2, verified in the manual); the CGWB/National Water Mission 2020 guide gives 0.75 to 0.95 for roofs. Most included roofs are RCC; 0.80 is the midpoint of the manual's range and the value used in the CGWB 2003 rooftop tables and the midway check-in. The 0.70 and 0.90 columns in Table 3 and the sensitivity figure below show that the tier medians move by a few points but the ordering does not change. Rainfall: site-year where reported (Goa 2010, Bhubaneswar 2004 to 2006); rain received on the roof divided by area where the source reports it that way (Ramtek, Warud); the source's own city average where that is what the source used (CSE pages); IMD 1991-2020 normals otherwise. No IMD normal was located for Chandigarh, so the CSE figure (1,059.3 mm) is used for both Chandigarh cases; using the CGWB district normal (742 mm) instead raises both ratios further above 100%, which does not change the conclusion that they are design-derived. For Ramtek the implied rainfall (215.7 m^3 over 360 m^2 = 599 mm) is well below the taluka average of 1,043.7 mm, so the monitored period was probably not a full year; using the taluka average would lower the Ramtek ratio from 11% to 6%. Catchment: for the CSE model projects the only area given is "total rooftop and surface area", so the design-calculated potentials apply a roof coefficient to roof plus paved and open ground; this is why that tier is used only as the design-share reference and not in the observed versus agency-reported comparison.

3 Reconciliation with the midway check-in

The midway check-in is the mid-term progress report submitted earlier in the course. The table below compares each of its six cases with the final dataset.

Table 2: Reconciliation of the six midway check-in cases with the final dataset. All six are retained; ratios were recomputed with one stated runoff coefficient and a stated rainfall basis.

Case	Midway check-in	Final	Reason for change
KITS Ramtek	about 9% (19 of 215.7 m^3 received)	11% (19 kL against potential 172.6 kL at C = 0.80); 8.8% of rain received; observed tier	Numbers verified word for word in CGWB (2011) p.73; the midway ratio divided by gross rain, the final divides by potential (rain × 0.8). The implied rainfall (599 mm) is below the Ramtek taluka average of 1,043.7 mm, so the monitored period was probably not a full year; with the taluka average the ratio would be 6%.
Goa University	61–76% (900 kL, about 400 m^2)	76% in 2010 (3.7 m rainfall); 28% in 2008 (partial year) and 19% in 2009 using the IMD Panaji normal	Midway used a range of runoff coefficients; final fixes C = 0.80 and adds the two earlier years reported in the same CSE table.
BEL Bengaluru	63–75% (about 1,100 kL recharged)	64% (1,100 kL, 2,000 m^2, IMD Bengaluru normal 1,077.6 mm)	Source says the water is harvested and stored for DM-water production, so the system type is storage, not recharge as the midway said.
Panjab University	about 79% (1,985 m^3)	112% (2,100 m^2, 1,059.3 mm, C = 0.80); agency-reported tier	The midway figure used an unstated area or rainfall. The Lok Sabha source gives the volume as "93% of total generated runoff", i.e. a design share, which is why the ratio exceeds 100%.
Warud	90% (derived from runoff assumption)	112% against potential; 90% of rain received; observed tier by the stated rule (flow meter named), with a sensitivity case grading it agency-reported	Confirmed in CGWB (2011) p.73 (500 m^2, 311.30 m^3 received, 280.17 m^3 recharged). The CGWB wording is identical to Ramtek, so one rule must grade both the same way; the round 90% suggests an assumed share, which is why both gradings are reported.
IIT Delhi Block VI	not comparable (rainfall NR)	82% using the IMD Safdarjung normal (762.3 mm); agency-reported tier, rainfall basis flagged	Confirmed in CGWB (2011) p.44 (830 m^3 from 1,660 m^2). Rainfall period still not reported, so the ratio is indicative.
Umaid Heritage, Sri Aurobindo Ashram, PAU Ludhiana	supplementary (mixed catchments or calculated)	Umaid Heritage excluded (110-acre mixed catchment, internal inconsistency 21.1 vs 2.1 ML); Sri Aurobindo Ashram included as a design calculation (43% of potential); PAU excluded (no roof area)	As the midway said: these are not comparable rooftop cases.

The midway text also said that "around ten studies reported only estimated or design-stage potential"; the final screening found 20 such records. It said "over 50 sources" were screened; the final count is 79 performance records plus 43 cost and 46 functionality records. Two items from the Milestone 1 plan were not carried through: system age as a regression predictor (year of installation is recorded where known, but too few observations have a stated study year to compute age), and campus sustainability reports as a data source (the Infosys and Wipro reports give portfolio totals without catchment areas and were excluded).

4 Performance dataset

Table 3: Performance dataset: all 35 observations. Evidence: Observed (instrument or tank record named), Agency (agency-reported, no instrument named), Design (design calculation). Potential uses runoff coefficient 0.80; the last column gives the ratio for C = 0.70 and 0.90. Rainfall basis: site-year (reported for that site and year), site-avg (average quoted by the source), city-avg (CSE city figure), city-normal (IMD normal), implied (rain received on roof divided by area).

No.	Site	Evidence	Type / system	Area m^2	Rain mm (basis)	Year	Vol. kL	Pot. kL	Ratio %	C 0.7–0.9
1	KITS Ramtek, IT building	Observed	institutional / recharge	360	599 (implied)	2001-02	19.0	173	11	13–10
2	Panchayat Samiti office, Warud	Observed	public / recharge	500	623 (implied)	2002-03	280.2	249	112	129–100
3	IIT Delhi, Block VI, New Delhi	Agency	institutional / recharge	1,660	762 (city-normal)	NR (c. 2001-05)	830.0	1,012	82	94–73
4	Goa University, 2010, Taleigao	Observed	institutional / both	400	3,700 (site-year)	2010	900.0	1,184	76	87–68
5	Goa University, 2008 (partial year), Taleigao	Observed	institutional / both	400	2,918 (city-normal)	2008	260.0	934	28	32–25
6	Goa University, 2009, Taleigao	Observed	institutional / both	400	2,918 (city-normal)	2009	180.0	934	19	22–17
7	Bharat Electronics Ltd, Bengaluru	Agency	industrial / storage	2,000	1,078 (city-normal)	annual (post 2003)	1,100.0	1,724	64	73–57
8	Panjab University, Basic Medical Sciences block, Chandigarh	Agency	institutional / recharge	2,100	1,059 (city-avg (CSE))	annual (c. 2004)	1,985.0	1,780	112	127–99
9	CSIO complex, Chandigarh	Agency	institutional / recharge	3,550	1,059 (city-avg (CSE))	annual (c. 2004)	3,812.0	3,008	127	145–113
10	Hydrology Project building, Bhubaneswar, 2004	Agency	public / recharge	2,250	1,096 (site-year)	2004	2,096.0	1,973	106	121–94
11	Hydrology Project building, Bhubaneswar, 2005	Agency	public / recharge	2,250	1,579 (site-year)	2005	2,566.9	2,842	90	103–80
12	Hydrology Project building, Bhubaneswar, 2006	Agency	public / recharge	2,250	1,881 (site-year)	2006	2,174.2	3,386	64	73–57
13	Escorts-Mahle-Goetze, Yelahanka, Bengaluru	Agency	industrial / storage	1,280	970 (city-avg (CSE))	annual (c. 2001)	1,050.0	993	106	121–94
14	K.G. Bhawan (CWC office), Hyderabad	Agency	public / recharge	1,578	800 (site-avg)	NR	1,000.0	1,010	99	113–88
15	R. Ramani residence, Koratur, Chennai	Observed	individual house / storage	50	1,200 (city-avg (CSE))	annual (c. 2001)	13.5	48	28	32–25
16	R.V. Ramamohan residence, Hyderabad	Observed	individual house / storage	100	804 (site-avg)	2012 (5 months)	15.0	64	23	27–21
17	Mira Model School, New Delhi	Design	school / recharge	16,200	611 (city-avg (CSE))	annual (design)	4,454.0	7,919	56	64–50
18	Janki Devi Memorial College, New Delhi	Design	institutional / recharge	32,170	611 (city-avg (CSE))	annual (design)	6,880.0	15,725	44	50–39
19	Jamia Hamdard University, New Delhi	Design	institutional / recharge	315,380	611 (city-avg (CSE))	annual (design)	67,444.0	154,158	44	50–39
20	The Shri Ram School, New Delhi	Design	school / both	6,000	611 (city-avg (CSE))	annual (design)	1,890.0	2,933	64	74–57
21	Tex Corp Ltd, Gurgaon	Design	industrial / storage	4,000	578 (city-avg (CSE))	annual (design)	1,962.0	1,849	106	121–94
22	Ford Foundation, New Delhi	Design	institutional / recharge	12,500	611 (city-avg (CSE))	annual (design)	6,491.8	6,110	106	121–94
23	Sri Aurobindo Ashram, New Delhi	Design	institutional / recharge	95,870	611 (city-avg (CSE))	annual (design)	16,366.0	46,861	35	40–31
24	Tihar Jail wards 1 and 13, New Delhi	Design	public / recharge	4,125	611 (city-avg (CSE))	annual (design)	1,280.0	2,016	63	73–56
25	Hero Honda Motors, Daruhera	Design	industrial / recharge	11,080	578 (city-avg (CSE))	annual (design)	5,441.0	5,122	106	121–94
26	Defence Colony C block, New Delhi	Design	residential colony / recharge	26,087	612 (city-avg (CSE))	annual (design)	9,563.0	12,768	75	86–67
27	Indian Spinal Injury Centre, New Delhi	Design	institutional / recharge	52,598	611 (city-avg (CSE))	annual (design)	27,317.0	25,710	106	121–94
28	New Friends Club, New Delhi	Design	institutional / recharge	3,700	611 (city-avg (CSE))	annual (design)	960.8	1,809	53	61–47
29	Delhi Gymkhana Club, New Delhi	Design	institutional / recharge	113,344	611 (city-avg (CSE))	annual (design)	28,765.8	55,403	52	59–46
30	Nizamuddin East colony, New Delhi	Design	residential colony / recharge	200,916	611 (city-avg (CSE))	annual (design)	48,531.0	98,208	49	56–44
31	Royal Bhutanese Embassy, New Delhi	Design	institutional / recharge	31,363	611 (city-avg (CSE))	annual (design)	7,867.0	15,330	51	59–46
32	Mother Dairy, Mangolpuri, New Delhi	Design	industrial / recharge	89,370	611 (city-avg (CSE))	annual (design)	32,250.0	43,684	74	84–66
33	Anup Ahuja farm house, New Delhi	Design	individual house / recharge	1,250	611 (city-avg (CSE))	annual (design)	272.0	611	45	51–40
34	Surya Vihar apartments, Kapas Hera, Gurgaon	Design	residential colony / recharge	44,029	611 (city-avg (CSE))	annual (design)	18,207.0	21,521	85	97–75
35	Shram Shakti Bhawan (CGWB design), New Delhi	Design	institutional / recharge	11,965	712 (city-avg (CSE))	annual (design)	3,000.0	6,817	44	50–39

Table 4: Measurement notes and sources for the observed and agency-reported observations (numbers as in Table 3). The design-calculated rows come from CSE model-project pages, https://www.cseindia.org/rainwater-harvesting-case-studies, with volumes stated as a fixed share of area × rainfall, and from the CGWB design note for Shram Shakti Bhawan.

No.	Measurement note	Source	URL
1	Flow meter in line. CGWB: 215.7 m3 received on roof, 19 m3 recharged (as per estimation). Rainfall = 215.7/360.	CGWB 2011, p.73	https://cgwb.gov.in/cgwbpnm/download/379
2	Flow meter in line (same CGWB wording as Ramtek). Volume stated as 280.17 m3 (90%) of 311.30 m3 received; the round 90% may be an assumption, see Results. Rainfall = 311.3/500.	CGWB 2011, p.73	https://cgwb.gov.in/cgwbpnm/download/379
3	CGWB: about 830 m3 recharged from 1660 m2 roof; period not stated. Rainfall = IMD Safdarjung normal 1991-2020.	CGWB 2011, p.44	https://cgwb.gov.in/cgwbpnm/download/379
4	Recharge measured with a device at the bore well (CSE); Chachadi 2013 gives 934 kL to 11 Nov 2010. 2010 monsoon rainfall 3.7 m reported.	CSE 2014 Urban RWH report; Chachadi 2013	https://cdn.cseindia.org/userfiles/Urban%20Rainwater%20harvesting%20report.pdf
5	System built July 2008, so 2008 is a partial season. Rainfall = IMD Panaji normal 1981-2010.	CSE 2014 Urban RWH report	https://cdn.cseindia.org/userfiles/Urban%20Rainwater%20harvesting%20report.pdf
6	Rainfall = IMD Panaji normal 1981-2010 (2009 site rainfall not reported).	CSE 2014 Urban RWH report	https://cdn.cseindia.org/userfiles/Urban%20Rainwater%20harvesting%20report.pdf
7	'Around 1100 KL of rainwater harvested annually' for DM water production; method not stated. Rainfall = IMD Bengaluru normal 1991-2020.	NWM/BEL case sheet	https://www.nwm.gov.in/sites/default/files/BEL_RWH-Ctr.pdf
8	Lok Sabha reply: 1985 m3 (93% of total generated runoff) recharged annually. Roof area from CSE 2010. No IMD normal located for Chandigarh; CSE city figure used.	Lok Sabha SQ 384 (23.08.2004); CSE 2010	https://eparlib.sansad.in/bitstream/123456789/520443/1/2310.pdf
9	Lok Sabha reply: 3812 m3 recharged annually. Roof area 3550 m2 from CSE 2010.	Lok Sabha SQ 384 (23.08.2004); CSE 2010	https://eparlib.sansad.in/bitstream/123456789/520443/1/2310.pdf
10	CGWB Table 34; project 'monitored weekly'; method of volume not stated; 2004 value equals 85% of gross rainfall exactly.	CGWB 2011, p.116	https://cgwb.gov.in/cgwbpnm/download/379
11	CGWB Table 34.	CGWB 2011, p.116	https://cgwb.gov.in/cgwbpnm/download/379
12	CGWB Table 34.	CGWB 2011, p.116	https://cgwb.gov.in/cgwbpnm/download/379
13	CSE: 'collects about 1.05 million litres per year'; method not stated.	CSE case study page	https://www.cseindia.org/node/1099
14	CWC: 'Water harvested : 10 lakh litres (approx)'; method not stated.	CWC RWH note	https://cwc.gov.in/sites/default/files/rwh.pdf
15	3 kL tank fills four to five times a year (12-15 kL); storage-limited, so a lower bound on roof yield. Midpoint used.	CSE case study page	https://www.cseindia.org/node/1099
16	About 15,000 L collected over 5 months with a 1 kL tank; storage-limited lower bound.	India Water Portal (Ramamohan 2012)	https://oldiwp.indiawaterportal.org/sites/default/files/iwp2/rooftop_rainwater_harvesting_system___hyderabad___rv_ramamohan___2012.pdf

Table 5: Performance ratio statistics by evidence tier (runoff coefficient 0.80 unless stated). Observed: instrument or tank record named; Agency: agency-reported, no instrument named; Design: design calculation.

Tier	n (sites)	Median	95% CI	Range	Share ≥ 99%	Median at C 0.70 / 0.90
Observed	7 (5)	27.8%	19–76%	11–112%	14%	32% / 25%
Agency	9 (7)	99.0%	64–112%	64–127%	56%	113% / 88%
Design	19 (19)	56.2%	49–75%	35–106%	21%	64% / 50%

Table 6: Exploratory OLS: performance ratio on evidence tier and catchment area, observed and agency-reported observations only (n = 16, R^2 = 0.53).

Term	Coefficient	95% CI	p
Intercept	-0.313	-2.343 to 1.718	0.745
Observed tier (1 = yes)	-0.172	-0.809 to 0.466	0.571
log10 catchment area	0.380	-0.231 to 0.991	0.202

Other tests: Mann-Whitney U (observed vs agency-reported) = 10, p = 0.0229; Spearman ρ of ratio against $\log_{10}$ area: all observed and agency-reported 0.67 (p = 0.004), observed only 0.41 (p = 0.36). The positive area correlation across the two tiers reflects the tier split (large agency-monitored institutional roofs are agency-reported, small metered houses are observed) and is not significant within the observed tier. With Warud in the observed tier the OLS tier coefficient is not significant; the regression is exploratory and is not used for any claim in the main report. Observed median excluding the partial-year Goa 2008 observation: 25.7% (n = 6); excluding the two storage-limited houses: 27.8% (n = 5); Warud graded agency-reported instead of observed: observed median 25.6% (n = 6), agency-reported median 102.4% (n = 10), Mann-Whitney U = 2, p = 0.0010; site-level medians (years averaged): observed 28.1%, agency-reported 99.0%.

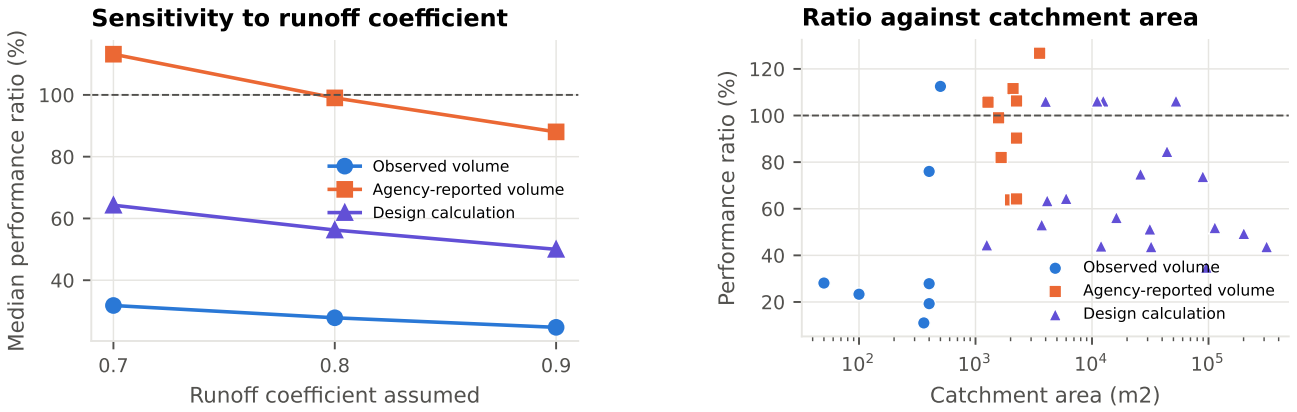


Figure 2: Left: median performance ratio by tier for runoff coefficients 0.70, 0.80 and 0.90. Right: performance ratio against catchment area; the ratio does not rise with size within the observed tier.

5 Cost dataset and benchmarks

Method. Capital cost in the source's cost year was converted to 2024 rupees with the CPI-IW annual averages (RBI Handbook Table 37; the 1982, 2001 and 2016 base series linked with the published factors 4.63 and 2.88). Annual cost = $\text{capital}_{2024} \times (\text{CRF} + 0.02)$, where CRF is the capital recovery factor for 15 years at 8% (0.1168) and 0.02 is the CGWB manual's O&M norm (2% of capital per year, Section 9). The 15-year life and the 2% O&M share are the manual's norms; the 8% discount rate is an assumption (a real rate typical for public infrastructure), and the manual's own worked example uses a 15% rate, which is reported as a sensitivity in Table 8. Cost per kL = annual cost divided by the annual yield stated by the source. For CSE Delhi projects that yield is already a design share (35 to 85%) of area $\times$ rainfall; for Shram Shakti Bhawan it is the CGWB design recharge; for the Bhubaneswar HP building it is the mean of three monitored years. The observed-yield case multiplies all costs by the design-calculated median divided by the observed median. This single multiplier is an approximation: individual design shares range from 35 to 85% of potential, and non-CSE rows state their yield differently, so per-row rescaling would move individual systems but not the median by much. The BEL Bengaluru project (Rs 57 lakh) was excluded because its cost includes a 170 ML surface reservoir, and SV Symphony because its reported volumes are physically implausible for its roof area.

Table 7: Cost dataset: 27 systems. Capital deflated to 2024 rupees with CPI-IW (RBI Table 37, linked series); levelised cost = capital $\times$ (CRF at 8%, 15 years + 0.02) / annual yield. The 8% rate is our assumption; the 15-year life and 2% O&M are the CGWB manual's norms and its example uses 15% (Table 8). Source URLs and quotes for every row are in the source extraction table supplied with the data.

No.	Site	Scale / system	Area m^2	Capital Rs	Year	Yield kL/yr	Yield basis	Capital 2024 Rs	Rs/kL
1	Mira Model School, New Delhi	institutional / recharge	16,200	120,000	2001	4,454	design	494,208	15.2
2	Janki Devi Memorial College, New Delhi	institutional / recharge	32,170	70,000	2001	6,880	design	288,288	5.7
3	Jamia Hamdard University, New Delhi	institutional / recharge	315,380	652,000	2001	67,444	design	2,685,197	5.4
4	The Shri Ram School, New Delhi	institutional / both	6,000	125,000	2000	1,890	design	536,809	38.9
5	Tex Corp Ltd, Gurgaon	industrial / storage	4,000	550,000	2002	1,962	design	2,175,908	151.7
6	Ford Foundation, New Delhi	institutional / recharge	12,500	400,000	2000	6,492	design	1,717,789	36.2
7	Sri Aurobindo Ashram, New Delhi	institutional / recharge	95,870	400,000	2003	16,366	design	1,525,475	12.8
8	Tihar Jail wards 1 and 13, New Delhi	institutional / recharge	4,125	20,000	2002	1,280	design	79,124	8.5
9	Indian Spinal Injury Centre, New Delhi	institutional / recharge	52,598	80,000	2002	27,317	design	316,496	1.6
10	BPCL housing complex, Noida	residential colony / recharge	13,910	450,000	2002	4,446	design	1,780,288	54.8
11	Defence Colony C block, New Delhi	residential colony / recharge	26,087	258,000	2003	9,563	design	983,931	14.1
12	Hero Honda Motors, Daruhera	industrial / recharge	11,080	1,200,000	2004	5,441	design	4,400,392	110.7
13	Anup Ahuja farm house, New Delhi	individual house / recharge	1,250	20,000	2002	272	design	79,124	39.8
14	Surya Vihar apartments, Gurgaon	residential colony / recharge	44,029	65,000	2002	18,207	design	257,153	1.9
15	Mother Dairy unit, New Delhi	industrial / recharge	89,370	435,000	2004	32,250	design	1,595,142	6.8
16	New Friends Club, New Delhi	institutional / recharge	3,700	40,000	2004	961	design	146,680	20.9
17	Delhi Gymkhana Club, New Delhi	institutional / recharge	113,344	80,000	2004	28,766	design	293,359	1.4
18	Nizamuddin East colony, New Delhi	residential colony / recharge	200,916	174,000	2004	48,531	design	638,057	1.8
19	Royal Bhutanese Embassy, New Delhi	institutional / recharge	31,363	110,000	2004	7,867	design	403,369	7.0
20	Brig. Jagdev Singh residence, New Delhi	individual house / recharge	285	7,500	2001	147	design	30,888	28.8
21	Shram Shakti Bhawan, New Delhi	institutional / recharge	11,965	410,000	2001	3,000	design	1,688,544	77.0
22	Nandi Deepa apartments, Bengaluru	residential colony / recharge	NR	2,220,000	2021	3,564	design	2,560,161	98.3

No.	Site	Scale / system	Area m^2	Capital Rs	Year	Yield kL/yr	Yield basis	Capital 2024 Rs	Rs/kL
23	Vidhani village houses, cluster (Saini et al.), Jaipur	residential colony / storage	2,466	131,498	2016	1,602	design	196,218	16.8
24	TSSM BSCOER and JSPM campuses (Varpe et al.), Pune	institutional / both	8,200	6,204,336	2019	5,453	design	7,910,817	198.5
25	Bearys modular house system (Raju et al.), Mangaluru	individual house / storage	81	5,710	2021	21	design	6,585	43.3
26	Punjab govt building standard design (900 m2)	institutional / recharge	900	281,079	2020	438	design	340,630	106.4
27	Hydrology Project building, Bhubaneswar	institutional / recharge	2,250	856,000	2004	2,279	agency-monitored, method not stated (mean 2004-06)	3,138,946	188.5

Table 8: Levelised cost per kL of design yield (2024 rupees) by group, and sensitivity.

Group	n	Median Rs/kL	Range
All	27	20.9 (95% CI 8.5–43.3)	IQR 6.9–65.9
individual house	3	39.8	28.8–43.3
residential colony	6	15.4	1.8–98.3
institutional	15	15.2	1.4–198.5
industrial	3	110.7	6.8–151.7
system: recharge	22	14.6	
system: storage	3	43.3	
system: both	2	118.7	
CSE Delhi-NCR rows (2000–2004)	20	13.4	
Rows with cost year $\geq$ 2010	5	98.3	
Sensitivity: 20-year life, 6% discount	27	16.4	
Sensitivity: 15% discount (the CGWB example rate)	27	29.2	
Rows with cost year $\geq$ 2010 at observed yield	5	199	
At observed yield (multiplier 2.02)	27	42	

Kruskal-Wallis across the four building-scale groups: H = 2.07, p = 0.559. Spearman ρ of $\log_{10}$ area against Rs/kL: -0.74 (p = 0.0000, n = 26); against capital per m^2: -0.73 (p = 0.0000). The delivered-yield multiplier is the design-calculated median ratio (design yield as a share of potential, 0.56) divided by the observed median (0.28).

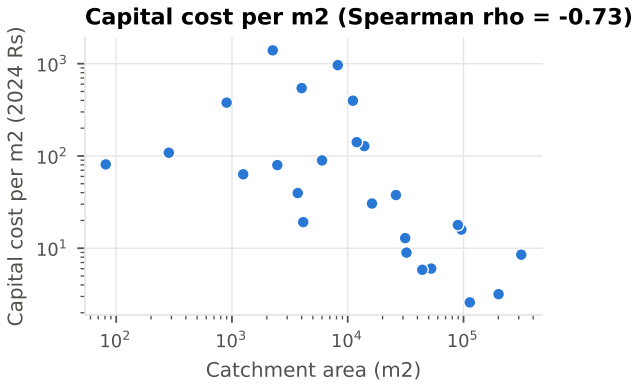


Figure 3: Capital cost per m^2 of catchment (2024 rupees) against catchment area, 26 systems with a stated area.

Table 9: Price benchmarks collected. The eight used in the cost figure of the main report are the Chennai lowest and top domestic slabs, the Delhi 20–30 kL slab, the BWSSB production cost, Chennai desalination O&M, the Metrowater 9 kL tanker, and the Bengaluru 2024 and Chennai 2019 private tanker prices (Rs per load divided by load volume).

Item	City	Value	Unit	Year	URL
CMWSSB domestic metered water tariff, up to 10 kL	Chennai	7.00	Rs/kL	2025 (w.e.f. 01-Apr-2025)	https://www.cmwssb.tn.gov.in/tariff
CMWSSB domestic metered water tariff, above 25 kL	Chennai	48.00	Rs/kL	2025	https://www.cmwssb.tn.gov.in/tariff
CMWSSB lorry (tanker) water supply, domestic 9,000 L	Chennai	700	Rs per 9,000 L load (= Rs 77.8/kL, derived)	2023/2025 (w.e.f. 01-Apr-2023 and 01-Apr-2025)	https://www.cmwssb.tn.gov.in/tariff

Item	City	Value	Unit	Year	URL
Chennai Metrowater tanker 9,000 L price hike	Chennai	735	Rs per 9,000 L load	2019 (Oct)	https://www.thenewsminute.com/amp/story/tamil-nadu/chennai-metrowater-hikes-tanker-price-without-any-public-announcement-110817
Chennai Metrowater public tanker 9,000 L during 2019 crisis	Chennai	700	Rs per 9,000 L load	2019 (June)	https://www.thenewsminute.com/amp/story/tamil-nadu/chennai-water-crisis-private-tankers-fleece-public-rates-increase-over-100-103859
Chennai private tanker 12,000 L, pre-crisis	Chennai	1500-2000	Rs per 12,000 L load	2018-2019 (April 2019 / 'last June')	https://www.thenewsminute.com/amp/story/tamil-nadu/chennai-water-crisis-private-tankers-fleece-public-rates-increase-over-100-103859
Chennai private tanker 12,000 L, June 2019 crisis peak	Chennai	3000-6000	Rs per 12,000 L load	2019 (May-June)	https://www.thenewsminute.com/amp/story/tamil-nadu/chennai-water-crisis-private-tankers-fleece-public-rates-increase-over-100-103859
Chennai Metrowater tanker 9,000 L (2004)	Chennai	650	Rs per 9,000 L load	2004	https://www.downtoearth.org.in/amp/story/environment/tanker-economics-12249
Chennai desalination (Minjur/Nemmeli) O&M cost	Chennai	68000	Rs per MLD (= Rs 68/kL, derived)	2019	https://citizenmatters.in/chennai-cost-of-desalination-vs-surface-water-treatment-plants-8405
Chennai surface water treatment O&M cost	Chennai	6000-23000 (avg 14500)	Rs per MLD (= Rs 6-23/kL, derived)	2019	https://citizenmatters.in/chennai-cost-of-desalination-vs-surface-water-treatment-plants-8405
Minjur desalination bulk purchase price paid by Metrowater	Chennai	48.66	Rs/kL	2012 (The Hindu, 19 Apr 2012) / 2010 agreement	https://en.wikipedia.org/wiki/Minjur_Seawater_Desalination_Plant
BWSSB Cauvery water production cost	Bengaluru	41	Rs/kL	2021	https://citizenmatters.in/rainwater-harvesting-in-bengaluru-webinar-64351
BWSSB domestic tariff revision - increase by slab	Bengaluru	0.15 paise/L (0-8000 L); 0.30 paise/L (8001-25000 L); 0.80 paise/L (25001-50000 L); 'Re 1 per litre' (>50000 L, likely paise)	increase per litre	2025 (order 10 Apr 2025, bills from May 2025)	https://www.theweek.in/wire-updates/national/2025/04/09/mes24-ka-water-hike.html
BWSSB minimum domestic slab after April 2025 revision	Bengaluru	0.30 paise per litre (= Rs 3/kL)	Rs/kL (derived)	2025	https://www.etvbharat.com/en/!bharat/bangalore-water-board-revises-tariff-residents-vent-anger-over-extra-burden-enn25041005877
Bengaluru private tanker, pre-crisis vs crisis (resident reports)	Bengaluru	600-1000 before; 1800-2000 during	Rs per 12,000 L load	2024 (Feb-Mar)	https://businesstoday.in/india/story/bengaluru-water-crisis-district-administration-fixes-rates-for-water-tanker-services-420502-2024-03-07
Bengaluru district administration capped tanker rates (within 5 km)	Bengaluru	600 (6,000 L); 700 (8,000 L); 1,000 (12,000 L)	Rs per load, incl GST	2024 (7 Mar 2024, for 4 months)	https://businesstoday.in/india/story/bengaluru-water-crisis-district-administration-fixes-rates-for-water-tanker-services-420502-2024-03-07
Delhi Jal Board domestic volumetric tariff (Category I)	Delhi	5.27 (up to 20 kL); 26.36 (20-30 kL); 43.93 (>30 kL)	Rs/kL plus service charge Rs 146.41 / 219.62 / 292.82 per month; sewer charge 60% of volumetric	NR (Citizen's Charter PDF, undated)	https://delhijalboard.delhi.gov.in/sites/default/files/inline-files/citizen_s_charter.pdf
Delhi private tanker prices (DTE investigation)	Delhi	2,500 (8,000 L); 5,000 (one full tanker, >=10,000 L)	Rs per load	2024 (June)	https://www.downtoearth.org.in/amp/story/water/tanker-economy-revealed-dte-uncovers-how-water-suppliers-openly-exploit-the-poor-across-delhi
BMC Mumbai residential piped water tariff	Mumbai	5.94 (up to 150 lpcd); 11.88 (150-200 lpcd); 17.82 (200-250 lpcd); 23.76 (>250 lpcd)	Rs/kL	2021	https://citizenmatters.in/how-much-do-mumbaikars-pay-for-water-28014
HMWSSB domestic tariff (approx., last revision)	Hyderabad	~10	Rs/kL	2026 (article 16 Sep 2026; refers to last revision)	https://www.siasat.com/water-bills-to-rise-by-5-pc-annually-as-hmwssb-looks-to-recover-revenue-3543386/
Hyderabad RWH pit cost (contractor example, HMWSSB mandate)	Hyderabad	100000-120000	Rs per pit (6 ft x 6 ft)	2024	https://thesouthfirst.com/telangana/hmwssb-mandates-rainwater-harvesting-pits-for-17000-buildings-to-tackle-water-crisis
Recharge structure cost table (CGWB-type norms as reproduced by IIT Bombay)	India (generic)	Recharge pit 2500-5000; Recharge trench 5000-10000; Recharge through hand pump 1500-2500; Recharge through dug well 5000-8000; Recharge well 50000-80000; Recharge shaft 60000-85000; Lateral shaft per m 2000-3000 + bore well 25000-35000	Rs per structure	NR (lecture undated; CGWB source c.2000s)	https://arctest.dev.nptel.ac.in/content/storage2/courses/105101010/downloads/Lecture11.pdf
CGWB dug well recharge average unit cost (XI Plan scheme)	India (7 states)	4000 (range 3600-5700)	Rs per dug well recharge structure	2011 (XI Plan)	https://www.cgwb.gov.in/cgwbpnm/download/379
CSE: RWH cost for existing building of ~300 sq m	Delhi (generic)	2000-30000	Rs per system	NR (page undated)	https://www.cseindia.org/costs-1097
CSE unit rates: PVC storage tank	Delhi / Mumbai / Pune / Rajkot / Chennai / Kolkata	6.00 / 4.25 / 7.00 / 6.00 / 3.00 / 5.4	Rs per litre of tank capacity	NR (undated)	https://www.cseindia.org/costs-1097
CSE unit rates: recharge bore drilling	Delhi / Mumbai / Pune / Rajkot / Chennai / Kolkata	600 (hand) / 1000 (mechanical); 10000; 197; 225; 160 (7in); 931 (first 50 m)	Rs per metre	NR (undated)	https://www.cseindia.org/costs-1097
Rain Centre Chennai: sump storage cost	Chennai	3.5-4	Rs per litre of storage	NR (c.2003; Rain Centre leaflet)	https://static.vikaspedia.in/media/files_en/energy/environment/three-main-design-concepts.pdf
Rain Centre Chennai: recharge well cost	Chennai	400	Rs per foot of depth	2003	https://www.indiatogether.org/environment/water/rwh-rkrish-may03.pdf
Rainwater Club Bengaluru: ferrocement tank unit cost	Bengaluru	1.20	Rs per litre of storage	NR (undated)	https://Sswm.info/sites/default/files/reference_attachments/RAINWATERCLUB%20Editor%20ny%20Rooftop%20Rainwater%20Harvesting%20for%20tackling%20Fluoride%20contaminated%20Ground%20Water.pdf
Typical Chennai house RWH retrofit cost (1000-1200 sq ft terrace)	Chennai	5000-8000	Rs per house	2017	https://citizenmatters.in/how-to-start-harvesting-rainwater-at-home-3181
Residential RWH with 10,000 L storage (Organiser op-ed)	India (generic)	20000	Rs per system; storage Rs 1.5-2 per litre	2009	https://organiser.org/2009/10/04/38978/general/r1419f082/
O&M cost statement - secondary guide (Studio Matrx)	India (generic)	2000-5000	Rs per year	2026	https://www.studiomatrx.org/guides/rainwater-harvesting
O&M cost statement - secondary guide (Houseyog)	India (generic)	2000-5000	Rs per year	2025/2026	https://www.houseyog.com/blog/rainwater-harvesting-cost-india/
O&M cost statement - vendor (NeeRain)	India (generic)	0-500	Rs per year	2026	https://neerain.com/rooftop-rainwater-harvesting-real-cost/

Item	City	Value	Unit	Year	URL
O&M frequency statements (CSE)	India (generic)	NR	frequency only	NR	https://www.cseindia.org/maintenance-monitoring-1125
Payback statement - campus RWH + reclamation (peer-reviewed)	India (hot semi-arid campus)	>250	years payback	2020	https://eprints.ncl.ac.uk/271510
Private tanker cost assumption used in Pune campus payback calc	Pune	1560000	Rs per year (campus tanker expenditure)	2019	http://www.ijirset.com/upload/2019/june/43_Rooftop.pdf

6 Functionality, installation and failure modes

Table 10: Functional-share estimates: 13 estimates from 12 surveys (the two Dwarka rows are two definitions applied to one DJB report). URLs and quotes are in the source extraction table supplied with the data.

No.	Survey	Population	n	Functional %	Definition used	Conducted by (reported via)
1	Chennai 2019, houses with a structure	houses with an RWH structure (1.62 lakh of 2.72 lakh inspected)	162,000	76	1.62 lakh houses had a structure; about 38,000 of them 'needed minor repairs', so 1.24 lakh (76.5%) needed no repair. Single GCC snapshot (Aug-Sep 2019).	Greater Chennai Corporation (The News Minute)
2	Chennai Gandhinagar 2003	independent houses	309	50	'only 50 per cent had implemented RWH properly'	Akashganga Trust (Down To Earth 2004)
3	Chennai approved buildings 2001-02	buildings approved under 1994 rule, with RWH	25	67	a third of installed systems 'not properly designed and therefore ineffective'	MIDS survey for CSE
4	Karnataka school structures 2007	government school rooftop structures	1,269	11	140 of 1,269 structures functional	Arghyam status report
5	Karnataka model schools 2009	school rooftop structures	27	59	16 of 27 functioning; 3 partial; 8 defunct	Arghyam
6	Faridabad public systems 2018-19	public RWH systems (parks, schools)	190	20	'around 80% had stopped working'	MCF survey (The Tribune)
7	Faridabad public systems 2024	public RWH systems	200	25	'around 70 to 80 per cent' non-functional (midpoint)	civic sources (The Tribune)
8	Gurugram pits 2023	RWH pits, public and private	500	30	'around 70%' lying defunct	MCG / GMDA (The Tribune)
9	Delhi Dwarka societies 2024 (structural)	group housing societies with RWH	176	84	176 minus 4 non-functional minus 25 dry pits	DJB action-taken report to NGT
10	Delhi Dwarka societies 2024 (clean recharge)	group housing societies with RWH	176	32	57 of 176 without faecal coliform in recharge water (119 contaminated)	DJB action-taken report to NGT
11	Bengaluru BWSSB connections 2023 (expert estimate)	BWSSB connections with RWH	NR	40	'only about 40% of BWSSB connections have functional RWH systems' (estimate)	S. Vishwanath (Deccan Herald)
12	Kaithal government buildings 2018	government-building systems	5	0	all five choked after the first year	The Tribune field report
13	Sirsa park systems 2025	municipal park systems	4	50	2 of 4 dead within two years	The Tribune field report

Notes on the Chennai 2019 figures. The Tamil Nadu government and the Greater Chennai Corporation reported the 2019 inspection drive in two snapshots: over 2 lakh buildings (1.36 lakh correctly installed and maintained, 37,131 needing maintenance, 60,461 without a structure) and, a few weeks later, about 2.72 lakh houses (1.62 lakh with a structure, about 38,000 of them needing minor repairs, almost 70,000 without). To avoid mixing snapshots, both steps of the Chennai cascade use the second one: 1.62 of 2.72 lakh houses had a structure (59.6%) and 1.24 of 1.62 lakh needed no repair (76.5%). The first snapshot gives a similar functional share (1.36 of 1.73 lakh, 78.6%). Both reach us through press reports (The News Minute); no original press release was located.

Table 11: Installation (compliance) figures used in the mandate-gap figure of the main report.

City	Population	Denominator	With RWH	%	Definition
Chennai 2019	houses inspected	272,000	162,000	59.6	1.62 lakh of 2.72 lakh inspected had a structure
Bengaluru 2023	properties with water connection	1,080,000	190,000	17.6	1.9 lakh of 10.8 lakh
Hyderabad 2024	properties >= 300 sq yd surveyed	39,000	22,000	56.4	22,000 installed; 17,000 non-compliant
Mumbai 2007-21	housing societies (estimate)	20,000	3,000	15.0	about 3,000 of about 20,000 societies
Pune FY2022-23	taxable properties (rebate claimants)	1,250,000	3,716	0.3	3,716 rebate claimants of about 12.5 lakh properties
Nagpur 2023	properties	636,000	1,123	0.2	1,123 of about 6.36 lakh properties
Delhi govt buildings 2019	government buildings (building class, not a city rate)	771	212	27.5	212 of 771 with RWH

Table 12: Failure-mode categories (11 after collapsing), number of distinct sources citing each (90 mentions from 40 distinct sources; these overlap with but are not the same set as the 46 survey records), and the cities concerned. Every mention with URL and quote is in the source extraction table supplied with the data.

Failure mode	Sources	Mentions	Cities
No maintenance, abandoned	18	18	Bengaluru, Chennai, Delhi, Delhi (Dwarka), Delhi (school), Delhi schools, Faridabad, Gurugram, Hyderabad, Karnataka schools
No awareness, incentive or enforcement	15	15	Bengaluru, Chennai, Delhi, Faridabad, Karnataka schools, Mumbai
Poor design or construction	15	17	Bengaluru, Chennai, Delhi, Delhi (Dwarka), Faridabad, Karnataka schools, Mumbai, Sirsa
Token structure, built for compliance only	10	10	Bengaluru, Chennai, Gurugram, Mumbai, Thane
Recharge pit or well silted or choked	7	7	Chennai, Delhi, Delhi (Dwarka), Faridabad, Kaithal

Failure mode	Sources	Mentions	Cities
Downpipes or conveyance broken or disconnected	6	6	Bengaluru, Chennai, Delhi schools, Karnataka schools
Other (space, vandalism, roof-only design)	6	7	Bengaluru, Chennai, Delhi, Karnataka schools
Storage undersized, overflow or leak	3	3	Bengaluru, Karnataka schools, Mumbai
Filter clogged or absent	3	3	Bengaluru, Chennai, Karnataka schools
First flush absent or not operated	2	2	Delhi, Karnataka schools
Water quality concerns, system disused	2	2	Delhi (Dwarka), Karnataka schools

Outcome evidence. The often-quoted 6 to 8 m rise in Chennai groundwater after 2003 traces only to the Rain Centre's own statements; the CMWSSB figure of a 4 m net rise and The Hindu's 2006 report of a 2 m monsoon rise are not attributed to rooftop RWH specifically. In 2019 the Rain Centre reported 90% of its 88 monitoring wells dry despite 16 years of the mandate. BWSSB data suggest about 10% of Bengaluru's runoff is harvested, around 3% of demand. Delhi Jal Board paid Rs 325.7 million in RWH rebates and collected Rs 567.1 million in penalties between 2017 and 2019 (details and URLs in the functionality search log supplied with the data).

7 Mandate instruments and reference inputs

Table 13: Mandate landscape, 24 jurisdictions. URLs and quotes are in the source extraction table supplied with the data.

Jurisdiction	Instrument	Year	Applies to	Penalty / incentive	Reported compliance
India (Model Building Bye-Laws 2016, MoHUA)	Model Building Bye-Laws 2016 (MBBL), MoHUA; rainwater harvesting chapter	2016	all buildings having a plot size of 100 sq.m or more (RWH proposal required with building plan)	NR (states decide)	33 States/UTs have adopted the RWH provisions (Lok Sabha reply 05.03.2020; also PIB 24.03.2020); 'barring Sikkim, Lakshadweep and Mizoram' all States/UTs adopted (Lok Sabha Q1241, 09.02.2023)
Tamil Nadu (incl. Chennai)	Tamil Nadu Municipal Laws (Second Amendment) Ordinance, 2003 (TN Ordinance No. 4 of 2003) inserting s.255-A Chennai City Municipal Corporation Act etc.	2003 (promulgated 19 July 2003)	every owner or occupier of a building (all buildings; Govt buildings too)	water supply connection disconnected until structure provided; Commissioner may build and recover cost as property tax; new buildings not given water/sewer connection without RWH (CSE)	Chennai 2019: GCC inspected ~2.72 lakh houses since Aug 2019, ~1.62 lakh had RWH (approx 60%), 9,000 notices issued; CMWSSB claims 8.9 lakh structures; Rain Centre 2015 audit of 1,100 samples found govt buildings worst violators
Bengaluru (BWSSB jurisdiction)	Bangalore Water Supply and Sewerage (Amendment) Act, 2011 (Karnataka Act No. 05 of 2011) substituting s.72A (bill introduced 2009); penalty regulations 2015; BWSS (Amendment) Bill 2021 extends to 30x40 sites	2009 (bill) / 2011 (Act assented 2 Feb 2011) / 2021 (extension)	existing buildings on sital area >= 2400 sq ft (60x40 ft site); new buildings on sital area >= 1200 sq ft (30x40 ft); 2021 Bill: sital area 2,325-10,763.9 sq ft RWH; >=10,763.9 sq ft dual piping + RWH	residential: 25% of water bill first 3 months then 50%; non-residential: 50% then 100% (Citizen Matters 2023); Rs 1.95-2 crore/month collected	Dec 2023: ~1.9 lakh of 10.8 lakh properties with water connections (~18%) have RWH; 39,703 properties paying monthly penalty; Mar 2024 (The Hindu): 1,96,591 residential buildings implemented, 39,213 have not
Karnataka (statewide, outside BWSSB)	Government orders for Government buildings; statewide urban mandate 'contemplated' (2011); 20% tax rebate (CSE)	NR	Government buildings; amendment of bye-laws in cities >20 lakh population under action (CSE); statewide new buildings contemplated 2011	20% rebate on tax payment for 5 years (CSE)	NR
Delhi (NCT)	Building Bye-laws 1983 modified by MoUD Gazette Notification 28.07.2001 (plots >=100 sq m); CGWA directive; DJB resolution No. 829; Unified Building Bye-Laws 2016	2001 (bye-law), 2012 (DJB mandatory per Outlook), 2016 (UBBL)	all buildings with plot area >= 100 sq m (2001 notification; DJB resolution 829); DJB compulsory for plots >= 500 sq m; UBBL 2016 features mandatory for new constructions above 105 sq m (DUAC)	building plans not sanctioned without RWH; DJB 10% rebate on water bill for plots >= 100 sq m with functional RWH; penalty 1.5 times water bill / '50% of the monthly bill amount' for non-compliance; proposed environmental compensation Rs 50,000 to Rs 5 lakh (2023 proposal); NGT Rs 5 lakh per educational institution (OA 217/2016)	Delhi: 1,560 consumers availed DJB rebate (NGT monitoring committee report OA 496/2016); 3,675 govt schools/colleges of 4,769 institutions had functional systems; only 1,869 private buildings as of 2021 (Citizen Matters); Dwarka: NGT expert committee inspected 354 CGHS, 'Most structures are found either completely defunct or just partially working' (2023)
Mumbai / Maharashtra (Greater Mumbai DCR; all towns non-gaothan)	Govt of Maharashtra UDD Notification No. TPB-4307/396/CR-124/2007/UD-11 dated 6 June 2007 modifying Development Control Regulations for Greater Mumbai 1991; BMC mandatory for new buildings since 2002 (Citizen Matters); Aug 2026 GR: old buildings seeking new water/drainage connections must have RWH	2002 (BMC, new buildings) / 2007 (300 sq m notification)	plots having area not less than 300 sq m in non-gaothan areas of all towns (2007); earlier 1,000 sq m (CSE)	no building permission unless provision for RWH; levy not exceeding Rs 1000 per annum per 100 sq m built-up area for failure to provide/maintain; completion/occupation certificate only after RWH (CSE)	approx. 3,000 societies of an estimated 20,000 had RWH 2007-2021 (Citizen Matters Jul 2024)
Hyderabad / Telangana (HMWSSB area)	AP Govt order (2000) for all municipal corporations/urban areas (IWP); APWALTA 2002; HMWSSB directive dated 14 Dec 2024 for 17,000 consumers	2000 (GO; tentative enforcement June 2001)	all new buildings with area 300 sq m or more irrespective of roof area; existing buildings within one year of GO (CSE); originally 200 sq m then 300 sq m (Realty Today)	50% subsidy on RWH structures was lifted due to poor response (CSE); 2024: double charges for HMWSSB water tankers for non-compliant (Rs 500 to Rs 1,000 per 5 kL)	Dec 2024: NGO survey of 39,000 properties; 42,000 buildings identified needing RWH; 22,000 installed; 17,000 non-compliant (Realty Today / South First)
Pune (PMC)	PMC requirement for new buildings (2006, per Realty Today); property-tax rebate scheme	2006	all new buildings (Realty Today)	property tax rebate up to 10% for RWH (PMC eco-housing rebate)	306 properties availed rebate in FY2012-13; 3,716 in FY2022-23, of approx. 12.5 lakh taxable properties (approx 0.3%)
Nagpur (NMC)	NMC regulation (mandatory from March 2005 per CSE; ruling effective 28 Aug 2009 per Propnewstime)	2005 (CSE) / 2009 (Propnewstime)	new constructions of area >= 300 sq m (non-gaothan); housing society layouts and amenity spaces	fine up to Rs 1,000 per annum per 100 sq m built-up area; NMC 5-10% property tax discount for eco initiatives	May 2023: only 1,123 buildings comply = 0.18% of ~6.36 lakh properties
Gujarat / Ahmedabad (AUDA/AMC GDCR)	Gujarat Development Control Regulations (GDCR); AUDA notification 2002 (1,500 sq m+); State Govt buildings after 2003	2002 (AUDA) / 2003 (Govt buildings)	GDCR: buildings 500-1,500 sq m follow RWH specifications; 1,500-4,000 sq m percolation wells (one per 4,000 sq m); AUDA 2002: constructions >= 1,500 sq m	Surat: 50% subsidy up to Rs 2,000 (CSE); no penalty stated	NR ('still not very popular', 2008 clipping)
Rajasthan / Jaipur (JDA)	Rajasthan state mandate (PHED certification); Jaipur/Rajasthan Building Bye-Laws 2020 Regulation 21.7 (rule 10.11.1); JDA enforcement 2011	2011 (JDA penalty) / 2020 (Bye-laws)	all buildings (any land use) with plot area over approx. 300 sq m (CSE legislation page) / over 500 sq m (CSE laws page); Bye-laws 2020: each plot owner (no minimum stated in translation)	water supply connection can be terminated if RWH completion certificate not submitted to PHED; JDA penalty Rs 25,000 (2011); Bye-laws 2020: annual maintenance fee levied by local authority	Jul 2023: only 220 buildings in Jaipur had complied (JDA)
Madhya Pradesh / Indore (also Jabalpur, Gwalior)	MP Government Order no. F 23 (107) 95 XXXII dated 7 July 2004; MP Bhumi Vikas Rules 1984; Jabalpur from 1 Apr 2005; Gwalior order 27 Jan 2006	2004	all new buildings with area >= 250 sq m; mandatory for G+3 structures	6% property tax rebate; Gwalior penalty Rs 7,000 for non-compliance	NR
Kerala	Kerala Municipality Building (Amendment) Rules, 2004 (Extraordinary Gazette No. 92/2004 dated 12-01-2004) inserting Chapter XVI-A, Rule 109A	2004	all new buildings: Residential A1 with floor area >= 100 sq m and plot area >= 200 sq m; A2, B, C, D, E and specified G1/G2 occupancies	no penalty stated; exemption where water logging common; KWA 90% / Jalanidhi 75% subsidies (CSE)	NR (CSE: flats with multiple owners maintained better than individual houses)
Haryana (incl. Gurugram, Faridabad)	HUDA rules (CSE); Haryana Building Code 2017	2017 (Code)	roof rainwater harvesting where rooftop area >= 100 sq m; recharge for Group Housing with plot > 500 sq m (Tribune); HUDA: all new buildings irrespective of roof area (CSE)	NR	Faridabad: MCF survey 2018-19 found ~75-80% of RWH systems defunct (Tribune Dec 2023)
Uttar Pradesh (Kanpur; Ghaziabad; statewide CM direction 2025)	Building bye-laws (Kanpur: >=1000 sq m per CSE legislation page / >=500 sq m per CSE laws page); Ghaziabad bye-laws >=300 sq m; CM direction 4 Oct 2025 for buildings >100 sq m	NR (bye-laws); 2025 (CM order)	Kanpur: new buildings >= 1000 sq m (or 500 sq m); Ghaziabad: plots >= 300 sq m; CM 2025: buildings larger than 100 sq m, urban and rural	Ghaziabad: penalties up to Rs 3 lakh depending on plot size and severity (2025)	Ghaziabad: of 100 identified locations, 20 visited (Jul 2025); no counts
Punjab (incl. Ludhiana)	Punjab building bylaws notified 2018 (Tribune): RWH for structures > 119 sq yards (earlier 250 sq yd residential / 500 sq yd commercial); RWH Cell in each ULB	2018	structures (residential, commercial, institutional) on more than 119 sq yards (approx 100 sq m); earlier residential > 250 sq yd and commercial > 500 sq yd	utility connections denied without RWH and completion certificate (Tribune Bathinda)	NR ('only several property owners ensured proper installation and functioning')
Chandigarh (UT)	Chandigarh Administration order (2008) for plots of one kanal or more; CSE: all new buildings	2008	plots of one kanal (approx 500 sq yd / 418 sq m) or more; administration mulling lower threshold (2019)	NR (no penalty stated; 'absence of any mechanism to keep a check')	'only a handful of private building owners complying with it' (Pioneer 2019)

Jurisdiction	Instrument	Year	Applies to	Penalty / incentive	Reported compliance
West Bengal / Kolkata	West Bengal Municipal (Building) Rules, 2007, Rule 171	2007	NR (threshold not stated in source)	NR	NR
Goa	Policy on Rainwater Harvesting, Govt of Goa (15/10/2008, WRD); RWH policy notified in gazette Dec 2022 with plot thresholds (Propnewstime)	2008 (policy) / 2022 (gazette policy)	2022 policy: residential complexes/apartments on plots >= 2,000 sq m; commercial complexes >= 1,500 sq m; industrial units >= 10,000 sq m; all government buildings; exempt if STP/greywater recycling exists; 2008: PWD govt buildings (CSE)	NR	NR
Himachal Pradesh	State direction: no building plan approved in urban areas without RWH (CSE)	NR	all buildings to be constructed in urban areas; schools, government buildings, rest houses, industries, bus stands	building plan not approved without RWH system	NR
Uttarakhand (Mussoorie-Dehradun Development Authority)	MDDA building byelaws provision (CSE); MDDA mandatory for new houses of 1,200 sq ft (Tribune 2011)	NR	all new houses of 1,200 sq ft (Tribune, Mussoorie)	NR	Mussoorie 2011: 'only 30% hoteliers have adopted roof-top rain water harvesting'
Andhra Pradesh (all municipal corporations/urban areas; Visakhapatnam)	AP Government Order (2000) making RWH mandatory in all municipal corporations, UDAs and municipalities; APWALTA 2002	2000	buildings of 300 sq m or more	NR	NR (no Visakhapatnam-specific data found)
Rajkot (Gujarat)	Rajkot General Development Control Regulations, Regulation No. 17.18	NR	buildings with built-up area >= 80 sq m; existing buildings > 1,000 sq m five-year moratorium	NR	NR
Ranchi (Jharkhand)	Ranchi Regional Development Authority byelaws	NR	NR	NR	NR

Table 14: Runoff coefficients and rainfall normals used. The CGWB (2007) Table 7.2 values were verified directly in the manual.

Item	Value	Unit	Period	Source document
Runoff coefficient - GI sheet roof	0.9	coefficient	CGWB 2007 (cited)	Table 7.2 'Runoff Coefficients of Common Types of Roofs', CGWB Manual on Artificial Recharge of Ground Water (2007), as reproduced in Journal of Rural Development 37(2), 2018 (NIRD&PR)
Runoff coefficient - Asbestos roof	0.8	coefficient	CGWB 2007 (cited)	same as above
Runoff coefficient - Tiled roof	0.75	coefficient	CGWB 2007 (cited)	same as above
Runoff coefficient - Concrete (RCC) roof	0.7	coefficient	CGWB 2007 (cited)	same as above
Runoff coefficient - Asphalt or concrete pavement	0.85	coefficient	2007	Table 4.4 'Usual Values of Runoff Coefficients (K)', Section 4.2.2.1.2, p.25, CGWB Manual on Artificial Recharge of Ground Water (Sept 2007) (fetched directly from cgwb.gov.in)
Runoff coefficient - Roof top (range)	0.75 - 0.95	coefficient	2020	'Catch the Rain - through Rain Water Harvesting', M.K. Garg, Scientist D, CGWB Faridabad (NWM/NYKS orientation ppt, 10 Dec 2020)
Runoff coefficient - Paved area (range)	0.50 - 0.85	coefficient	2020	same CGWB/NWM ppt
Runoff coefficient - Bare ground / unpaved (range)	0.10 - 0.20	coefficient	2020	same CGWB/NWM ppt
Runoff coefficient - Green area (range)	0.05 - 0.10	coefficient	2020	same CGWB/NWM ppt
Runoff coefficient - roof (worked example)	0.8	coefficient	2019	Down To Earth, 'Catch water where it falls: urban rainwater harvesting', 3 Jul 2019
Normal annual rainfall - Chennai (Nungambakkam)	1376.8	mm	1991-2020	IMD RMC Chennai climatological table
Normal annual rainfall - Bengaluru City	1077.6	mm	1991-2020	IMD Met Centre Bengaluru climatological table
Normal annual rainfall - New Delhi (Safdarjung)	762.3	mm	1991-2020	IMD RMC New Delhi climatological table
Normal annual rainfall - Mumbai (Santacruz)	2502.3	mm	1991-2020	IMD city.imd.gov.in Climatological Table (station id 43003 = Mumbai-Santacruz)
Normal annual rainfall - Hyderabad	859.6	mm	1991-2020	IMD city.imd.gov.in Climatological Table
Normal annual rainfall - Pune (Shivajinagar)	841.2	mm	1991-2020	IMD city.imd.gov.in Climatological Table (station id 43063 = Pune-Shivajinagar)
Normal annual rainfall - Nagpur (Sonegaon Airport)	1151.2	mm	1991-2020	IMD city.imd.gov.in Climatological Table (station id 42867 = Nagpur-Sonegaon)
Normal annual rainfall - Nagpur district (range)	1000-1200	mm	1901-1992	CGWB Ground Water Information, Nagpur District, Maharashtra (1770/DBR/2013), CGWB Central Region Nagpur
Average annual rainfall - Ramtek taluka (Nagpur district)	1043.7	mm	2002-2011 average	CGWB Ground Water Information, Nagpur District (2013), Table 3 Rainfall data
Normal annual rainfall - Panaji / Goa	2918.0	mm	1981-2010	IMD city.imd.gov.in Climatological Table (Panjim)
Normal annual rainfall - Chandigarh	NR	mm	NR	Not found on an official page (IMD Chandigarh pages give only monthly/seasonal normals)
Normal annual rainfall - Jodhpur	374.3	mm	1981-2010	IMD city.imd.gov.in Climatological Table
Normal annual rainfall - Ludhiana	NR	mm	NR	Not found on an official page
Normal annual rainfall - Patiala (nearest IMD table to Ludhiana/Chandigarh found)	875.6	mm	1991-2020	IMD city.imd.gov.in Climatological Table (station id 42101 = Patiala)
Normal annual rainfall - Ahmedabad	824	mm	2010-2019	IMD Met Centre Ahmedabad 'climate' note
Normal annual rainfall - Jaipur	600.3	mm	1991-2020	IMD Met Centre Jaipur climatological table
Normal annual rainfall - Kolkata (Alipore)	1813.3	mm	1991-2020	IMD city.imd.gov.in Climatological Table
Normal annual rainfall - Thiruvananthapuram	1835.3	mm	1991-2020	IMD city.imd.gov.in Climatological Table
Normal annual rainfall - Indore	955.1	mm	1991-2020	IMD city.imd.gov.in Climatological Table (station id 42754 = Indore)
Normal annual rainfall - Lucknow	990.1	mm	1971-2000	IMD Met Centre Lucknow 'Climatology of Lucknow' Table 3
Normal annual rainfall - Dehradun	2161.5	mm	1991-2020	IMD Met Centre Dehradun climatological table
Normal annual rainfall - Guwahati	1695.3	mm	1991-2020	IMD city.imd.gov.in Climatological Table (station id 42410 = Guwahati)
Normal annual rainfall - Coimbatore	618.5	mm	1991-2020	IMD RMC Chennai climatological table
Normal annual rainfall - Madurai Airport	843.3	mm	1991-2020	IMD RMC Chennai climatological table
Normal annual rainfall - Visakhapatnam	1113.1	mm	1991-2020	IMD city.imd.gov.in Climatological Table

Item	Value	Unit	Period	Source document
Normal annual rainfall - Amravati district (for Warud)	886.4	mm	NR (IMD normal; period not stated; monthly chart period 1941-1990)	Model Agriculture Contingency Plan (Rainfed), Amravati district, ICAR-CRIDA / PDKV Akola, Section 1.2 Rainfall, 'Source: IMD'
Normal annual rainfall - Kochi	3014.8	mm	1981-2010	IMD city.imd.gov.in Climatological Table

8 Sources

Table 15: All 168 distinct source URLs used, with the strand and the first record each supports.

URL	Strand	First use
http://www.ijirset.com/upload/2019/june/43_Rooftop.pdf	Cost	TSSM BSCOER & JSPM NTC campuses, Narhe (Varpe et al.)
https://Sswm.info/sites/default/files/reference_attachments/RAINWATERCLUB%20Editor%20ny%20Rooftop%20Rainwater%20Harvesting%20for%20tackling%20Fluoride%20contaminated%20Ground%20Water.pdf	Cost	Rainwater Club ASHRAYA small house model (ferrocement tank)
https://admin.indiawaterportal.org/articles/rainwater-harvesting-made-mandatory-andhra-pradesh-all-municipal-corporations-and-urban	Mandates	Hyderabad / Telangana (HMWSSB area)
https://admin.indiawaterportal.org/sites/default/files/iwp2/Model_School_Report_Suvarna_Jala__Arghyam_2009.pdf	Functionality	Arghyam
https://admin.indiawaterportal.org/sites/default/files/iwp2/rainwater_harvesting_for_aquifer_storage_recovery.pdf	Performance	Goa University rooftop RWH + recharge bore well (Case study
https://agris.fao.org/search/en/records/65df11d2b766d82b1800245a	Performance	TNAU Coimbatore SWCE workshop roof (Ray et al. 2014, J Progr
https://arctest.dev.nptel.ac.in/content/storage2/courses/105101010/downloads/Lecture11.pdf	Performance	Shram Shakti Bhawan, New Delhi rooftop/pavement recharge (CG
https://base.ac.in/wp-content/uploads/2021/11/BASE-University-WP-Series-12-2021.pdf	Functionality	BASE University WP 12/2021 (Harshita Bhat, Pleasa Serin Abra
https://businesstoday.in/india/story/bengaluru-water-crisis-district-administration-fixes-rates-for-water-tanker-services-420502-2024-03-07	Benchmarks	Bengaluru private tanker, pre-crisis vs crisis (resident rep
https://cag.gov.in/uploads/download_audit_report/2019/Report_no_3_of_2019_Social_General_and_Economic_Sector_(Non_PSU)_Government_of_Haryana.pdf	Functionality	CAG performance audit
https://cdn.cseindia.org/userfiles/Urban%20Rainwater%20harvesting%20report.pdf	Performance	Goa University rooftop RWH + recharge bore well (Case study
https://citizenmatters.in/bengaluru-outer-ring-road-apartment-water-scarcity-bill-solution-rainwater-harvesting-monsoon-44485	Cost	The Greens, Doddanekundi (ORR)
https://citizenmatters.in/bengaluru-rainwater-harvesting-3-lakh-annual-savings-borewell-recharge-37514	Performance	SV Symphony apartments, Chikkabellandur (Sarjapur Rd)
https://citizenmatters.in/bengaluru-rainwater-harvesting-bwssb-kscst-water-36374	Functionality	BWSSB data (Citizen Matters)
https://citizenmatters.in/bengaluru-rwh-plans-problems-solutions/	Functionality	BWSSB count as quoted by S Vishwanath (Biome) in Citizen Mat
https://citizenmatters.in/chennai-cost-of-desalination-vs-surface-water-treatment-plants-8405	Benchmarks	Chennai desalination (Minjur/Nemmeli) O&M cost
https://citizenmatters.in/chennai-household-rainwater-harvesting-guidelines-3059	Functionality	Akash Ganga Trust (Sekhar Raghavan, Rain Centre) for CMDA -
https://citizenmatters.in/chennai-rainwater-harvesting-infrastructure-government-buildings-2909	Functionality	Public Works Department (PWD), Tamil Nadu
https://citizenmatters.in/how-much-do-mumbaikars-pay-for-water-28014	Benchmarks	BMC Mumbai residential piped water tariff
https://citizenmatters.in/how-to-start-harvesting-rainwater-at-home-3181	Benchmarks	Typical Chennai house RWH retrofit cost (1000-1200 sq ft ter
https://citizenmatters.in/mumbai-rainwater-harvesting-lags-behind/	Functionality	Indian Express report quoted by Citizen Matters
https://citizenmatters.in/rainwater-harvesting-delhi-legislation-vs-practice-30914	Functionality	Delhi Jal Board (administrative claim)
https://citizenmatters.in/rainwater-harvesting-in-bengaluru-webinar-64351	Cost	Aditi Elite, Bellandur
https://cmwssb.tn.gov.in/sites/default/files/rainwater_harvesting/muncipal_ord_2003.pdf	Mandates	Tamil Nadu (incl. Chennai)
https://connect.iisc.ac.in/2019/09/why-drain-the-rivers-when-you-can-catch-the-rain/	Performance	A.R. Shivakumar residence 'Sourabha'
https://crmapp.mcgm.gov.in/irj/go/km/docs/documents/HomePage%20Data/Quick%20Links/Eco%20Housing/RainWaterHarvesting.pdf	Mandates	Mumbai / Maharashtra (Greater Mumbai DCR; all towns non-gaot
https://cseindia.org/content/downloadreports/1569	Performance	Panjab University Chandigarh, Basic (Medical) Sciences Block
https://cwc.gov.in/sites/default/files/rwh.pdf	Performance	K.G. Bhawan (Central Water Commission office), Hyderabad
https://delhijalboard.delhi.gov.in/sites/default/files/inline-files/citizen_s_charter.pdf	Benchmarks	Delhi Jal Board domestic volumetric tariff (Category I)
https://delhiplanning.delhi.gov.in/sites/default/files/scheme_of_rainwater_harvesting.pdf	Functionality	Planning Department, Govt of NCT Delhi - Evaluation Study of
https://dgde.gov.in/?p=17733	Cost	Lucknow Cantonment Board (4 sites)
https://doaj.org/article/356fcd240ae44afe9b683585b2909ebd	Performance	Gurgaon residential area UAV study (Puppala et al. 2023, Wat
https://dste.py.gov.in/sites/default/files/water-conservation.pdf	Performance	Hindustan Unilever Pondicherry plant (factory-wide)
https://duac.org.in/u/BuildingByeLaws	Mandates	Delhi (NCT)
https://en.wikipedia.org/wiki/Minjur_Seawater_Desalination_Plant	Benchmarks	Minjur desalination bulk purchase price paid by Metrowater
https://eparlib.nic.in/bitstream/123456789/1470313/1/AU1241.pdf	Mandates	India (Model Building Bye-Laws 2016, MoHUA)
https://eparlib.nic.in/bitstream/123456789/970953/1/AU2478.pdf	Mandates	India (Model Building Bye-Laws 2016, MoHUA)
https://eparlib.sansad.in/bitstream/123456789/520443/1/2310.pdf	Performance	Panjab University Chandigarh, Basic (Medical) Sciences Block
https://eparlib.sansad.in/bitstream/123456789/523489/1/2608.pdf	Performance	KITS Ramtek rooftop RWH - CGWB impact assessment (Lok Sabha
https://eprints.ncl.ac.uk/271510	Cost	Higher-education campus decentralised system (Zang, Kumar, W
https://epubs.icar.org.in/index.php/JAE/article/view/14641	Performance	Punjab Agricultural University (PAU) Ludhiana library roofto
https://frontdesk.co.in/building-bye-laws/regulation-21-7-rainwater-harvesting-structures-and-tree-plantation-jaipur-building-bye-laws-2020/	Mandates	Rajasthan / Jaipur (JDA)
https://gmrgroup.in/media/from-water-stressed-to-water-positive-delhi-airport-sets-a-new-benchmark-in-sustainable-aviation	Performance	Delhi IGI Airport (DIAL)
https://gondwanags.org.in/wp-content/uploads/2021/11/7-JGSR-Vol-1-2-LR-Abstracts-Panigrahi-Panigrahi.pdf	Performance	Bhubaneswar official building (Panigrahi & Panigrahi 2016, J
https://i-scholar.in/index.php/Ijae/article/view/48491	Performance	KVK Hoshiarpur rooftop recharge structure (Ajaib Singh, IJAE
https://ideas.repec.org/p/sch/wpaper/302.html	Functionality	ISEC - K S Umamani & S Manasi, Working Paper 302
https://idr.nitk.ac.in/items/76ba2754-1538-45e4-95a9-791fe4ddacf6	Performance	South Indian University campus (NITK, Cleaner Eng & Tech 4,
https://ijg.journals.publicknowledgeproject.org/index.php/journal/article/view/2153	Performance	Pune urban area city-wide (Vaddadi et al. 2022, Int J Geoinf
https://indiaenvironmentportal.org.in/news-clippings/auda-rule-not-enough-to-harvest-water	Mandates	Gujarat / Ahmedabad (AUDA/AMC GDCR)
https://indiatogether.org/2007/mar/env-rwhchen.htm	Functionality	Sekhar Raghavan / Rain Centre
https://irrigation.punjab.gov.in/sites/default/files/Proposal%20RWH.pdf	Cost	Punjab govt buildings standard RWH-recharge structure (propo
https://kscst.org.in/spp/49_series/49s_spp/Poster_Presentation_Synopsis/49_49S_BE_3551.pdf	Performance	PES College of Engineering Mandya, Civil Dept (KSCST student
https://madrasmusings.com/vol-29-no-6/back-to-harvesting-but-no-rain/	Functionality	CMWSSB (Metrowater) administrative count; Anna University Ph
https://naviiina.iiitb.ac.in/?p=3884	Performance	IIIT Bangalore campus RWH
https://neerain.com/rooftop-rainwater-harvesting-real-cost/	Cost	Green Valley Apartments (vendor case, NeeRain)

URL	Strand	First use
https://nl.ircwash.org/sites/default/files/CSE-2001-Managing.pdf	Functionality	Madras Institute of Development Studies (MIDS) for CSE
https://ojs.pphouse.org/index.php/IJBSM/article/view/102	Performance	TNAU campus office/department buildings (Manikandan et al. 2
https://oldiwp.indiawaterportal.org/articles/rooftop-rain-water-harvesting-arrangements-kerala-municipality-building-amendment-rules	Mandates	Kerala
https://oldiwp.indiawaterportal.org/sites/default/files/iwp2/Bangalore_Water_Supply_Sewerage_Amendment_Act_2011.pdf	Mandates	Bengaluru (BWSSB jurisdiction)
https://oldiwp.indiawaterportal.org/sites/default/files/iwp2/rooftop_rainwater_harvesting_system___hyderabad___rv_ramamohan___2012.pdf	Performance	R.V. Ramamohan residence, Hyderabad (author's own system)
https://organiser.org/2009/10/04/38978/general/r1419f082/	Benchmarks	Residential RWH with 10,000 L storage (Organiser op-ed)
https://propnewstime.com/latestnewsstories/MTI3MQ==/nagpur-rainwater-harvesting-makes-a-significant-leap-yet-only-0-18--of-buildings-comply	Functionality	Nagpur Municipal Corporation records
https://propnewstime.com/latestnewsstories/MTk1MTg=/ghaziabad-to-fine-housing-societies-inr-3-lakh-for-skipping-rainwater-harvesting	Mandates	Uttar Pradesh (Kanpur; Ghaziabad; statewide CM direction 202
https://propnewstime.com/latestnewsstories/MjA5MA==/jaipur-s-jda-reports-rainwater-harvesting-in-only-220-buildings	Functionality	Jaipur Development Authority (JDA) records
https://propnewstime.com/latestnewsstories/MjUxOTg=/goa-makes-rooftop-rainwater-harvesting-mandatory-for-large-buildings	Mandates	Goa
https://sg.news.yahoo.com/33-states-uts-adopt-rainwater-233017149.html	Mandates	India (Model Building Bye-Laws 2016, MoHUA)
https://sswm.info/node/6007	Functionality	Observer Research Foundation (ORF) expert study, reported by
https://static.vikaspedia.in/media/files_en/energy/environment/three-main-design-concepts.pdf	Benchmarks	Rain Centre Chennai: sump storage cost
https://thebetterindia.com/254751/ias-hero-rainwater-harvesting-system-schools-water-shortage-conservation-gujarat-cloudburst-floods-vadodara-district-collector-cost-effective-inspiring-women-him16	Cost	Vadodara district schools (Varsha Kal Nidhi, 963 schools)
https://theleaflet.in/chennai-on-a-rainwater-harvesting-spree-are-we-on-the-right-track	Functionality	PWD official statement (The Leaflet)
https://thelogicalindian.com/environment/sabari-terrace-rainwater/	Performance	Sabari Terrace apartments, Sholinganallur
https://theprint.in/india/maharashtra-makes-structural-fitness-provision-for-rainwater-harvesting-a-must-for-old-buildings/3010494/	Mandates	Mumbai / Maharashtra (Greater Mumbai DCR; all towns non-gaot
https://therealtytoday.com/news/over-17000-buildings-in-hyderabad-to-face-penalties-for-non-compliance-with-rainwater-harvesting-rules	Functionality	HMWS&SB survey (with 18 NGOs)
https://therealtytoday.com/news/pune-sees-tenfold-increase-in-properties-claiming-tax-rebates-for-rainwater-harvesting/	Mandates	Pune (PMC)
https://thesouthfirst.com/telangana/hmwssb-mandates-rainwater-harvesting-pits-for-17000-buildings-to-tackle-water-crisis	Mandates	Hyderabad / Telangana (HMWSSB area)
https://urbanwaters.in/uw_resource/rainwater-harvesting-groundwater-recharge-project-at-apd-bengaluru/	Performance	APD campus, Kothanur (Biome / Urban Waters)
https://urbanwaters.webindia.com/?p=3429	Cost	Nandi Deepa Apartments, Bannerghatta Rd (Hulimavu)
https://www.academia.edu/27170072/Rooftop_Rainwater_Harvesting_a_case_study	Performance	SSJCET campus Asangaon, Thane (Dagwal et al. 2016, IOSR-JMCE
https://www.amarujala.com/lucknow/up-cm-yogi-orders-rainwater-harvesting-to-be-mandatory-in-buildings-larger-than-100-square-meters-potters-ex-2025-10-04	Mandates	Uttar Pradesh (Kanpur; Ghaziabad; statewide CM direction 202
https://www.cgwb.gov.in/cgwbpnm/download/379	Performance	KITS Ramtek (Kavikulguru Institute of Technology and Science
https://www.cmwssb.tn.gov.in/tariff	Benchmarks	CMWSSB domestic metered water tariff, up to 10 kL
https://www.cseindia.org/catching-water-where-it-falls-1113	Performance	Centre for Science and Environment (CSE) building, Tughlakab
https://www.cseindia.org/costs-1097	Cost	CSE building, Tughlakabad
https://www.cseindia.org/laws-and-policy-1161	Mandates	Gujarat / Ahmedabad (AUDA/AMC GDCR)
https://www.cseindia.org/legislation-on-rainwater-harvesting-1111	Mandates	Karnataka (statewide, outside BWSSB)
https://www.cseindia.org/maintenance-monitoring-1125	Benchmarks	O&M frequency statements (CSE)
https://www.cseindia.org/node/1099	Performance	Escorts-Mahle-Goetze (Yelahanka) pilot rooftop RWH
https://www.cseindia.org/node/1101	Performance	Mira Model School, Janakpuri (CSE model project)
https://www.cseindia.org/node/1102	Performance	Janki Devi Memorial College (CSE model project)
https://www.cseindia.org/node/1970	Cost	Mira Model School
https://www.cseindia.org/node/1979	Cost	Anup Ahuja's Farm House, Sultanpur Estate
https://www.cseindia.org/node/1980	Cost	BPCL Housing Complex
https://www.cseindia.org/node/1981	Performance	Defence Colony C Block lanes (CSE model project)
https://www.cseindia.org/node/1982	Cost	Delhi Gymkhana Club
https://www.cseindia.org/node/1983	Performance	Ford Foundation (UNDP building), Lodhi Estate (CSE model pro
https://www.cseindia.org/node/1984	Performance	Hero Honda Motors, Daruhera (Rewari) (CSE model project)
https://www.cseindia.org/node/1985	Performance	Indian Spinal Injury Centre, Vasant Kunj (CSE model project)
https://www.cseindia.org/node/1986	Performance	Jamia Hamdard University (CSE model project)
https://www.cseindia.org/node/1987	Cost	Janki Devi Memorial College
https://www.cseindia.org/node/1988	Performance	Sri Aurobindo Ashram (incl. Mother's International School, M
https://www.cseindia.org/node/1989	Cost	New Friends Club
https://www.cseindia.org/node/1990	Cost	Nizamuddin East Colony
https://www.cseindia.org/node/1991	Cost	Royal Bhutanese Embassy
https://www.cseindia.org/node/1992	Performance	The Shri Ram School, Vasant Vihar (CSE model project)
https://www.cseindia.org/node/1993	Cost	Surya Vihar Apartments, Kapas Hera
https://www.cseindia.org/node/1994	Performance	Tex Corp Limited factory, Gurgaon (CSE model project)
https://www.cseindia.org/node/1995	Performance	Tihar Jail Wards 1 and 13 (CSE model project)
https://www.cseindia.org/node/1996	Cost	Mother Dairy F&V Unit, Mangolpuri
https://www.cseindia.org/panchsheel-park-colonys-rainwater-harvesting-system-747	Performance	Panchsheel Park Colony (CSE model project)
https://www.cseindia.org/water-harvesters-730	Performance	Rashtrapati Bhavan / President's Estate (CSE water harvester
https://www.dailypioneer.com/2019/state-editions/chandigarh-for-strict-enforcement-of-water-harvesting.html	Mandates	Chandigarh (UT)
https://www.deccanchronicle.com/southern-states/telangana/rain-harvest-drive-to-boost-hyderabads-groundwater-1974009	Functionality	HMWS&SB ('Intinta Inkudu Gunta' 99-day plan)
https://www.deccanherald.com/amp/story/india%2Fkarnataka%2Fbengaluru%2Fcity-fails-make-use-record-2162613	Functionality	BWSSB (administrative count)
https://www.deccanherald.com/amp/story/india%2Fkarnataka%2Fbengaluru%2Fmonth-collected-in-penalties-746109.html	Functionality	BWSSB (administrative count)
https://www.deccanherald.com/amp/story/india%2Fkarnataka%2Fbengaluru%2Frainwater-harvesting-must-for-large-properties-in-bengaluru-assembly-passes-bill-1031208.html	Mandates	Bengaluru (BWSSB jurisdiction)
https://www.deccanherald.com/amp/story/india%2Fkarnataka%2Fbengaluru%2Fwanted-a-plan-to-quench-bengalurus-thirst-1205627.html	Functionality	BWSSB count + expert estimate (Suchethana, WELL Labs) in Dec
https://www.deccanherald.com/amp/story/india%2Fkarnataka%2Fstate-may-make-rainwater-harvesting-2385947	Mandates	Karnataka (statewide, outside BWSSB)
https://www.downtoearth.org.in/amp/story/environment/tanker-economics-12249	Benchmarks	Chennai Metrowater tanker 9,000 L (2004)
https://www.downtoearth.org.in/amp/story/water/tanker-economy-revealed-dte-uncovers-how-water-suppliers-openly-exploit-the-poor-across-delhi	Benchmarks	Delhi private tanker prices (DTE investigation)
https://www.downtoearth.org.in/coverage/raintapping-12250	Functionality	MIDS sample survey commissioned by CSE (as reported by Down
https://www.dtnext.in/news/chennai/omr-residents-save-lakhs-by-collecting-rainwater-from-terrace-723983	Cost	OMR residents' welfare associations (c.40 RWAs)
https://www.etvbharat.com/en/!bharat/bangalore-water-board-revises-tariff-residents-vent-anger-over-extra-burden-enn25041005877	Benchmarks	BWSSB minimum domestic slab after April 2025 revision
https://www.freepressjournal.in/mumbai/thanes-rainwater-harvesting-mandate-fails-in-practice-amid-lack-of-post-completion-monitoring	Functionality	Thane Municipal Corporation records (Free Press Journal)
https://www.greentribunal.gov.in/sites/default/files/news_updates/PROGRESS%20REPORT%20BY%20MONITORING%20COMMITTEE%20IN%20OA%20NO.%20496%20of%202016.pdf	Mandates	Delhi (NCT)
https://www.greentribunal.gov.in/sites/default/files/news_updates/REPORT%20BY%20DJB%20in%20EA%20NO.%2004%20OF%202023%20IN%20OA%20NO.%20147%20OF%202021%20%20Mahesh%20Chandra%20Saxena%20Vs%20The%20Chief%20Secretary%20Gort.%20of%20NCT%20of%20Delhi%20&%20Ors.pdf	Functionality	Delhi Jal Board (survey + lab testing) - Action Taken Report

URL	Strand	First use
https://www.greentribunal.gov.in/sites/default/files/news_updates/Report%20by%20CGWA%20in%20EA%20No.%2034%20of%202023%20IN%20OA%20No.526%20of%202019%20(Mahesh%20Chandra%20Saxena%20Vs.%20CPCB%20&%20Ors.).pdf	Functionality	Joint Committee chaired by CGWB (with CPCB, MoHUA) for NGT O
https://www.greentribunal.gov.in/sites/default/files/news_updates/Report%20by%20DPCC%20in%20EA%20No.%2034%20of%202023%20IN%20OA%20No.%20526%20of%202019%20(Sh.%20Mahesh%20Chandra%20Saxena%20Vs%20CPCC%20&%20Anr.).pdf	Mandates	Delhi (NCT)
https://www.greentribunal.gov.in/sites/default/files/news_updates/Status%20Report%20by%20DPCC%20in%20OA%20No.%20528%20of%202022%20(Amit%20Sharma%20Vs%20Vivekanand%20School).pdf	Functionality	Joint Committee DPCC/DJB/Ground Water Authority/DM Shahdara
https://www.houseyog.com/blog/rainwater-harvesting-cost-india/	Benchmarks	O&M cost statement - secondary guide (Houseyog)
https://www.iamrenew.com/environment/after-states-ngt-comes-after-schools-over-rainwater-harvesting/	Functionality	Directorate of Education / NGT compliance (after NGT Nov 201
https://www.iamrenew.com/environment/rainwater-harvesting-systems-in-delhi-pollute-water-table-finds-expert-committee/	Mandates	Delhi (NCT)
https://www.iimb.ac.in/sites/default/files/2022-10/Green-IIMB.pdf	Performance	IIM Bangalore rooftop RWH from 7 buildings
https://www.ijert.org/roof-top-rain-water-harvesting-in-rural-areas-a-case-study-of-vidhani-village-jaipur	Cost	Vidhani village houses (Saini et al.)
https://www.ijraset.com/research-paper/rainwater-harvesting-systems-in-urban-and-rural-india	Functionality	IJRASET review paper (Reddy et al. 2025) - NO SOURCE GIVEN
https://www.inderscience.com/filter.php?aid=76372	Cost	Residential township (techno-economic tool) - Bapat, Mhaisal
https://www.indiaenvironmentportal.org.in/reports-and-documents/policy-on-rainwater-harvesting-goa	Mandates	Goa
https://www.indiatogether.org/environment/water/rwh-rkrish-may03.pdf	Performance	Akash Ganga Rain Centre, Chennai
https://www.infosys.com/sustainability/documents/esg-2023-24/water.pdf	Performance	Infosys India campuses (aggregate) rooftop + lake rainwater
https://www.internationalairportreview.com/news/161865/kempegowda-international-airport-becomes-water-positive	Performance	Kempegowda International Airport (BIAL)
https://www.longdom.org/open-access/rooftop-rainwater-harvesting-for-recharging-shallow-groundwater-39697.html	Performance	JNTU Hyderabad (Kukatpally) RWH structure Near New IST build
https://www.nwm.gov.in/sites/default/files/BEL_RWH-Ctr.pdf	Performance	Bharat Electronics Ltd (BEL) Bengaluru unit rooftop RWH
https://www.orfonline.org/research/rainwater-harvesting-strategy-delhi-government	Functionality	Delhi Jal Board survey (reported by ORF, Rumi Aijaz)
https://www.outlookindia.com/national/delhi-govt-suggests-slapping-environmental-compensation-of-up-to-rs-51-on-properties-without-functional-rainwater-harvesting-system-news-300610	Mandates	Delhi (NCT)
https://www.phytojournal.com/archives/2020/vol9issue2S/PartG/S-9-2-72-108.pdf	Performance	RK University SOE building, Rajkot (Gohel et al. 2020)
https://www.pressreader.com/india/the-hindu-bangalore-9WW1/20240321/281556590815663	Functionality	BWSSB data (The Hindu)
https://www.siasat.com/rainwater-harvesting-pits-made-mandatory-for-homes-near-hyderabads-orr-3246709/	Functionality	HMWS&SB survey (Siasat report)
https://www.siasat.com/water-bills-to-rise-by-5-pc-annually-as-hmwssb-looks-to-recover-revenue-3543386/	Benchmarks	HMWSSB domestic tariff (approx., last revision)
https://www.studiomatrx.org/guides/rainwater-harvesting	Benchmarks	O&M cost statement - secondary guide (Studio Matrx)
https://www.technoarete.org/common_abstract/pdf/IJERMCE/v8/i11/Ext_92503.pdf	Cost	Bearys Institute of Technology modular RWH, Area-1 single re
https://www.thenewsminute.com/amp/story/tamil-nadu/16-yrs-after-landmark-rainwater-harvesting-rule-tn-chennai-thirsts-water-103979	Functionality	Rain Centre (Sekhar Raghavan, Indukanth Ragade)
https://www.thenewsminute.com/amp/story/tamil-nadu/9000-houses-issued-notices-chennai-corp-install-rainwater-harvesting-systems-108206	Functionality	Greater Chennai Corporation
https://www.thenewsminute.com/amp/story/tamil-nadu/chennai-metrowater-hikes-tanker-price-without-any-public-announcement-110817	Benchmarks	Chennai Metrowater tanker 9,000 L price hike
https://www.thenewsminute.com/amp/story/tamil-nadu/chennai-water-crisis-private-tankers-fleece-public-rates-increase-over-100-103859	Benchmarks	Chennai Metrowater public tanker 9,000 L during 2019 crisis
https://www.thenewsminute.com/amp/story/tamil-nadu/install-rainwater-harvesting-systems-3-months-or-face-action-tn-govt-107476	Functionality	Tamil Nadu Government / Greater Chennai Corporation (200 war
https://www.thenewsminute.com/tamil-nadu/16-yrs-after-landmark-rainwater-harvesting-rule-tn-chennai-thirsts-water-103979	Mandates	Tamil Nadu (incl. Chennai)
https://www.thestatesman.com/india/govt-to-revive-rainwater-harvesting-systems-in-75-schools-aiming-to-save-50-crore-liters-of-water-1503596585.html	Functionality	Delhi Government technical teams (audit of CM Shri Schools)
https://www.theweek.in/wire-updates/national/2025/04/09/mes24-ka-water-hike.html	Benchmarks	BWSSB domestic tariff revision - increase by slab
https://www.tribuneindia.com/2003/20030404/cth2.htm	Performance	DAV Senior Secondary School Sector 8 Chandigarh (Tribune 200
https://www.tribuneindia.com/2011/20110716/dplus.htm	Mandates	Uttarakhand (Mussoorie-Dehradun Development Authority)
https://www.tribuneindia.com/news/archive/bathinda/state-govt-notifies-new-building-bylaws-693575	Mandates	Punjab (incl. Ludhiana)
https://www.tribuneindia.com/news/archive/haryana/rainwater-harvesting-system-a-mirage-in-kaithal-district-614691	Functionality	The Tribune field reporting (government buildings)
https://www.tribuneindia.com/news/archive/news-detail-690641	Mandates	Punjab (incl. Ludhiana)
https://www.tribuneindia.com/news/haryana/70-rainwater-harvesting-pits-lie-defunct-audit-ordered-516641/amp	Functionality	Municipal Corporation Gurugram / GMDA
https://www.tribuneindia.com/news/haryana/75-rainwater-harvesting-systems-in-faridabad-defunct-571708/amp	Functionality	Municipal Corporation Faridabad (MCF) survey team
https://www.tribuneindia.com/news/haryana/majority-of-rainwater-harvesting-systems-defunct-in-faridabad-629993/amp	Functionality	civic administration sources / survey for civic body (A K Ga
https://www.tribuneindia.com/news/haryana/rainwater-harvesting-systems-built-to-save-water-recharge-underground-water-lying-defunct	Functionality	The Tribune field reporting
https://www.unishivaji.ac.in/uploads/journal/Journal_42/22.pdf	Performance	Dahiwadi College, Man tahsil, Satara (Khilare et al. 2012)
https://www.wipro.com/content/dam/nexus/en/investor/annual-reports/2024-2025/esg-wipro-dashboard-fy24-25.pdf	Performance	Wipro (all India, aggregate)

9 Statement of contributions

Contributions. Vedant Abhangrao: search strategy, screening log and the mandate strand. Nilay Shahapure: data harmonisation, analysis code, figures and this supplementary document. Arjun Jhunjhunwala: cost and benchmark strand, deflation and levelised cost. Yash Krishnani: functionality and failure-mode strand, report drafting and the presentation. All four members extracted records and cross-checked each other's rows against the source quotes.

10 Limitations in full

- The observed tier has seven observations at five sites. Two are single-house tank records that count only what the tank could hold, so they are lower bounds. The Goa 2008 value is a partial season. Warud (112%) is graded observed by the stated rule but its volume is a round 90% of rain received and may be an assumed share; grading it agency-reported gives an observed median of 26%. The result (about a quarter of potential) is therefore indicative and should be read with the sensitivity cases in Table 5. At Ramtek the low ratio reflects the intake capacity of the dug well (19 of 215.7 m^3 accepted, the rest overflowing), a technical limit rather than neglect.
- Publication bias: demonstration and showcase systems are more likely to be written up than failed ones. This biases the measured ratio upward and the estimated gap downward.
- Rainfall basis varies: four observed or agency-reported observations use an IMD city normal (1991-2020, or 1981-2010

for Panaji) and four use the city average quoted by the source (CSE), rather than the rainfall of the year the volume was measured, so individual ratios can be off by the year's rainfall anomaly; two (Ramtek, Warud) use rain received on the roof divided by area. No IMD normal for Chandigarh was found on an official page.

- For Panjab University and CSIO the roof area (CSE, 2010) and the recharged volume (Lok Sabha, 2004) come from different documents about the same structure.
- Functionality surveys use different definitions ("correctly installed and maintained", "needing no repair", "functional", "defunct", "dry pit") and sample different populations (households, public buildings, schools). Two of the 13 estimates (Dwarka structural and Dwarka clean recharge) are two definitions applied to one DJB report, and the Bengaluru figure is an expert estimate without a sample. Several reach us only through press reports; where the original was found it is cited.
- Cost rows are dominated by one programme (CSE Delhi model projects, 2000 to 2004) whose costs are the marginal cost of recharge structures on large campuses. The five post-2010 rows are far more expensive. The levelised cost depends on the assumed life and discount rate (Table 8).
- The regression is exploratory and shows association only; with 16 observations and two predictors it cannot support causal claims, and with Warud graded observed the tier coefficient is not significant, so no claim in the main report rests on it.
- An economic explanation competes with the institutional one for recharge systems: at observed yield the owner's cost per kL exceeds the piped tariff while the recharge benefit is shared, so weak private incentives, not only weak enforcement, can explain neglect. The two explanations lead to the same recommendation (tie rebates to periodic functional inspection).
- Several government documents (on the older CGWB website and the Rajya Sabha archive) and journal full texts could not be accessed; 17 performance records are logged as not accessible.

11 Code used in this work

All numbers, tables and figures in the main report and in this document are produced by the scripts listed in this section. The scripts are supplied in the submission folder (`code` and `slides`) and are printed in full in Sections 11.4 to 11.8. Nothing in the report was computed by hand.

11.1 What each script does and where its output is used

Table 16: The five scripts, their inputs and outputs, and the parts of the report that depend on them.

Script	What it does	Reads and writes	Where the output appears
`build_dataset.py` (Python)	Builds the harmonised dataset from the source extraction tables. For every performance record it computes theoretical potential ($A \times R \times C/1000$ with $C = 0.80$, and again with 0.70 and 0.90), the performance ratio, and the share of gross rain. It assigns the evidence tier with the one grading rule (observed, agency-reported, design-calculated). For every cost record it deflates the capital cost to 2024 rupees with the CPI-IW series, annualises it (capital recovery factor for 15 years at 8 percent, plus 2 percent per year maintenance; a 15 percent run is kept as a sensitivity) and divides by the design yield. It collapses the 90 failure-mode mentions into eleven categories and writes the screening counts.	Reads the seven source extraction tables in the `research` folder. Writes the eight tables in the `data` folder (performance, cost, functionality, compliance, failure modes in two forms, mandates, the CPI series and the screening log; the codebook in that folder describes every column).	Every potential, ratio and cost-per-kL value in the performance and cost tables of this document; the tier labels used throughout; the screening counts in Section 2; the eleven failure-mode categories in Section 6.
`analysis.py` (Python)	Computes all statistics and draws all figures. Statistics: medians with bootstrap 95 percent confidence intervals (5,000 resamples), Mann-Whitney U between the observed and agency-reported tiers, the Warud regrading check, the runoff-coefficient sensitivity, site-level medians, an exploratory regression, Spearman correlation of cost against catchment area, Kruskal-Wallis across building scales, the observed-yield cost multiplier, and the Chennai and Bengaluru cascades.	Reads the `data` folder. Writes the four main and four supplementary figures (PNG and PDF) to the `figures` folder, and the results file (JSON and a readable summary) to the `results` folder.	Figures 1 to 3 of the main report and Figures 1 to 4 of this document; every statistic quoted in the Abstract, Results and Discussion (28 percent, 99 percent, 56 percent, p = 0.023 and 0.001, Rs 21, 29, 42, 98 and 200 per kL, the 13 percent cascade).
`make_supplementary_tables.py` (Python)	Converts the data tables into the LaTeX tables of this document, with sequential row numbers, the tier words instead of codes, and the reconciliation table for the six cases from the mid-term progress report.	Reads the `data` and `research` folders and the results file. Writes the `tables` folder inside `report`.	All tables in Sections 2 to 7 of this document, and the list of 168 source URLs in Section 8.

Script	What it does	Reads and writes	Where the output appears
style.py (Python)	Sets the colours, fonts and line widths shared by all figures, so that the three tiers keep the same colour in every plot.	Imported by analysis.py. Writes nothing.	The appearance of every figure.
build_deck.js (JavaScript, with the pptxgenjs library and the helper apply_theme.js)	Builds the presentation from the figures and the results file, so the numbers on the slides are read from the same results file as the report. Speaker notes are written into each slide.	Reads the figures and results folders. Writes the presentation file.	The presentation.

11.2 How to rerun everything

From the submission folder, with Python 3 (pandas, numpy, scipy, statsmodels, matplotlib) and a LaTeX installation:

```
python3 code/build_dataset.py
python3 code/analysis.py
python3 code/make_supplementary_tables.py
cd report && pdflatex report.tex && pdflatex report.tex
pdflatex supplementary.tex && pdflatex supplementary.tex
```

The first three steps take under one minute. The presentation is rebuilt with node build_deck.js inside the slides folder after npm install pptxgenjs. Two files are not generated by code: the two files in research/sources, which are verbatim extracts typed from the CGWB 2007 manual and the CGWB 2011 compilation after reading them page by page, and the source extraction tables in research, which were filled by hand from the sources listed in Section 8.

11.3 Where the key numbers are computed

For a reader who wants to check one number without reading the whole script:

- Theoretical potential and the performance ratio: build_dataset.py, the lines that create the columns potential_kL and performance_ratio (and the _c070 and _c090 variants).
- Evidence tier: build_dataset.py, the tier column in the curated performance list, with the grading rule stated in the comment above it; the observation notes record what each source says about measurement.
- Deflation and levelised cost: build_dataset.py, the constants LIFE, OM_SHARE and DISC, the capital recovery factor, and the columns capital_2024_inr, annual_cost_2024_inr and cost_per_kL_design_2024.
- Tier medians, confidence intervals and the Mann-Whitney test: analysis.py, Part 1 (Performance gap).
- The Warud regrading check and the runoff sensitivity: analysis.py, Part 1, the blocks labelled T1_excl_warud, T2_incl_warud and the median_c070 and median_c090 entries.
- The Chennai and Bengaluru cascades: analysis.py, Part 2 (Mandate gap), the mandate_gap dictionary.
- Failure-mode counts by distinct source: build_dataset.py (category mapping) and analysis.py, Part 3.
- Cost medians, the Spearman and Kruskal-Wallis tests, the observed-yield multiplier: analysis.py, Part 4 (Cost-effectiveness).

In the code the three tiers are stored as the codes T1, T2 and T3; these mean observed, agency-reported and design-calculated respectively, and the tier_name column in the performance table carries the words.

11.4 build_dataset.py

```python
"""Build the harmonised MIRAGE dataset from the raw research tables.

Inputs : research/*.csv (raw extraction, one row per source record)
Outputs: data/mirage_performance.csv   performance observations (yield vs potential)
         data/mirage_cost.csv          cost rows with levelised cost per kL
         data/mirage_functionality.csv functionality surveys with a measured split
         data/mirage_failure_modes.csv failure-mode mentions, collapsed to 11 categories
         data/mirage_mandates.csv      cleaned mandate table
         data/screening_log.csv        PRISMA-style counts

Every curated value below is copied from the research tables; the research_id
column points to the raw row, which carries the URL and the verbatim quote.
Run:  python3 code/build_dataset.py   (from the project root)
"""
from pathlib import Path
import pandas as pd
import numpy as np

ROOT = Path(__file__).resolve().parents[1]
RES = ROOT / "research"
DATA = ROOT / "data"
DATA.mkdir(exist_ok=True)

# ---------------------------------------------------------------------------
# 1. Performance observations
# ---------------------------------------------------------------------------
# tier: T1 = the source names a measuring instrument (flow meter, recharge-measuring
#            device) or a tank-filling record for the site. Applied mechanically: Ramtek and
#            Warud (identical CGWB wording, both with a flow meter) are both T1.
#       T2 = annual volume reported by an implementing agency with no instrument named,
#            or stated as a share of estimated runoff
#       T3 = design calculation (reported as harvested but equal to a fixed
#            share of area x rainfall)
# rainfall_basis: site-year = rainfall for the site and year of the volume;
#       site-avg = average annual rainfall quoted by the source for the site;
#       city-normal = IMD normal for the city (period in note);
#       implied = rainfall received on the roof reported by the source, divided by roof area.
PERF = [
 # obs_id, site, city, state, bldg, sys, tier, area, rain, rain_basis, year, vol, vol_type, storage_kL, installed, note, source, url, research_id, midway_ratio
 ("P01","KITS Ramtek, IT building","Ramtek (Nagpur dist.)","Maharashtra","institutional","recharge","T1",360,599.2,"implied","2001-02",19.0,"recharged",np.nan,2001,
  "Flow meter in line. CGWB: 215.7 m3 received on roof, 19 m3 recharged (as per estimation). Rainfall = 215.7/360.","CGWB 2011, p.73","https://cgwb.gov.in/cgwbpnm/download/379","A1 (verified from PDF p.73)","9%"),
 ("P02","Panchayat Samiti office, Warud","Warud (Amravati dist.)","Maharashtra","public","recharge","T1",500,622.6,"implied","2002-03",280.17,"recharged",np.nan,2002,
  "Flow meter in line (same CGWB wording as Ramtek). Volume stated as 280.17 m3 (90%) of 311.30 m3 received; the round 90% may be an assumption, see Results. Rainfall = 311.3/500.","CGWB 2011, p.73","https://cgwb.gov.in/cgwbpnm/download/379","A5 (verified from PDF p.73)","90%"),
 ("P03","IIT Delhi, Block VI","New Delhi","Delhi","institutional","recharge","T2",1660,762.3,"city-normal","NR (c. 2001-05)",830.0,"recharged",np.nan,2001,
  "CGWB: about 830 m3 recharged from 1660 m2 roof; period not stated. Rainfall = IMD Safdarjung normal 1991-2020.","CGWB 2011, p.44","https://cgwb.gov.in/cgwbpnm/download/379","A6 (verified from PDF p.44)","NR (rainfall not reported)"),
 ("P05","Goa University, 2010","Taleigao","Goa","institutional","both","T1",400,3700,"site-year","2010",900.0,"recharged",100,2008,
  "Recharge measured with a device at the bore well (CSE); Chachadi 2013 gives 934 kL to 11 Nov 2010. 2010 monsoon rainfall 3.7 m reported.","CSE 2014 Urban RWH report; Chachadi 2013","https://cdn.cseindia.org/userfiles/Urban Rainwater harvesting report.pdf","A2","61-76%"),
 ("P06","Goa University, 2008 (partial year)","Taleigao","Goa","institutional","both","T1",400,2918,"city-normal","2008",260.0,"recharged",100,2008,
  "System built July 2008, so 2008 is a partial season. Rainfall = IMD Panaji normal 1981-2010.","CSE 2014 Urban RWH report","https://cdn.cseindia.org/userfiles/Urban Rainwater harvesting report.pdf","A2",""),
 ("P07","Goa University, 2009","Taleigao","Goa","institutional","both","T1",400,2918,"city-normal","2009",180.0,"recharged",100,2008,
  "Rainfall = IMD Panaji normal 1981-2010 (2009 site rainfall not reported).","CSE 2014 Urban RWH report","https://cdn.cseindia.org/userfiles/Urban Rainwater harvesting report.pdf","A2",""),
 ("P08","Bharat Electronics Ltd, Bengaluru","Bengaluru","Karnataka","industrial","storage","T2",2000,1077.6,"city-normal","annual (post 2003)",1100.0,"harvested (stored)",np.nan,2003,
  "'Around 1100 KL of rainwater harvested annually' for DM water production; method not stated. Rainfall = IMD Bengaluru normal 1991-2020.","NWM/BEL case sheet","https://www.nwm.gov.in/sites/default/files/BEL_RWH-Ctr.pdf","A3","63-75%"),
 ("P09","Panjab University, Basic Medical Sciences block","Chandigarh","Chandigarh","institutional","recharge","T2",2100,1059.3,"city-avg (CSE)","annual (c. 2004)",1985.0,"recharged",np.nan,2001,
  "Lok Sabha reply: 1985 m3 (93% of total generated runoff) recharged annually. Roof area from CSE 2010. No IMD normal located for Chandigarh; CSE city figure used.","Lok Sabha SQ 384 (23.08.2004); CSE 2010","https://eparlib.sansad.in/bitstream/123456789/520443/1/2310.pdf","A4","79%"),
 ("P10","CSIO complex","Chandigarh","Chandigarh","institutional","recharge","T2",3550,1059.3,"city-avg (CSE)","annual (c. 2004)",3812.0,"recharged",np.nan,1999,
  "Lok Sabha reply: 3812 m3 recharged annually. Roof area 3550 m2 from CSE 2010.","Lok Sabha SQ 384 (23.08.2004); CSE 2010","https://eparlib.sansad.in/bitstream/123456789/520443/1/2310.pdf","B2",""),
 ("P11","Hydrology Project building, Bhubaneswar, 2004","Bhubaneswar","Odisha","public","recharge","T2",2250,1096,"site-year","2004",2096.0,"harvested",np.nan,2004,
  "CGWB Table 34; project 'monitored weekly'; method of volume not stated; 2004 value equals 85% of gross rainfall exactly.","CGWB 2011, p.116","https://cgwb.gov.in/cgwbpnm/download/379","new (PDF p.116)",""),
 ("P12","Hydrology Project building, Bhubaneswar, 2005","Bhubaneswar","Odisha","public","recharge","T2",2250,1579,"site-year","2005",2566.86,"harvested",np.nan,2004,
  "CGWB Table 34.","CGWB 2011, p.116","https://cgwb.gov.in/cgwbpnm/download/379","new (PDF p.116)",""),
 ("P13","Hydrology Project building, Bhubaneswar, 2006","Bhubaneswar","Odisha","public","recharge","T2",2250,1881,"site-year","2006",2174.23,"harvested",np.nan,2004,
  "CGWB Table 34.","CGWB 2011, p.116","https://cgwb.gov.in/cgwbpnm/download/379","new (PDF p.116)",""),
 ("P14","Escorts-Mahle-Goetze, Yelahanka","Bengaluru","Karnataka","industrial","storage","T2",1280,970,"city-avg (CSE)","annual (c. 2001)",1050.0,"collected",4.2,2000,
  "CSE: 'collects about 1.05 million litres per year'; method not stated.","CSE case study page","https://www.cseindia.org/node/1099","B25",""),
```

```
 ("P15","K.G. Bhawan (CWC office)","Hyderabad","Telangana","public","recharge","T2",1577.8,800,"site-avg","NR",1000.0,"harvested",np.nan,np.nan,
  "CWC: 'Water harvested : 10 lakh litres (approx)'; method not stated.","CWC RWH note","https://cwc.gov.in/sites/default/files/rwh.pdf","B29",""),
 ("P16","R. Ramani residence, Korattur","Chennai","Tamil Nadu","individual house","storage","T1",50,1200,"city-avg (CSE)","annual (c. 2001)",13.5,"collected (tank record)",3,1994,
  "3 kL tank fills four to five times a year (12-15 kL); storage-limited, so a lower bound on roof yield. Midpoint used.","CSE case study page","https://www.cseindia.org/node/1099","B26",""),
 ("P17","R.V. Ramamohan residence","Hyderabad","Telangana","individual house","storage","T1",100,804,"site-avg","2012 (5 months)",15.0,"collected (tank record)",1,2012,
  "About 15,000 L collected over 5 months with a 1 kL tank; storage-limited lower bound.","India Water Portal (Ramamohan 2012)","https://oldiwp.indiawaterportal.org/sites/default/files/iwp2/rooftop_rainwater_harvesting_system___hyderabad___rv_ramamohan___2012.pdf","B27",""),
]
# Design-calculated rows (T3): CSE model projects where "total volume harvested" is a
# fixed share of area x rainfall. Volume and area from research/cost_cases.csv rows C01-C19.
T3 = [
 ("P18","Mira Model School","New Delhi","school","recharge",16200,611,4454,"C01/B13","45% of potential"),
 ("P19","Janki Devi Memorial College","New Delhi","institutional","recharge",32170,611,6880,"C02/B14","35% of potential"),
 ("P20","Jamia Hamdard University","New Delhi","institutional","recharge",315380,611,67444,"C03/B15","35% of potential"),
 ("P21","The Shri Ram School","New Delhi","school","both",6000,611,1890,"C04/B16","55% of potential"),
 ("P22","Tex Corp Ltd, Gurgaon","Gurgaon","industrial","storage",4000,577.8,1962,"C05/B17","85% of potential"),
 ("P23","Ford Foundation","New Delhi","institutional","recharge",12500,611,6491.8,"C06/B18","85% of potential"),
 ("P24","Sri Aurobindo Ashram","New Delhi","institutional","recharge",95870,611,16366,"C07/A7b","43% of potential"),
 ("P25","Tihar Jail wards 1 and 13","New Delhi","public","recharge",4125,611,1280,"C08/B19","50.78% of potential"),
 ("P26","Hero Honda Motors, Daruhera","Daruhera","industrial","recharge",11080,577.8,5441,"C12/B20","85% of potential"),
 ("P27","Defence Colony C block","New Delhi","residential colony","recharge",26087,611.8,9563,"C11/B21","share not stated"),
 ("P28","Indian Spinal Injury Centre","New Delhi","institutional","recharge",52598,611,27317,"C09/B22","share not stated"),
 ("P29","New Friends Club","New Delhi","institutional","recharge",3700,611,960.79,"C16","42% of runoff"),
 ("P30","Delhi Gymkhana Club","New Delhi","institutional","recharge",113344,611,28765.8,"C17","43% of runoff"),
 ("P31","Nizamuddin East colony","New Delhi","residential colony","recharge",200916,611,48531,"C18","39.6% of runoff"),
 ("P32","Royal Bhutanese Embassy","New Delhi","institutional","recharge",31363,611,7867,"C19","26% of runoff"),
 ("P33","Mother Dairy, Mangolpuri","New Delhi","industrial","recharge",89370,611,32250,"C15","59% of runoff"),
 ("P34","Anup Ahuja farm house","New Delhi","individual house","recharge",1250,611,272,"C13","60% of runoff"),
 ("P35","Surya Vihar apartments, Kapas Hera","Gurgaon","residential colony","recharge",44029,611,18207,"C14","share not stated"),
 ("P36","Shram Shakti Bhawan (CGWB design)","New Delhi","institutional","recharge",11965,712.2,3000,"B1 / CGWB 2011 p.44","expected recharge 3000 m3/yr from runoff 3325 m3"),
]
cols = ["obs_id","site","city","state","building_type","system_type","tier","catchment_area_m2","rainfall_mm","rainfall_basis","year_of_volume","reported_volume_kL","volume_type","storage_capacity_kL","year_installed","measurement_note","source","url","research_id","midway_ratio"]
perf = pd.DataFrame(PERF, columns=cols)
t3rows = []
for o in T3:
    t3rows.append(dict(obs_id=o[0], site=o[1], city=o[2], state=("Delhi" if o[0]=="P36" else "Delhi/Haryana"), building_type=o[3], system_type=o[4], tier="T3",
        catchment_area_m2=o[5], rainfall_mm=o[6], rainfall_basis="city-avg (CSE)", year_of_volume="annual (design)", reported_volume_kL=o[7],
        volume_type="harvested (calculated)", storage_capacity_kL=np.nan, year_installed=np.nan,
        measurement_note=(f"CGWB design note: {o[9]}; campus catchment (roof + paved)." if o[0]=="P36" else f"CSE model project page: volume is a calculation ({o[9]}); whole-site catchment (roof + paved + open)."),
        source=("CGWB 2011 p.44; NPTEL design note" if o[0]=="P36" else "CSE model project pages (2000-2004)"), url=("https://cgwb.gov.in/cgwbpnm/download/379" if o[0]=="P36" else "https://www.cseindia.org/rainwater-harvesting-case-studies"), research_id=o[8], midway_ratio=""))
perf = pd.concat([perf, pd.DataFrame(t3rows)], ignore_index=True)

C_CENTRAL, C_LOW, C_HIGH = 0.80, 0.70, 0.90   # CGWB 2007 Table 7.2: concrete 0.7 ... GI sheet 0.9
for tag, c in [("", C_CENTRAL), ("_c070", C_LOW), ("_c090", C_HIGH)]:
    perf["potential_kL"+tag] = perf.catchment_area_m2 * perf.rainfall_mm * c / 1000
    perf["performance_ratio"+tag] = perf.reported_volume_kL / perf["potential_kL"+tag]
perf["share_of_gross_rain"] = perf.reported_volume_kL / (perf.catchment_area_m2 * perf.rainfall_mm / 1000)
perf["log10_area"] = np.log10(perf.catchment_area_m2)
perf["runoff_coefficient_central"] = C_CENTRAL
perf.insert(perf.columns.get_loc("tier") + 1, "tier_name", perf.tier.map({"T1": "observed", "T2": "agency-reported", "T3": "design-calculated"}))
perf.to_csv(DATA / "mirage_performance.csv", index=False)

# ---------------------------------------------------------------------------
# 2. Cost rows
# ---------------------------------------------------------------------------
# capital cost (Rs, nominal in cost_year), annual yield (kL/yr) as reported by the source,
# yield_basis: design = computed by the source; observed = reported from monitoring.
COST = [
 # id, site, city, scale, system, area, capital, year, yield, basis, research_id, url
 ("K01","Mira Model School","New Delhi","institutional","recharge",16200,120000,2001,4454,"design","C01","https://www.cseindia.org/rainwater-harvesting-case-studies"),
 ("K02","Janki Devi Memorial College","New Delhi","institutional","recharge",32170,70000,2001,6880,"design","C02",""),
 ("K03","Jamia Hamdard University","New Delhi","institutional","recharge",315380,652000,2001,67444,"design","C03",""),
 ("K04","The Shri Ram School","New Delhi","institutional","both",6000,125000,2000,1890,"design","C04",""),
 ("K05","Tex Corp Ltd","Gurgaon","industrial","storage",4000,550000,2002,1962,"design","C05",""),
 ("K06","Ford Foundation","New Delhi","institutional","recharge",12500,400000,2000,6491.8,"design","C06",""),
 ("K07","Sri Aurobindo Ashram","New Delhi","institutional","recharge",95870,400000,2003,16366,"design","C07",""),
 ("K08","Tihar Jail wards 1 and 13","New Delhi","institutional","recharge",4125,20000,2002,1280,"design","C08",""),
 ("K09","Indian Spinal Injury Centre","New Delhi","institutional","recharge",52598,80000,2002,27317,"design","C09",""),
 ("K10","BPCL housing complex","Noida","residential colony","recharge",13910,450000,2002,4446,"design","C10",""),
 ("K11","Defence Colony C block","New Delhi","residential colony","recharge",26087,258000,2003,9563,"design","C11",""),
 ("K12","Hero Honda Motors","Daruhera","industrial","recharge",11080,1200000,2004,5441,"design","C12",""),
 ("K13","Anup Ahuja farm house","New Delhi","individual house","recharge",1250,20000,2002,272,"design","C13",""),
 ("K14","Surya Vihar apartments","Gurgaon","residential colony","recharge",44029,65000,2002,18207,"design","C14",""),
 ("K15","Mother Dairy unit","New Delhi","industrial","recharge",89370,435000,2004,32250,"design","C15",""),
 ("K16","New Friends Club","New Delhi","institutional","recharge",3700,40000,2004,960.79,"design","C16",""),
 ("K17","Delhi Gymkhana Club","New Delhi","institutional","recharge",113344,80000,2004,28765.8,"design","C17",""),
 ("K18","Nizamuddin East colony","New Delhi","residential colony","recharge",200916,174000,2004,48531,"design","C18",""),
 ("K19","Royal Bhutanese Embassy","New Delhi","institutional","recharge",31363,110000,2004,7867,"design","C19",""),
```

```
 ("K20","Brig. Jagdev Singh residence","New Delhi","individual house","recharge",285,7500,2001,147,"design","C22",""),
 ("K21","Shram Shakti Bhawan","New Delhi","institutional","recharge",11965,410000,2001,3000,"design","C25 / CGWB 2011 p.44","https://cgwb.gov.in/cgwbpnm/download/379"),
 ("K22","Nandi Deepa apartments","Bengaluru","residential colony","recharge",np.nan,2220000,2021,3564,"design","C27",""),
 ("K23","Vidhani village houses, cluster (Saini et al.)","Jaipur","residential colony","storage",2466,131498,2016,1602,"design","C34",""),
 ("K24","TSSM BSCOER and JSPM campuses (Varpe et al.)","Pune","institutional","both",8200,6204336,2019,5453,"design","C35",""),
 ("K25","Bearys modular house system (Raju et al.)","Mangaluru","individual house","storage",81.3,5710,2021,20.83,"design","C36",""),
 ("K26","Punjab govt building standard design (900 m2)","Punjab","institutional","recharge",900,281079,2020,438,"design","C37","",""),
 ("K27","Hydrology Project building, Bhubaneswar","Bhubaneswar","institutional","recharge",2250,856000,2004,2279.0,"agency-monitored, method not stated (mean 2004-06)","CGWB 2011 p.116","https://cgwb.gov.in/cgwbpnm/download/379"),
]
cost = pd.DataFrame(COST, columns=["cost_id","site","city","building_scale","system_type","catchment_area_m2","capital_cost_inr","cost_year","annual_yield_kL","yield_basis","research_id","url"])
cpi = pd.read_csv(DATA / "cpi_iw_deflator.csv").set_index("fy_start_year")["linked_2001_100"]
BASE_YEAR = 2024
cost["cpi_factor_to_2024"] = cost.cost_year.map(lambda y: cpi[BASE_YEAR] / cpi[int(y)])
cost["capital_2024_inr"] = cost.capital_cost_inr * cost.cpi_factor_to_2024
LIFE, OM_SHARE, DISC = 15, 0.02, 0.08       # CGWB 2007 example: 15-year life, O&M 2 % of capital; 8 % discount is our assumption (CGWB example uses 15 %)
crf = DISC * (1 + DISC) ** LIFE / ((1 + DISC) ** LIFE - 1)
cost["annual_cost_2024_inr"] = cost.capital_2024_inr * (crf + OM_SHARE)
cost["cost_per_kL_design_2024"] = cost.annual_cost_2024_inr / cost.annual_yield_kL
crf15 = 0.15 * 1.15 ** LIFE / (1.15 ** LIFE - 1)
cost["cost_per_kL_design_2024_disc15"] = cost.capital_2024_inr * (crf15 + OM_SHARE) / cost.annual_yield_kL
cost["capital_per_m2_2024"] = cost.capital_2024_inr / cost.catchment_area_m2
cost["capital_per_kL_yield_2024"] = cost.capital_2024_inr / cost.annual_yield_kL
cc_raw = pd.read_csv(RES / "cost_cases.csv").set_index("case_id")["url"]
cost["url"] = [u if isinstance(u, str) and u.strip() else cc_raw.get(str(rid).split(" ")[0].split("/")[0], "") for u, rid in zip(cost.url, cost.research_id)]
cost.to_csv(DATA / "mirage_cost.csv", index=False)

# ---------------------------------------------------------------------------
# 3. Functionality surveys with a measured split
# ---------------------------------------------------------------------------
FUNC = [
 # id, label, city, year, population, n, pct_functional, definition, who, research_id
 ("F02","Chennai 2019, houses with a structure","Chennai",2019,"houses with an RWH structure (1.62 lakh of 2.72 lakh inspected)",162000,76.5,"1.62 lakh houses had a structure; about 38,000 of them 'needed minor repairs', so 1.24 lakh (76.5%) needed no repair. Single GCC snapshot (Aug-Sep 2019).","Greater Chennai Corporation (The News Minute)","S10"),
 ("F03","Chennai Gandhinagar 2003","Chennai",2003,"independent houses",309,50.0,"'only 50 per cent had implemented RWH properly'","Akashganga Trust (Down To Earth 2004)","S03"),
 ("F04","Chennai approved buildings 2001-02","Chennai",2002,"buildings approved under 1994 rule, with RWH",25,67.0,"a third of installed systems 'not properly designed and therefore ineffective'","MIDS survey for CSE","S01/S02"),
 ("F05","Karnataka school structures 2007","Karnataka (rural)",2007,"government school rooftop structures",1269,11.0,"140 of 1,269 structures functional","Arghyam status report","S21"),
 ("F06","Karnataka model schools 2009","Karnataka (rural)",2009,"school rooftop structures",27,59.3,"16 of 27 functioning; 3 partial; 8 defunct","Arghyam","S22"),
 ("F07","Faridabad public systems 2018-19","Faridabad",2019,"public RWH systems (parks, schools)",190,20.0,"'around 80% had stopped working'","MCF survey (The Tribune)","S30"),
 ("F08","Faridabad public systems 2024","Faridabad",2024,"public RWH systems",200,25.0,"'around 70 to 80 per cent' non-functional (midpoint)","civic sources (The Tribune)","S31"),
 ("F09","Gurugram pits 2023","Gurugram",2023,"RWH pits, public and private",500,30.0,"'around 70%' lying defunct","MCG / GMDA (The Tribune)","S32"),
 ("F10","Delhi Dwarka societies 2024 (structural)","Delhi",2024,"group housing societies with RWH",176,83.5,"176 minus 4 non-functional minus 25 dry pits","DJB action-taken report to NGT","S27"),
 ("F11","Delhi Dwarka societies 2024 (clean recharge)","Delhi",2024,"group housing societies with RWH",176,32.4,"57 of 176 without faecal coliform in recharge water (119 contaminated)","DJB action-taken report to NGT","S27"),
 ("F12","Bengaluru BWSSB connections 2023 (expert estimate)","Bengaluru",2023,"BWSSB connections with RWH",np.nan,40.0,"'only about 40% of BWSSB connections have functional RWH systems' (estimate)","S. Vishwanath (Deccan Herald)","S17"),
 ("F13","Kaithal government buildings 2018","Kaithal",2018,"government-building systems",5,0.0,"all five choked after the first year","The Tribune field report","S33"),
 ("F14","Sirsa park systems 2025","Sirsa",2025,"municipal park systems",4,50.0,"2 of 4 dead within two years","The Tribune field report","S34"),
]
func = pd.DataFrame(FUNC, columns=["func_id","label","city","year","population","sample_n","pct_functional","definition","conducted_by","research_id"])
func.to_csv(DATA / "mirage_functionality.csv", index=False)

# Installation (compliance) figures: share of mandated/connected properties with a structure
COMPL = [
 ("Chennai 2019","Chennai",2019,"houses inspected",272000,162000,59.6,"1.62 lakh of 2.72 lakh inspected had a structure","mandates.csv / S10"),
 ("Bengaluru 2023","Bengaluru",2023,"properties with water connection",1080000,190000,17.6,"1.9 lakh of 10.8 lakh","mandates.csv / S18"),
 ("Hyderabad 2024","Hyderabad",2024,"properties >= 300 sq yd surveyed",39000,22000,56.4,"22,000 installed; 17,000 non-compliant","mandates.csv / S36"),
 ("Mumbai 2007-21","Mumbai",2021,"housing societies (estimate)",20000,3000,15.0,"about 3,000 of about 20,000 societies","mandates.csv / S39"),
 ("Pune FY2022-23","Pune",2023,"taxable properties (rebate claimants)",1250000,3716,0.3,"3,716 rebate claimants of about 12.5 lakh properties","mandates.csv"),
 ("Nagpur 2023","Nagpur",2023,"properties",636000,1123,0.18,"1,123 of about 6.36 lakh properties","mandates.csv / S43"),
 ("Delhi govt buildings 2019","Delhi",2019,"government buildings (building class, not a city rate)",771,212,27.5,"212 of 771 with RWH","S24"),
]
compl = pd.DataFrame(COMPL, columns=["label","city","year","population","denominator","numerator","pct_with_structure","definition","research_id"])
compl.to_csv(DATA / "mirage_compliance.csv", index=False)

# ---------------------------------------------------------------------------
# 4. Failure modes: collapse the raw categories to 11 groups (10 named + other), count distinct sources
```

```python
# ---------------------------------------------------------------------------
fm = pd.read_csv(RES / "failure_modes.csv")
def collapse(c):
    c = str(c).lower()
    if c.startswith("poor construction"): return "Poor design or construction"
    if c.startswith("no maintenance"): return "No maintenance, abandoned"
    if c.startswith("lack of awareness"): return "No awareness, incentive or enforcement"
    if c.startswith("structure built only"): return "Token structure, built for compliance only"
    if c.startswith("silted"): return "Recharge pit or well silted or choked"
    if c.startswith("disconnected"): return "Downpipes or conveyance broken or disconnected"
    if c.startswith("clogged"): return "Filter clogged or absent"
    if c.startswith("undersized"): return "Storage undersized, overflow or leak"
    if c.startswith("first-flush"): return "First flush absent or not operated"
    if c.startswith("water quality"): return "Water quality concerns, system disused"
    return "Other (space, vandalism, roof-only design)"
fm["category"] = fm.failure_mode_category.map(collapse)
fm_counts = fm.groupby("category").agg(mentions=("fm_id","count"), distinct_sources=("survey_or_source_id","nunique"), cities
    ↪ =("city", lambda s: ", ".join(sorted(set(map(str, s)))))).reset_index().sort_values("distinct_sources", ascending=
    ↪ False)
fm_counts.to_csv(DATA / "mirage_failure_modes.csv", index=False)
fm[["fm_id","survey_or_source_id","city","category","failure_mode_category","raw_description","count_or_pct","url"]].to_csv(
    ↪ DATA / "mirage_failure_modes_long.csv", index=False)

# ---------------------------------------------------------------------------
# 5. Mandates: copy with trimmed text
# ---------------------------------------------------------------------------
m = pd.read_csv(RES / "mandates.csv")
m.to_csv(DATA / "mirage_mandates.csv", index=False)

# ---------------------------------------------------------------------------
# 6. Screening log (PRISMA-style). Counts from the research notes.
# ---------------------------------------------------------------------------
perf_raw = pd.read_csv(RES / "performance_cases.csv")
n_perf_records = len(perf_raw)
LOG = [
 ("Performance strand: site records identified (CGWB, CSE, Lok Sabha replies, academic, campus and company reports)", 89, "72
    ↪  rows in research/performance_cases.csv plus 17 records logged in performance_notes.md as not accessible"),
 ("Performance strand: records not accessible (documents that could not be opened or were paywalled)", 17, "performance_notes
    ↪ .md access log"),
 ("Performance strand: records screened in detail (72 from the performance search, 6 CSE model projects from the cost search,
    ↪  1 case read directly from the CGWB 2011 compilation)", 79, "performance_cases.csv; cost_cases.csv C13, C14, C16-C19;
    ↪  CGWB 2011 p.116"),
 ("Excluded: potential only (no actual volume reported)", 20, "rows P1-P19 and B24"),
 ("Excluded: no catchment area or no volume", 9, "rows N1-N9"),
 ("Excluded: modelled volumes (JNTU Hyderabad, 3 structures)", 3, "rows B30a-c"),
 ("Excluded: agency volumes with no roof area (Lok Sabha 2004 list)", 10, "rows B3-B12"),
 ("Excluded: mixed or non-roof catchment, aggregate, or internally inconsistent", 5, "rows B28 (SV Symphony), B31 (Infosys
    ↪ aggregate), B32 (Goa surface runoff), A7a (Umaid Heritage), A7c (PAU, no area)"),
 ("Included site records", 32, "25 rows from performance_cases.csv (A1/A1b count as one site; B1 Shram Shakti as a design
    ↪ calculation) + 6 from cost_cases.csv + 1 from CGWB PDF"),
 ("Included observations (some sites have several years)", int(len(perf)), "data/mirage_performance.csv: 35 observations at
    ↪ 31 sites"),
 ("Included as T1 observed volumes (instrument or tank record named)", int((perf.tier=="T1").sum()), "P01, P02, P05-P07, P16,
    ↪  P17 (5 sites)"),
 ("Included as T2 agency-reported volumes", int((perf.tier=="T2").sum()), "P03, P08-P15 (7 sites)"),
 ("Included as T3 design calculations (CSE model projects and CGWB design note)", int((perf.tier=="T3").sum()), "P18-P36 (19
    ↪ sites)"),
 ("Cost strand: case records screened", 43, "research/cost_cases.csv"),
 ("Cost strand: rows with both capital cost and annual yield and a cost year, used for levelised cost", len(cost), "data/
    ↪ mirage_cost.csv"),
 ("Cost strand: benchmark price rows", 37, "research/cost_benchmarks.csv"),
 ("Functionality strand: survey records screened", 46, "research/functionality_surveys.csv"),
 ("Functionality strand: surveys with a measured functional / defunct split", len(func), "data/mirage_functionality.csv"),
 ("Functionality strand: city compliance (installation) figures", len(compl), "data/mirage_compliance.csv"),
 ("Failure-mode mentions extracted", len(fm), "research/failure_modes.csv"),
 ("Failure-mode distinct sources", int(fm.survey_or_source_id.nunique()), "research/failure_modes.csv, survey_or_source_id"),
 ("Failure-mode categories after collapsing", int(fm.category.nunique()), "data/mirage_failure_modes.csv"),
 ("Mandate jurisdictions tabulated", len(m), "data/mirage_mandates.csv"),
]
pd.DataFrame(LOG, columns=["step","count","where"]).to_csv(DATA / "screening_log.csv", index=False)

print("performance obs:", len(perf), perf.tier.value_counts().to_dict())
print("cost rows:", len(cost))
print("functionality rows:", len(func), "| compliance rows:", len(compl))
print(fm_counts.to_string(index=False))
```

11.5 analysis.py

```python
"""MIRAGE analysis: statistics and figures.

Reads  data/mirage_*.csv
Writes figures/fig*.png (300 dpi) and figures/fig*.pdf, results/results.json, results/results.md
Run:   python3 code/analysis.py   (from the project root)
"""
import json
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.lines import Line2D
from scipy import stats
import statsmodels.api as sm
import sys

ROOT = Path(__file__).resolve().parents[1]
sys.path.insert(0, str(ROOT / "code"))
import style  # noqa: E402

style.apply()
DATA, FIG, RESD = ROOT / "data", ROOT / "figures", ROOT / "results"
FIG.mkdir(exist_ok=True); RESD.mkdir(exist_ok=True)
rng = np.random.default_rng(2206)

BLUE, ORANGE, VIOLET, INK, INK2, MUTED, GRID = "#2a78d6", "#eb6834", "#6250d6", "#0b0b0b", "#52514e", "#8a8984", "#e6e5e1"
TIER = {"T1": ("Observed volume (instrument or tank record)", BLUE, "o"),
        "T2": ("Agency-reported volume", ORANGE, "s"),
        "T3": ("Design calculation (CSE, CGWB design notes)", VIOLET, "^")}

perf = pd.read_csv(DATA / "mirage_performance.csv")
cost = pd.read_csv(DATA / "mirage_cost.csv")
func = pd.read_csv(DATA / "mirage_functionality.csv")
compl = pd.read_csv(DATA / "mirage_compliance.csv")
fm = pd.read_csv(DATA / "mirage_failure_modes.csv")
fm_long = pd.read_csv(DATA / "mirage_failure_modes_long.csv")
N_FM_SOURCES = int(fm_long.survey_or_source_id.nunique()); N_FM_MENTIONS = int(len(fm_long)); N_FM_CATS = int(fm.category.nunique())
R = {}

def r0(v):
    """Round half up to an integer, for labels (Python rounds half to even)."""
    import math
    return int(math.floor(v + 0.5))

def save(fig, name, tight=True):
    bb = "tight" if tight else None
    fig.savefig(FIG / f"{name}.png", bbox_inches=bb); fig.savefig(FIG / f"{name}.pdf", bbox_inches=bb); plt.close(fig)

def boot_median(x, n=5000):
    x = np.asarray(x, float); meds = [np.median(rng.choice(x, len(x), replace=True)) for _ in range(n)]
    return float(np.median(x)), float(np.percentile(meds, 2.5)), float(np.percentile(meds, 97.5))

# ---------------------------------------------------------------------------
# 1. Performance gap
# ---------------------------------------------------------------------------
pr = {}
for t in ["T1", "T2", "T3"]:
    x = perf.loc[perf.tier == t, "performance_ratio"]
    med, lo, hi = boot_median(x)
    pr[t] = dict(n=int(len(x)), sites=int(perf.loc[perf.tier == t, "site"].str.replace(r", \d{4}.*", "", regex=True).str.replace(" (partial year)", "").nunique()),
                 median=med, ci_lo=lo, ci_hi=hi, min=float(x.min()), max=float(x.max()),
                 share_ge_1=float((x >= 0.99).mean()),
                 median_c070=float(perf.loc[perf.tier == t, "performance_ratio_c070"].median()),
                 median_c090=float(perf.loc[perf.tier == t, "performance_ratio_c090"].median()))
t1, t2 = perf.loc[perf.tier == "T1", "performance_ratio"], perf.loc[perf.tier == "T2", "performance_ratio"]
u = stats.mannwhitneyu(t1, t2, alternative="two-sided")
pr["T1_vs_T2_mannwhitney"] = dict(U=float(u.statistic), p=float(u.pvalue))
t1x = perf.loc[(perf.tier == "T1") & (perf.obs_id != "P06"), "performance_ratio"]   # drop partial-year Goa 2008
pr["T1_excl_partial_year"] = dict(n=int(len(t1x)), median=float(t1x.median()))
t1s = perf.loc[(perf.tier == "T1") & (~perf.obs_id.isin(["P16", "P17"])), "performance_ratio"]  # drop storage-limited houses
pr["T1_excl_storage_limited"] = dict(n=int(len(t1s)), median=float(t1s.median()))
t1w = perf.loc[(perf.tier == "T1") & (perf.obs_id != "P02"), "performance_ratio"]   # Warud graded T2 instead
pr["T1_excl_warud"] = dict(n=int(len(t1w)), median=float(t1w.median()))
t2w = pd.concat([perf.loc[perf.tier == "T2", "performance_ratio"], perf.loc[perf.obs_id == "P02", "performance_ratio"]])
uw = stats.mannwhitneyu(t1w, t2w, alternative="two-sided")
pr["T2_incl_warud"] = dict(n=int(len(t2w)), median=float(t2w.median()), mannwhitney_U=float(uw.statistic), mannwhitney_p=float(uw.pvalue))
# site-level (average the Goa years) so one site does not dominate
site = perf[perf.tier.isin(["T1", "T2"])].copy()
site["site_key"] = site.site.str.replace(r", \d{4}.*", "", regex=True).str.replace(" (partial year)", "")
site_level = site.groupby(["site_key", "tier"], as_index=False).performance_ratio.mean()
pr["site_level"] = {t: dict(n=int((site_level.tier == t).sum()), median=float(site_level.loc[site_level.tier == t, "performance_ratio"].median())) for t in ["T1", "T2"]}
# exploratory regression (T1 + T2 only)
d = perf[perf.tier.isin(["T1", "T2"])].copy()
d["observed"] = (d.tier == "T1").astype(int)
X = sm.add_constant(d[["observed", "log10_area"]])
ols = sm.OLS(d.performance_ratio, X).fit()
rho = stats.spearmanr(d.log10_area, d.performance_ratio)
```

```
rho_t1 = stats.spearmanr(perf.loc[perf.tier == "T1", "log10_area"], perf.loc[perf.tier == "T1", "performance_ratio"])
pr["regression"] = dict(n=int(ols.nobs), r2=float(ols.rsquared), params={k: float(v) for k, v in ols.params.items()},
                        pvalues={k: float(v) for k, v in ols.pvalues.items()}, conf_int={k: [float(a), float(b)] for k, (a, b) in ols.conf_int().iterrows()},
                        spearman_area_all=dict(rho=float(rho.statistic), p=float(rho.pvalue)), spearman_area_T1=dict(rho=float(rho_t1.statistic), p=float(rho_t1.pvalue)))
R["performance"] = pr

# Figure 1: scatter + strip
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7.2, 1.85), gridspec_kw=dict(width_ratios=[1.25, 1]))
lim = [5, 3e5]
ax1.plot(lim, lim, color=INK2, lw=0.9, ls="--", zorder=1)
ax1.plot(lim, [v * 0.25 for v in lim], color=GRID, lw=0.9, ls=":", zorder=1)
ax1.text(1.6e5, 1.75e5, "1:1", fontsize=7, color=INK2, ha="right")
ax1.text(4e4, 6.5e3, "1:4", fontsize=7, color=INK2, ha="right")
for t, (lab, col, mk) in TIER.items():
    s = perf[perf.tier == t]
    ax1.scatter(s.potential_kL, s.reported_volume_kL, s=34, c=col, marker=mk, edgecolor="white", linewidth=0.6, label=lab, zorder=3)
ax1.set_xscale("log"); ax1.set_yscale("log"); ax1.set_xlim(lim); ax1.set_ylim(lim)
ax1.set_xlabel("Potential, catchment area x rainfall x 0.8 (kL/yr)"); ax1.set_ylabel("Reported volume (kL/yr)")
ax1.set_title("a. Reported yield against potential", loc="left")
h, l = ax1.get_legend_handles_labels(); fig.legend(h, l, loc="lower center", ncol=3, fontsize=6.5, handletextpad=0.3, columnspacing=1.2, bbox_to_anchor=(0.5, -0.1), frameon=False)
# strip plot
xpos = {"T1": 0, "T2": 1, "T3": 2}
for t, (lab, col, mk) in TIER.items():
    s = perf[perf.tier == t]
    jitter = rng.uniform(-0.16, 0.16, len(s))
    ax2.scatter(xpos[t] + jitter, s.performance_ratio * 100, s=30, c=col, marker=mk, edgecolor="white", linewidth=0.6, zorder=3)
    ax2.hlines(pr[t]["median"] * 100, xpos[t] - 0.3, xpos[t] + 0.3, color=INK, lw=1.6, zorder=4)
    ax2.text(xpos[t] + 0.33, pr[t]["median"] * 100, f"{pr[t]['median']*100:.0f}%", fontsize=7.5, va="center", color=INK)
ax2.axhline(100, color=INK2, lw=0.9, ls="--")
ax2.text(2.58, 96, "100% at C = 0.80", fontsize=6.3, color=INK2, ha="right", va="top")
ax2.set_xticks([0, 1, 2]); ax2.set_xticklabels([f"Observed\n(n={pr['T1']['n']})", f"Agency-\nreported (n={pr['T2']['n']})", f"Design\ncalc. (n={pr['T3']['n']})"], fontsize=7.5)
ax2.set_ylabel("Ratio (% of potential)"); ax2.set_ylim(0, 140); ax2.set_xlim(-0.5, 2.6)
ax2.set_title("b. Ratio by evidence tier (bar = median)", loc="left")
fig.tight_layout(w_pad=1.5); save(fig, "fig1_performance_gap")

# ---------------------------------------------------------------------------
# 2. Mandate gap: installation, functionality, cascade
# ---------------------------------------------------------------------------
chennai_install = float(compl.loc[compl.label == "Chennai 2019", "pct_with_structure"].iloc[0]) / 100
chennai_func = float(func.loc[func.func_id == "F02", "pct_functional"].iloc[0]) / 100
t1_med = pr["T1"]["median"]
cascade = dict(mandated=1.0, with_structure=chennai_install, functional=chennai_install * chennai_func, delivered=chennai_install * chennai_func * t1_med)
blr_install = float(compl.loc[compl.label == "Bengaluru 2023", "pct_with_structure"].iloc[0]) / 100
blr_func = float(func.loc[func.func_id == "F12", "pct_functional"].iloc[0]) / 100
R["mandate_gap"] = dict(chennai=cascade, bengaluru=dict(with_structure=blr_install, functional=blr_install * blr_func, delivered=blr_install * blr_func * t1_med),
                        functionality_median=float(func.pct_functional.median()), functionality_n=int(len(func)),
                        functionality_surveys=int(func.research_id.str.replace(r"/.*", "", regex=True).nunique()),
                        functionality_min=float(func.pct_functional.min()), functionality_max=float(func.pct_functional.max()),
                        functionality_structural_only_median=float(func[~func.func_id.isin(["F11"])].pct_functional.median()),
                        chennai_excl_warud_delivered=float(chennai_install * chennai_func * pr["T1_excl_warud"]["median"]))

SHORT = {"Chennai 2019 city inspection": "Chennai 2019, all inspected (2.7 lakh)", "Chennai 2019, houses with a structure": "Chennai 2019, needing no repair (1.6 lakh)",
         "Chennai Gandhinagar 2003": "Chennai Gandhinagar 2003 (309)", "Chennai approved buildings 2001-02": "Chennai approved bldgs 2002 (25)",
         "Karnataka school structures 2007": "Karnataka schools 2007 (1,269)", "Karnataka model schools 2009": "Karnataka model schools 2009 (27)",
         "Faridabad public systems 2018-19": "Faridabad public 2018-19 (190)", "Faridabad public systems 2024": "Faridabad public 2024 (200)",
         "Gurugram pits 2023": "Gurugram pits 2023 (500)", "Delhi Dwarka societies 2024 (structural)": "Dwarka 2024, structural (176)",
         "Delhi Dwarka societies 2024 (clean recharge)": "Dwarka 2024, clean water (176)", "Bengaluru BWSSB connections 2023 (expert estimate)": "Bengaluru 2023, expert estimate",
         "Kaithal government buildings 2018": "Kaithal govt bldgs 2018 (5)", "Sirsa park systems 2025": "Sirsa parks 2025 (4)"}
fig = plt.figure(figsize=(7.2, 1.85))
axes = [fig.add_axes([0.11, 0.2, 0.14, 0.68]), fig.add_axes([0.52, 0.2, 0.20, 0.68]), fig.add_axes([0.83, 0.2, 0.16, 0.68])]
ax = axes[0]
c = compl.sort_values("pct_with_structure")
ax.barh(c.label, c.pct_with_structure, color=BLUE, height=0.62)
for y, v in enumerate(c.pct_with_structure):
    ax.text(v + 1.5, y, f"{v:.1f}%" if v < 1 else f"{v:.0f}%", va="center", fontsize=6.5, color=INK)
ax.set_xlim(0, 85); ax.set_xlabel("Properties with a structure (%)"); ax.set_title("a. Installation gap", loc="left"); ax.tick_params(axis="y", labelsize=6.5)
ax = axes[1]
f = func.sort_values("pct_functional")
ax.barh([SHORT.get(l, l) for l in f.label], f.pct_functional, color=ORANGE, height=0.62)
for y, v in enumerate(f.pct_functional):
    ax.text(v + 1.2, y, f"{r0(v)}%", va="center", fontsize=6.5, color=INK)
ax.axvline(func.pct_functional.median(), color=INK2, lw=0.9, ls="--"); ax.text(func.pct_functional.median() + 1.5, 12.6, f"median {r0(func.pct_functional.median())}%", fontsize=6.3, color=INK2, va="bottom")
ax.set_xlim(0, 100); ax.set_xlabel("Installed systems functional (%)"); ax.set_title("b. Functionality gap", loc="left"); ax.tick_params(axis="y", labelsize=6.8)
ax = axes[2]
labels = ["Man-\ndated", "With\nstruc-\nture", "Func-\ntional", "Deli-\nvered"]
```

```
vals = [cascade["mandated"], cascade["with_structure"], cascade["functional"], cascade["delivered"]]
ax.bar(range(4), [v * 100 for v in vals], color=[MUTED, BLUE, ORANGE, INK], width=0.62)
for i, v in enumerate(vals):
    ax.text(i, v * 100 + 2, f"{r0(v*100)}%", ha="center", fontsize=6.8, color=INK)
ax.set_xticks(range(4)); ax.set_xticklabels(labels, fontsize=5.8); ax.set_ylim(0, 115); ax.set_ylabel("Share of mandated potential (%)", fontsize=7.5)
ax.set_title("c. Chennai cascade", loc="left")
save(fig, "fig2_mandate_gap")

# ---------------------------------------------------------------------------
# 3. Failure modes
# ---------------------------------------------------------------------------
FMSHORT = {"No maintenance, abandoned": "No maintenance, abandoned", "No awareness, incentive or enforcement": "No incentive or enforcement", "Poor design or construction": "Poor design or construction",
           "Token structure, built for compliance only": "Token structure for approval", "Recharge pit or well silted or choked": "Recharge pit silted or choked", "Downpipes or conveyance broken or disconnected": "Downpipes disconnected",
           "Other (space, vandalism, roof-only design)": "Other (space, vandalism)", "Storage undersized, overflow or leak": "Storage undersized or leaking", "Filter clogged or absent": "Filter clogged or absent",
           "First flush absent or not operated": "First flush not operated", "Water quality concerns, system disused": "Water quality, system disused"}
fig, ax = plt.subplots(figsize=(3.2, 1.8))
f = fm.sort_values("distinct_sources")
ax.barh(f.category.map(FMSHORT), f.distinct_sources, color=BLUE, height=0.62)
for y, v in enumerate(f.distinct_sources):
    ax.text(v + 0.3, y, str(v), va="center", fontsize=7.5, color=INK)
ax.set_xlabel(f"Sources citing the reason (of {N_FM_SOURCES})"); ax.set_xlim(0, 21); ax.tick_params(axis="y", labelsize=7)
ax.set_title("Reported reasons systems fail", loc="left")
fig.subplots_adjust(left=0.46, right=0.97, top=0.89, bottom=0.22); save(fig, "fig3_failure_modes", tight=False)
R["failure_modes"] = fm.to_dict(orient="records"); R["failure_mode_counts"] = dict(sources=N_FM_SOURCES, mentions=N_FM_MENTIONS, categories=N_FM_CATS)

# ---------------------------------------------------------------------------
# 4. Cost-effectiveness
# ---------------------------------------------------------------------------
cst = {}
cst["overall"] = dict(n=int(len(cost)), **dict(zip(["median", "ci_lo", "ci_hi"], boot_median(cost.cost_per_kL_design_2024))), q25=float(cost.cost_per_kL_design_2024.quantile(.25)), q75=float(cost.cost_per_kL_design_2024.quantile(.75)))
for sc, s in cost.groupby("building_scale"):
    cst[sc] = dict(n=int(len(s)), median=float(s.cost_per_kL_design_2024.median()), min=float(s.cost_per_kL_design_2024.min()), max=float(s.cost_per_kL_design_2024.max()))
for sy, s in cost.groupby("system_type"):
    cst["system_" + sy] = dict(n=int(len(s)), median=float(s.cost_per_kL_design_2024.median()))
kw = stats.kruskal(*[s.cost_per_kL_design_2024.values for _, s in cost.groupby("building_scale")])
cst["kruskal_scale"] = dict(H=float(kw.statistic), p=float(kw.pvalue))
cc = cost.dropna(subset=["catchment_area_m2"])
sp = stats.spearmanr(np.log10(cc.catchment_area_m2), cc.cost_per_kL_design_2024)
cst["spearman_area_cost"] = dict(n=int(len(cc)), rho=float(sp.statistic), p=float(sp.pvalue))
sp2 = stats.spearmanr(np.log10(cc.catchment_area_m2), cc.capital_per_m2_2024)
cst["spearman_area_capital_per_m2"] = dict(n=int(len(cc)), rho=float(sp2.statistic), p=float(sp2.pvalue))
design_ratio = pr["T3"]["median"]; obs_ratio = pr["T1"]["median"]
mult = design_ratio / obs_ratio
cst["delivered_yield_multiplier"] = float(mult)
cst["overall_delivered_median"] = float(cost.cost_per_kL_design_2024.median() * mult)
cst["cse_delhi_only"] = dict(n=int(cost.research_id.str.match(r"C(0\d|1\d|22)").sum()), median=float(cost[cost.research_id.str.match(r"C(0\d|1\d|22)")].cost_per_kL_design_2024.median()))
cst["post2010"] = dict(n=int((cost.cost_year >= 2010).sum()), median=float(cost[cost.cost_year >= 2010].cost_per_kL_design_2024.median()))
# sensitivity: 20-year life, 6 % discount
crf20 = 0.06 * 1.06 ** 20 / (1.06 ** 20 - 1)
alt = cost.capital_2024_inr * (crf20 + 0.02) / cost.annual_yield_kL
cst["sensitivity_20yr_6pct_median"] = float(alt.median())
cst["sensitivity_15pct_median"] = float(cost.cost_per_kL_design_2024_disc15.median())
cst["post2010_delivered_median"] = float(cost[cost.cost_year >= 2010].cost_per_kL_design_2024.median() * mult)
R["cost"] = cst
BENCH = [("Chennai piped water, lowest slab (2025)", 7), ("Chennai piped water, top slab (2025)", 48), ("Delhi piped water, 20-30 kL slab", 26.4),
         ("BWSSB production cost, Cauvery (2021)", 41), ("Chennai desalination O&M (2019)", 68), ("Chennai Metrowater tanker, 9 kL (2025)", 77.8),
         ("Bengaluru private tanker, 12 kL, 2024 crisis", 158), ("Chennai private tanker, 12 kL, 2019 crisis", 375)]
R["cost_benchmarks"] = BENCH

fig, ax = plt.subplots(figsize=(4.2, 1.8))
order = ["individual house", "residential colony", "institutional", "industrial"]
for i, sc in enumerate(order):
    s = cost[cost.building_scale == sc]
    for sy, mk in [("recharge", "o"), ("storage", "s"), ("both", "D")]:
        ss = s[s.system_type == sy]
        if len(ss):
            ax.scatter(i + rng.uniform(-0.18, 0.18, len(ss)), ss.cost_per_kL_design_2024, s=30, c=BLUE, marker=mk, edgecolor="white", linewidth=0.6, zorder=3)
    ax.hlines(s.cost_per_kL_design_2024.median(), i - 0.3, i + 0.3, color=INK, lw=1.6, zorder=4)
    ax.text(i + 0.33, s.cost_per_kL_design_2024.median(), f"Rs {s.cost_per_kL_design_2024.median():.0f}", fontsize=6.5, va="center")
ax.axhspan(7, 48, color="#e6e5e1", alpha=0.6, zorder=0)
ax.axhspan(77.8, 375, color=ORANGE, alpha=0.10, zorder=0)
ax.axhline(68, color=INK2, lw=0.8, ls="--", zorder=1)
ax.text(3.98, 11, "Piped tariffs\nRs 7-48/kL", fontsize=6.3, color=INK2, ha="right", va="bottom")
ax.text(-0.45, 71, "Desalination\nO&M Rs 68/kL", fontsize=6, color=INK2, ha="left", va="bottom")
ax.text(3.98, 400, "Tanker water\nRs 78-375/kL", fontsize=6.3, color=INK2, ha="right", va="bottom")
ax.set_yscale("log"); ax.set_ylim(1, 900); ax.set_xlim(-0.5, 4.0)
ax.set_xticks(range(4)); ax.set_xticklabels([f"{lab}\n(n={int((cost.building_scale==o).sum())})" for o, lab in zip(order, ["House", "Colony", "Institutional", "Industrial"])], fontsize=7)
ax.set_ylabel("Rs per kL of design yield (2024)")
ax.set_title("Cost of harvested water by building scale", loc="left")
```

```
handles = [Line2D([], [], marker=m, color=BLUE, ls="", markeredgecolor="white", label=l) for m, l in [("o", "recharge"), ("s", "storage"), ("D", "both")]]
ax.legend(handles=handles, loc="upper left", fontsize=6.5, ncol=3, handletextpad=0.2, columnspacing=0.8, bbox_to_anchor=(0.0, 0.97))
fig.subplots_adjust(left=0.12, right=0.98, top=0.89, bottom=0.22); save(fig, "fig4_cost_per_kL", tight=False)

# ---------------------------------------------------------------------------
# Supplementary figures
# ---------------------------------------------------------------------------
# S1 sensitivity to runoff coefficient
fig, ax = plt.subplots(figsize=(3.6, 2.6))
for t, (lab, col, mk) in TIER.items():
    meds = [perf.loc[perf.tier == t, c].median() * 100 for c in ["performance_ratio_c070", "performance_ratio", "performance_ratio_c090"]]
    ax.plot([0.70, 0.80, 0.90], meds, marker=mk, color=col, label=lab.split(" (")[0])
ax.axhline(100, color=INK2, lw=0.8, ls="--")
ax.set_xlabel("Runoff coefficient assumed"); ax.set_ylabel("Median performance ratio (%)"); ax.set_xticks([0.7, 0.8, 0.9]); ax.legend(fontsize=6.5)
ax.set_title("Sensitivity to runoff coefficient", loc="left"); fig.tight_layout(); save(fig, "figS1_sensitivity_runoff")

# S2 ratio vs catchment area
fig, ax = plt.subplots(figsize=(3.6, 2.6))
for t, (lab, col, mk) in TIER.items():
    s = perf[perf.tier == t]
    ax.scatter(s.catchment_area_m2, s.performance_ratio * 100, s=28, c=col, marker=mk, edgecolor="white", linewidth=0.6, label=lab.split(" (")[0])
ax.set_xscale("log"); ax.set_xlabel("Catchment area (m2)"); ax.set_ylabel("Performance ratio (%)"); ax.axhline(100, color=INK2, lw=0.8, ls="--")
ax.legend(fontsize=6.5); ax.set_title("Ratio against catchment area", loc="left"); fig.tight_layout(); save(fig, "figS2_ratio_vs_area")

# S3 capital cost per m2 vs area (economies of scale)
fig, ax = plt.subplots(figsize=(3.6, 2.6))
ax.scatter(cc.catchment_area_m2, cc.capital_per_m2_2024, s=28, c=BLUE, edgecolor="white", linewidth=0.6)
ax.set_xscale("log"); ax.set_yscale("log"); ax.set_xlabel("Catchment area (m2)"); ax.set_ylabel("Capital cost per m2 (2024 Rs)")
ax.set_title(f"Capital cost per m2 (Spearman rho = {sp2.statistic:.2f})", loc="left"); fig.tight_layout(); save(fig, "figS3_capital_per_m2")

# S4 screening flow diagram
fig, ax = plt.subplots(figsize=(7.4, 4.8)); ax.axis("off")
def box(x, y, w, h, text, fc="#ffffff", ec=INK2):
    ax.add_patch(plt.Rectangle((x, y), w, h, fc=fc, ec=ec, lw=0.8)); ax.text(x + w / 2, y + h / 2, text, ha="center", va="center", fontsize=6.3, color=INK, linespacing=1.25)
def arrow(x1, y1, x2, y2):
    ax.annotate("", xy=(x2, y2), xytext=(x1, y1), arrowprops=dict(arrowstyle="->", color=INK2, lw=0.8))
ax.set_xlim(0, 100); ax.set_ylim(0, 100)
box(1, 82, 32, 16, "Performance strand\n89 site records identified\n(CGWB, CSE, Lok Sabha replies,\nacademic, campus, company reports)")
box(1, 61, 32, 13, "79 screened in detail\n(17 not accessible; 7 added from the\ncost search and the CGWB compilation)")
box(1, 31, 32, 23, "Excluded 47\n20 potential only\n9 no area or no volume\n3 modelled\n10 agency volumes without roof area\n5 mixed catchment, aggregate\nor inconsistent")
box(1, 4, 32, 22, "Included 32 site records\n= 35 observations at 31 sites\nObserved: 7 obs, 5 sites\nAgency-reported: 9 obs, 7 sites\nDesign calculation: 19 sites", fc="#eef3fb")
arrow(17, 82, 17, 74); arrow(17, 61, 17, 54); arrow(17, 33, 17, 26)
box(35, 82, 30, 16, "Cost strand\n43 case records with a cost figure\n37 benchmark price rows")
box(35, 52, 30, 22, "27 rows with capital cost,\ncost year and annual yield\nlevelised to 2024 Rs\n(15-year life, 8% discount,\nO&M 2% of capital)", fc="#eef3fb")
arrow(50, 82, 50, 74)
box(35, 4, 30, 22, "Benchmarks used\npiped tariffs (Rs 7-48/kL)\ndesalination O&M (Rs 68/kL)\ntanker water (Rs 78-375/kL)")
box(67, 82, 32, 16, "Functionality strand\n46 survey and audit records\n(Chennai, Bengaluru, Delhi, Haryana,\nHyderabad, Mumbai, Karnataka)")
box(67, 52, 32, 22, "13 functional-share estimates\nfrom 12 surveys and audits\n7 installation (compliance) figures\n90 failure-mode mentions\nfrom 40 distinct sources", fc="#eef3fb")
arrow(83, 82, 83, 74)
box(67, 4, 32, 22, "Mandate strand\n24 jurisdictions tabulated\nprimary instruments located for\nTamil Nadu, Karnataka, Maharashtra,\nKerala, Delhi and MBBL 2016")
save(fig, "figS4_screening_flow")

# ---------------------------------------------------------------------------
# write results
# ---------------------------------------------------------------------------
(RESD / "results.json").write_text(json.dumps(R, indent=2, default=float))
lines = ["# MIRAGE results summary (generated by code/analysis.py)", ""]
lines += ["## Performance ratio by tier (runoff coefficient 0.80)"]
for t in ["T1", "T2", "T3"]:
    p = pr[t]; lines.append(f"- {TIER[t][0]}: n={p['n']} obs at {p['sites']} sites; median {p['median']*100:.1f}% (bootstrap 95% CI {p['ci_lo']*100:.0f}-{p['ci_hi']*100:.0f}%), range {p['min']*100:.0f}-{p['max']*100:.0f}%; share >= 99%: {p['share_ge_1']*100:.0f}%; median at C=0.70: {p['median_c070']*100:.0f}%, at C=0.90: {p['median_c090']*100:.0f}%")
lines.append(f"- Mann-Whitney observed vs agency-reported: U={pr['T1_vs_T2_mannwhitney']['U']:.0f}, p={pr['T1_vs_T2_mannwhitney']['p']:.4f}")
lines.append(f"- Observed tier excluding partial-year Goa 2008: n={pr['T1_excl_partial_year']['n']}, median {pr['T1_excl_partial_year']['median']*100:.1f}%")
lines.append(f"- Observed tier excluding storage-limited houses: n={pr['T1_excl_storage_limited']['n']}, median {pr['T1_excl_storage_limited']['median']*100:.1f}%")
lines.append(f"- Warud graded agency-reported instead: observed n={pr['T1_excl_warud']['n']}, median {pr['T1_excl_warud']['median']*100:.1f}%; agency-reported n={pr['T2_incl_warud']['n']}, median {pr['T2_incl_warud']['median']*100:.1f}%; Mann-Whitney U={pr['T2_incl_warud']['mannwhitney_U']:.0f}, p={pr['T2_incl_warud']['mannwhitney_p']:.4f}")
lines.append(f"- Site-level medians (years averaged): observed {pr['site_level']['T1']['median']*100:.1f}% (n={pr['site_level']['T1']['n']}), agency-reported {pr['site_level']['T2']['median']*100:.1f}% (n={pr['site_level']['T2']['n']})")
rg = pr["regression"]
lines.append(f"- OLS ratio ~ observed + log10(area), n={rg['n']}, R2={rg['r2']:.2f}: observed coef {rg['params']['observed']:.2f} (p={rg['pvalues']['observed']:.3f}, 95% CI {rg['conf_int']['observed'][0]:.2f} to {rg['conf_int']['observed'][1]:.2f}); log10 area coef {rg['params']['log10_area']:.2f} (p={rg['pvalues']['log10_area']:.3f})")
```

```
lines.append(f"- Spearman ratio vs log area: observed and agency-reported rho={rg['spearman_area_all']['rho']:.2f} (p={rg['spearman_area_all']['p']:.2f}); observed only rho={rg['spearman_area_T1']['rho']:.2f} (p={rg['spearman_area_T1']['p']:.2f})")
lines += ["", "## Mandate gap"]
mg = R["mandate_gap"]
lines.append(f"- Chennai cascade: with structure {mg['chennai']['with_structure']*100:.1f}% -> functional {mg['chennai']['functional']*100:.1f}% -> delivered {mg['chennai']['delivered']*100:.1f}% of mandated potential")
lines.append(f"- Bengaluru cascade: with structure {mg['bengaluru']['with_structure']*100:.1f}% -> functional {mg['bengaluru']['functional']*100:.1f}% -> delivered {mg['bengaluru']['delivered']*100:.1f}%")
lines.append(f"- Functionality: {mg['functionality_n']} estimates from {mg['functionality_surveys']} surveys, median functional {mg['functionality_median']:.0f}% (range {mg['functionality_min']:.0f}-{mg['functionality_max']:.0f}%; structural-only definition: {mg['functionality_structural_only_median']:.0f}%); Chennai delivered with Warud as agency-reported: {mg['chennai_excl_warud_delivered']*100:.1f}%")
lines.append(f"- Failure modes: {N_FM_MENTIONS} mentions from {N_FM_SOURCES} distinct sources in {N_FM_CATS} categories")
lines += ["", "## Failure modes (distinct sources)"] + [f"- {r['category']}: {r['distinct_sources']} sources, {r['mentions']} mentions" for r in R["failure_modes"]]
lines += ["", "## Cost per kL (2024 Rs, design yield, 15 yr, 8%, O&M 2%)"]
o = cst["overall"]; lines.append(f"- All rows n={o['n']}: median Rs {o['median']:.1f}/kL (95% CI {o['ci_lo']:.1f}-{o['ci_hi']:.1f}), IQR {o['q25']:.1f}-{o['q75']:.1f}")
for sc in order: lines.append(f"- {sc}: n={cst[sc]['n']}, median Rs {cst[sc]['median']:.1f}, range {cst[sc]['min']:.1f}-{cst[sc]['max']:.1f}")
for sy in ["recharge", "storage", "both"]: lines.append(f"- system {sy}: n={cst['system_'+sy]['n']}, median Rs {cst['system_'+sy]['median']:.1f}")
lines.append(f"- Kruskal-Wallis across scales: H={cst['kruskal_scale']['H']:.2f}, p={cst['kruskal_scale']['p']:.3f}")
lines.append(f"- Spearman log area vs cost/kL: rho={cst['spearman_area_cost']['rho']:.2f}, p={cst['spearman_area_cost']['p']:.4f} (n={cst['spearman_area_cost']['n']}); log area vs capital/m2: rho={cst['spearman_area_capital_per_m2']['rho']:.2f}, p={cst['spearman_area_capital_per_m2']['p']:.4f}")
lines.append(f"- Delivered-yield multiplier (design-calculated median {design_ratio:.2f} / observed median {obs_ratio:.2f}) = {mult:.2f}; overall median at delivered yield Rs {cst['overall_delivered_median']:.0f}/kL")
lines.append(f"- CSE Delhi-NCR rows only: n={cst['cse_delhi_only']['n']}, median Rs {cst['cse_delhi_only']['median']:.1f}; rows with cost year >= 2010: n={cst['post2010']['n']}, median Rs {cst['post2010']['median']:.1f}")
lines.append(f"- Sensitivity 20-year life, 6% discount: median Rs {cst['sensitivity_20yr_6pct_median']:.1f}/kL; 15% discount (CGWB example): median Rs {cst['sensitivity_15pct_median']:.1f}/kL; post-2010 rows at observed yield: Rs {cst['post2010_delivered_median']:.0f}/kL")
lines += ["", "## Benchmarks (Rs/kL)"] + [f"- {l}: {v}" for l, v in BENCH]
(RESD / "results.md").write_text("\n".join(lines))
print("\n".join(lines))
```

11.6 make_supplementary_tables.py

```python
"""Generate LaTeX table fragments for the supplementary document from data/*.csv and research/*.csv.
Writes report/tables/*.tex. Run from the project root: python3 code/make_supplementary_tables.py
"""
from pathlib import Path
import json
import pandas as pd
import numpy as np

ROOT = Path(__file__).resolve().parents[1]
DATA, RES, OUT = ROOT / "data", ROOT / "research", ROOT / "report" / "tables"
OUT.mkdir(exist_ok=True)
R = json.load(open(ROOT / "results" / "results.json"))
TIERNAME = {"T1": "Observed", "T2": "Agency", "T3": "Design"}

def esc(x):
    if x is None or (isinstance(x, float) and np.isnan(x)): return "NR"
    s = str(x)
    for a, b in [("\u20b9", "Rs "), ("\u2265", ">="), ("\u2264", "<="), ("\u2192", "->"), ("\u00d7", "x"), ("\u00b0", " deg"), ("\u2013", "-"), ("\u2014", "-"), ("\u2019", "'"), ("\u2018", "'"), ("\u201c", '"'), ("\u201d", '"'), ("\u00a0", " "), ("\u2082", "2"), ("\u00b2", "2"), ("\u00b3", "3"), ("\u2248", "approx. ")]:
        s = s.replace(a, b)
    for a, b in [("\\", "\\textbackslash "), ("&", "\\&"), ("%", "\\%"), ("$", "\\$"), ("#", "\\#"), ("_", "\\_"), ("{", "\\{"), ("}", "\\}"), ("~", "\\textasciitilde "), ("^", "\\textasciicircum ")]:
        s = s.replace(a, b)
    return s

def url(u):
    if not isinstance(u, str) or not u.strip() or u == "NR": return "NR"
    u = u.split(";")[0].strip().replace(" ", "%20")
    return "\\url{" + u + "}"

def num(x, nd=0):
    try:
        v = float(x)
        if np.isnan(v): return "NR"
        return f"{v:,.{nd}f}"
    except Exception:
        return esc(x)

def longtable(cols, rows, caption, label, colspec, fontsize="\\scriptsize"):
    out = [f"{{{fontsize}\\setlength{{\\tabcolsep}}{{3pt}}", f"\\begin{{longtable}}{{{colspec}}}", f"\\caption{{{caption}}}\\label{{{label}}}\\\\", "\\toprule"]
    out.append(" & ".join(f"\\textbf{{{c}}}" for c in cols) + "\\\\ \\midrule\\endfirsthead")
    out.append("\\toprule " + " & ".join(f"\\textbf{{{c}}}" for c in cols) + "\\\\ \\midrule\\endhead")
    out.append("\\bottomrule\\endfoot")
    for r in rows: out.append(" & ".join(r) + "\\\\")
    out += ["\\end{longtable}}", ""]
    return "\n".join(out)

# ---------------- S-A performance dataset ----------------
p = pd.read_csv(DATA / "mirage_performance.csv")
rows = []
for i, (_, r) in enumerate(p.iterrows()):
    rows.append([str(i + 1), esc(r.site) if str(r.city).split(" (")[0] in str(r.site) else esc(r.site) + ", " + esc(r.city), TIERNAME[r.tier], esc(r.building_type) + " / " + esc(r.system_type), num(r.catchment_area_m2), num(r.rainfall_mm) + " (" + esc(r.rainfall_basis) + ")",
                 esc(r.year_of_volume), num(r.reported_volume_kL, 1), num(r.potential_kL), f"{r.performance_ratio*100:.0f}", f"{r.performance_ratio_c070*100:.0f}--{r.performance_ratio_c090*100:.0f}"])
(OUT / "perf_table.tex").write_text(longtable(["No.", "Site", "Evidence", "Type / system", "Area m$^2$", "Rain mm (basis)", "Year", "Vol. kL", "Pot. kL", "Ratio \\%", "C 0.7--0.9"], rows,
    "Performance dataset: all 35 observations. Evidence: Observed (instrument or tank record named), Agency (agency-reported, no instrument named), Design (design calculation). Potential uses runoff coefficient 0.80; the last column gives the ratio for C = 0.70 and 0.90. Rainfall basis: site-year (reported for that site and year), site-avg (average quoted by the source), city-avg (CSE city figure), city-normal (IMD normal), implied (rain received on roof divided by area).",
    "tab:perf", "l p{2.7cm} l >{\\raggedright\\arraybackslash}p{1.6cm} r p{1.6cm} p{1.1cm} r r r l"))
rows = [[str(i + 1), esc(r.measurement_note), esc(r.source), url(r.url)] for i, (_, r) in enumerate(p.iterrows()) if r.tier != "T3"]
(OUT / "perf_notes.tex").write_text(longtable(["No.", "Measurement note", "Source", "URL"], rows, "Measurement notes and sources for the observed and agency-reported observations (numbers as in Table~\\ref{tab:perf}). The design-calculated rows come from CSE model-project pages, \\url{https://www.cseindia.org/rainwater-harvesting-case-studies}, with volumes stated as a fixed share of area $\\times$ rainfall, and from the CGWB design note for Shram Shakti Bhawan.", "tab:perfnotes", "l p{7.5cm} p{3cm} p{5cm}"))

# ---------------- reconciliation with the mid-term progress report ----------------
recon = [
 ("KITS Ramtek", "about 9\\% (19 of 215.7 m$^3$ received)", "11\\% (19 kL against potential 172.6 kL at C = 0.80); 8.8\\% of rain received; observed tier", "Numbers verified word for word in CGWB (2011) p.73; the midway ratio divided by gross rain, the final divides by potential (rain $\\times$ 0.8). The implied rainfall (599 mm) is below the Ramtek taluka average of 1,043.7 mm, so the monitored period was probably not a full year; with the taluka average the ratio would be 6\\%."),
 ("Goa University", "61--76\\% (900 kL, about 400 m$^2$)", "76\\% in 2010 (3.7 m rainfall); 28\\% in 2008 (partial year) and 19\\% in 2009 using the IMD Panaji normal", "Midway used a range of runoff coefficients; final fixes C = 0.80 and adds the two earlier years reported in the same CSE table."),
 ("BEL Bengaluru", "63--75\\% (about 1,100 kL recharged)", "64\\% (1,100 kL, 2,000 m$^2$, IMD Bengaluru normal 1,077.6 mm)", "Source says the water is harvested and stored for DM-water production, so the system type is storage, not recharge as the midway said."),
 ("Panjab University", "about 79\\% (1,985 m$^3$)", "112\\% (2,100 m$^2$, 1,059.3 mm, C = 0.80); agency-reported tier", "The midway figure used an unstated area or rainfall. The Lok Sabha source gives the volume as ''93\\% of total generated runoff'', i.e. a design share, which is why the ratio exceeds 100\\%."),
 ("Warud", "90\\% (derived from runoff assumption)", "112\\% against potential; 90\\% of rain received; observed tier by the stated rule (flow meter named), with a sensitivity case grading it agency-reported", "Confirmed in CGWB (2011) p.73 (500 m$^2$, 311.30 m$^3$ received, 280.17 m$^3$ recharged). The CGWB wording is identical to Ramtek, so one rule
```

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    ("must grade both the same way; the round 90\\% suggests an assumed share, which is why both gradings are reported."),
 ("IIT Delhi Block VI", "not comparable (rainfall NR)", "82\\% using the IMD Safdarjung normal (762.3 mm); agency-reported tier, rainfall basis flagged", "Confirmed in CGWB (2011) p.44 (830 m$^3$ from 1,660 m$^2$). Rainfall period still not reported, so the ratio is indicative."),
 ("Umaid Heritage, Sri Aurobindo Ashram, PAU Ludhiana", "supplementary (mixed catchments or calculated)", "Umaid Heritage excluded (110-acre mixed catchment, internal inconsistency 21.1 vs 2.1 ML); Sri Aurobindo Ashram included as a design calculation (43\\% of potential); PAU excluded (no roof area)", "As the midway said: these are not comparable rooftop cases."),
]
rows = [[esc(a), b, c, d] for a, b, c, d in recon]
(OUT / "recon_table.tex").write_text(longtable(["Case", "Midway check-in", "Final", "Reason for change"], rows, "Reconciliation of the six midway check-in cases with the final dataset. All six are retained; ratios were recomputed with one stated runoff coefficient and a stated rainfall basis.", "tab:recon", "p{2.3cm} p{3.4cm} p{4.6cm} p{5.6cm}"))

# ---------------- S-C screening log ----------------
log = pd.read_csv(DATA / "screening_log.csv")
STEPFIX = lambda t: t.replace("T1 observed volumes (instrument or tank record named)", "observed volumes (instrument or tank record named)").replace("T2 agency-reported volumes", "agency-reported volumes").replace("T3 design calculations", "design calculations")
rows = [[esc(STEPFIX(r["step"])), num(r["count"])] for _, r in log.iterrows()]
(OUT / "screening_table.tex").write_text(longtable(["Step", "Count"], rows, "Screening and inclusion log (counts behind the flow diagram above).", "tab:screen", "p{13cm} r"))

# ---------------- S-D statistics ----------------
pr = R["performance"]; rg = pr["regression"]
stats_tex = r"""
\begin{table}[H]\centering\small
\caption{Performance ratio statistics by evidence tier (runoff coefficient 0.80 unless stated). Observed: instrument or tank record named; Agency: agency-reported, no instrument named; Design: design calculation.}\label{tab:stats}
\begin{tabular}{lrrrrrr}\toprule
Tier & n (sites) & Median & 95\% CI & Range & Share $\geq$ 99\% & Median at C 0.70 / 0.90\\ \midrule
""" + "\n".join(f"{TIERNAME[t]} & {pr[t]['n']} ({pr[t]['sites']}) & {pr[t]['median']*100:.1f}\\% & {pr[t]['ci_lo']*100:.0f}--{pr[t]['ci_hi']*100:.0f}\\% & {pr[t]['min']*100:.0f}--{pr[t]['max']*100:.0f}\\% & {pr[t]['share_ge_1']*100:.0f}\\% & {pr[t]['median_c070']*100:.0f}\\% / {pr[t]['median_c090']*100:.0f}\\%\\\\" for t in ["T1", "T2", "T3"]) + r"""
\bottomrule\end{tabular}
\end{table}

\begin{table}[H]\centering\small
\caption{Exploratory OLS: performance ratio on evidence tier and catchment area, observed and agency-reported observations only (n = """ + str(rg["n"]) + ", R$^2$ = " + f"{rg['r2']:.2f}" + r""").}\label{tab:ols}
\begin{tabular}{lrrr}\toprule
Term & Coefficient & 95\% CI & p\\ \midrule
""" + "\n".join(f"{ {'const': 'Intercept', 'observed': 'Observed tier (1 = yes)', 'log10_area': 'log10 catchment area'}[k] } & {rg['params'][k]:.3f} & {rg['conf_int'][k][0]:.3f} to {rg['conf_int'][k][1]:.3f} & {rg['pvalues'][k]:.3f}\\\\" for k in ["const", "observed", "log10_area"]) + r"""
\bottomrule\end{tabular}
\end{table}
Other tests: Mann-Whitney U (observed vs agency-reported) = """ + f"{pr['T1_vs_T2_mannwhitney']['U']:.0f}, p = {pr['T1_vs_T2_mannwhitney']['p']:.4f}" + r"""; Spearman $\rho$ of ratio against $\log_{10}$ area: all observed and agency-reported """ + f"{rg['spearman_area_all']['rho']:.2f} (p = {rg['spearman_area_all']['p']:.3f})" + r""", observed only """ + f"{rg['spearman_area_T1']['rho']:.2f} (p = {rg['spearman_area_T1']['p']:.2f})" + r""". The positive area correlation across the two tiers reflects the tier split (large agency-monitored institutional roofs are agency-reported, small metered houses are observed) and is not significant within the observed tier. With Warud in the observed tier the OLS tier coefficient is not significant; the regression is exploratory and is not used for any claim in the main report. Observed median excluding the partial-year Goa 2008 observation: """ + f"{pr['T1_excl_partial_year']['median']*100:.1f}\\% (n = {pr['T1_excl_partial_year']['n']})" + r"""; excluding the two storage-limited houses: """ + f"{pr['T1_excl_storage_limited']['median']*100:.1f}\\% (n = {pr['T1_excl_storage_limited']['n']})" + r"""; Warud graded agency-reported instead of observed: observed median """ + f"{pr['T1_excl_warud']['median']*100:.1f}\\% (n = {pr['T1_excl_warud']['n']}), agency-reported median {pr['T2_incl_warud']['median']*100:.1f}\\% (n = {pr['T2_incl_warud']['n']}), Mann-Whitney U = {pr['T2_incl_warud']['mannwhitney_U']:.0f}, p = {pr['T2_incl_warud']['mannwhitney_p']:.4f}" + r"""; site-level medians (years averaged): observed """ + f"{pr['site_level']['T1']['median']*100:.1f}\\%" + r""", agency-reported """ + f"{pr['site_level']['T2']['median']*100:.1f}\\%" + r""".
"""
(OUT / "stats.tex").write_text(stats_tex)

# ---------------- S-E cost dataset ----------------
c = pd.read_csv(DATA / "mirage_cost.csv")
rows = [[str(i + 1), esc(r.site) if str(r.city) in str(r.site) else esc(r.site) + ", " + esc(r.city), esc(r.building_scale) + " / " + esc(r.system_type), num(r.catchment_area_m2), num(r.capital_cost_inr), str(int(r.cost_year)), num(r.annual_yield_kL), esc(r.yield_basis), num(r.capital_2024_inr), num(r.cost_per_kL_design_2024, 1)] for i, (_, r) in enumerate(c.iterrows())]
(OUT / "cost_table.tex").write_text(longtable(["No.", "Site", "Scale / system", "Area m$^2$", "Capital Rs", "Year", "Yield kL/yr", "Yield basis", "Capital 2024 Rs", "Rs/kL"], rows,
    "Cost dataset: 27 systems. Capital deflated to 2024 rupees with CPI-IW (RBI Table 37, linked series); levelised cost = capital $\\times$ (CRF at 8\\%, 15 years + 0.02) / annual yield. The 8\\% rate is our assumption; the 15-year life and 2\\% O\\&M are the CGWB manual's norms and its example uses 15\\% (Table~\\ref{tab:coststats}). Source URLs and quotes for every row are in the source extraction table supplied with the data.",
    "tab:cost", "l p{3.2cm} p{2.3cm} r r l r p{1.6cm} r r"))
cst = R["cost"]
cost_stats = r"""
\begin{table}[H]\centering\small
\caption{Levelised cost per kL of design yield (2024 rupees) by group, and sensitivity.}\label{tab:coststats}
\begin{tabular}{lrrr}\toprule
Group & n & Median Rs/kL & Range\\ \midrule
All & """ + f"{cst['overall']['n']} & {cst['overall']['median']:.1f} (95\\% CI {cst['overall']['ci_lo']:.1f}--{cst['overall']['ci_hi']:.1f}) & IQR {cst['overall']['q25']:.1f}--{cst['overall']['q75']:.1f}" + r"""\\
""" + "\n".join(f"{g} & {cst[g]['n']} & {cst[g]['median']:.1f} & {cst[g]['min']:.1f}--{cst[g]['max']:.1f}\\\\" for g in ["individual house", "residential colony", "institutional", "industrial"]) + "\n" + "\n".join(f"system: {g} & {cst['system_'+g]['n']} & {cst['system_'+g]['median']:.1f} & \\\\" for g in ["recharge", "storage", "both"]) + r"""
CSE Delhi-NCR rows (2000--2004) & """ + f"{cst['cse_delhi_only']['n']} & {cst['cse_delhi_only']['median']:.1f} &" + r"""\\
Rows with cost year $\geq$ 2010 & """ + f"{cst['post2010']['n']} & {cst['post2010']['median']:.1f} &" + r"""\\
Sensitivity: 20-year life, 6\% discount & 27 & """ + f"{cst['sensitivity_20yr_6pct_median']:.1f} &" + r"""\\
Sensitivity: 15\% discount (the CGWB example rate) & 27 & """ + f"{cst['sensitivity_15pct_median']:.1f} &" + r"""\\
```

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Rows with cost year $\geq$ 2010 at observed yield & 5 & """ + f"{cst['post2010_delivered_median']:.0f} &" + r"""\\
At observed yield (multiplier """ + f"{cst['delivered_yield_multiplier']:.2f}" + r""") & 27 & """ + f"{cst['overall_delivered_median']:.0f} &" + r"""\\
\bottomrule\end{tabular}
\end{table}
Kruskal-Wallis across the four building-scale groups: H = """ + f"{cst['kruskal_scale']['H']:.2f}, p = {cst['kruskal_scale']['p']:.3f}" + r""". Spearman $\rho$ of $\log_{10}$ area against Rs/kL: """ + f"{cst['spearman_area_cost']['rho']:.2f} (p = {cst['spearman_area_cost']['p']:.4f}, n = {cst['spearman_area_cost']['n']})" + r"""; against capital per m$^2$: """ + f"{cst['spearman_area_capital_per_m2']['rho']:.2f} (p = {cst['spearman_area_capital_per_m2']['p']:.4f})" + r""". The delivered-yield multiplier is the design-calculated median ratio (design yield as a share of potential, """ + f"{pr['T3']['median']:.2f}" + r""") divided by the observed median (""" + f"{pr['T1']['median']:.2f}" + r""").
"""
(OUT / "cost_stats.tex").write_text(cost_stats)
b = pd.read_csv(RES / "cost_benchmarks.csv")
rows = [[esc(r["item"]), esc(r["city"]), esc(r["value"]), esc(r["unit"]), esc(r["year"]), url(r["url"])] for _, r in b.iterrows()]
(OUT / "bench_table.tex").write_text(longtable(["Item", "City", "Value", "Unit", "Year", "URL"], rows, "Price benchmarks collected. The eight used in the cost figure of the main report are the Chennai lowest and top domestic slabs, the Delhi 20--30 kL slab, the BWSSB production cost, Chennai desalination O\\&M, the Metrowater 9 kL tanker, and the Bengaluru 2024 and Chennai 2019 private tanker prices (Rs per load divided by load volume).", "tab:bench", "p{4cm} p{1.5cm} p{2.5cm} p{2.1cm} p{1.7cm} p{3.4cm}", "\\tiny"))

# ---------------- S-F functionality ----------------
f = pd.read_csv(DATA / "mirage_functionality.csv")
rows = [[str(i + 1), esc(r.label), esc(r.population), num(r.sample_n), f"{r.pct_functional:.0f}", esc(r.definition), esc(r.conducted_by)] for i, (_, r) in enumerate(f.iterrows())]
(OUT / "func_table.tex").write_text(longtable(["No.", "Survey", "Population", "n", "Functional \\%", "Definition used", "Conducted by (reported via)"], rows, "Functional-share estimates: 13 estimates from 12 surveys (the two Dwarka rows are two definitions applied to one DJB report). URLs and quotes are in the source extraction table supplied with the data.", "tab:func", "l p{2.8cm} p{2.6cm} r r p{4.6cm} p{2.9cm}"))
cm = pd.read_csv(DATA / "mirage_compliance.csv")
rows = [[esc(r.label), esc(r.population), num(r.denominator), num(r.numerator), f"{r.pct_with_structure:.1f}", esc(r.definition)] for _, r in cm.iterrows()]
(OUT / "compl_table.tex").write_text(longtable(["City", "Population", "Denominator", "With RWH", "\\%", "Definition"], rows, "Installation (compliance) figures used in the mandate-gap figure of the main report.", "tab:compl", "p{2.5cm} p{3.4cm} r r r p{5.5cm}"))
fm = pd.read_csv(DATA / "mirage_failure_modes.csv")
rows = [[esc(r.category), num(r.distinct_sources), num(r.mentions), esc(r.cities)] for _, r in fm.iterrows()]
(OUT / "fm_table.tex").write_text(longtable(["Failure mode", "Sources", "Mentions", "Cities"], rows, f"Failure-mode categories ({len(fm)} after collapsing), number of distinct sources citing each (90 mentions from 40 distinct sources; these overlap with but are not the same set as the 46 survey records), and the cities concerned. Every mention with URL and quote is in the source extraction table supplied with the data.", "tab:fm", "p{5cm} r r p{7.5cm}"))

# ---------------- S-G mandates ----------------
m = pd.read_csv(DATA / "mirage_mandates.csv")
rows = [[esc(r.jurisdiction), esc(r.instrument), esc(r.year_enacted), esc(r.applicability_threshold), esc(r.penalty_or_incentive), esc(r.reported_compliance)] for _, r in m.iterrows()]
(OUT / "mandate_table.tex").write_text(longtable(["Jurisdiction", "Instrument", "Year", "Applies to", "Penalty / incentive", "Reported compliance"], rows, "Mandate landscape, 24 jurisdictions. URLs and quotes are in the source extraction table supplied with the data.", "tab:mandates", "p{2.1cm} p{3cm} p{1.3cm} p{3cm} p{2.9cm} p{3.4cm}", "\\tiny"))
ri = pd.read_csv(RES / "reference_inputs.csv")
rows = [[esc(r["item"]), esc(r["value"]), esc(r["unit"]), esc(r["period_or_year"]), esc(r["source_document"])] for _, r in ri.iterrows() if str(r["item"]).startswith(("Runoff", "Normal", "Average"))]
(OUT / "refinputs_table.tex").write_text(longtable(["Item", "Value", "Unit", "Period", "Source document"], rows, "Runoff coefficients and rainfall normals used. The CGWB (2007) Table 7.2 values were verified directly in the manual.", "tab:refinputs", "p{4.8cm} p{2.3cm} p{1.4cm} p{2.1cm} p{5.2cm}"))

# ---------------- S-H source list ----------------
srcs = []
for fn, idcol, namecol in [("performance_cases.csv", "case_id", "case_name"), ("cost_cases.csv", "case_id", "case_name"), ("functionality_surveys.csv", "survey_id", "conducted_by"), ("mandates.csv", None, "jurisdiction"), ("cost_benchmarks.csv", None, "item")]:
    d = pd.read_csv(RES / fn)
    for _, r in d.iterrows():
        u = str(r.url) if "url" in d.columns else ""
        for uu in u.replace("|", ";").split(";"):
            uu = uu.strip()
            if uu and uu != "NR" and uu.startswith("http"):
                srcs.append((uu.replace(" ", "%20"), {"performance_cases": "Performance", "cost_cases": "Cost", "functionality_surveys": "Functionality", "mandates": "Mandates", "cost_benchmarks": "Benchmarks"}[fn.replace(".csv", "")], str(r[namecol])[:60]))
S = pd.DataFrame(srcs, columns=["url", "strand", "used_for"]).drop_duplicates("url")
S = S.sort_values("url")
rows = [["\\url{" + r.url + "}", esc(r.strand), esc(r.used_for)] for _, r in S.iterrows()]
(OUT / "sources_table.tex").write_text(longtable(["URL", "Strand", "First use"], rows, f"All {len(S)} distinct source URLs used, with the strand and the first record each supports.", "tab:sources", "p{9.2cm} p{2.5cm} p{4.2cm}", "\\tiny"))
print("tables written; sources:", len(S))
```

11.7 style.py

```python
"""Shared plotting style for the MIRAGE figures.

Palette follows one rule set: one dominant blue for the main series,
orange as the single contrasting hue (storage vs recharge, residential vs
institutional), grey for reference lines and context. Text is always ink,
never series colour.
"""
import matplotlib as mpl

BLUE = "#2a78d6"
ORANGE = "#eb6834"
AQUA = "#1baf7a"
INK = "#0b0b0b"
INK2 = "#52514e"
MUTED = "#8a8984"
GRID = "#e6e5e1"
SURFACE = "#ffffff"

SERIES = [BLUE, ORANGE, AQUA]


def apply():
    mpl.rcParams.update({
        "font.family": "DejaVu Sans",
        "font.size": 9,
        "axes.titlesize": 10,
        "axes.titleweight": "bold",
        "axes.labelsize": 9,
        "axes.labelcolor": INK2,
        "axes.edgecolor": GRID,
        "axes.linewidth": 0.8,
        "axes.spines.top": False,
        "axes.spines.right": False,
        "axes.grid": True,
        "grid.color": GRID,
        "grid.linewidth": 0.6,
        "axes.axisbelow": True,
        "xtick.color": INK2,
        "ytick.color": INK2,
        "xtick.labelsize": 8,
        "ytick.labelsize": 8,
        "legend.fontsize": 8,
        "legend.frameon": False,
        "figure.dpi": 150,
        "savefig.dpi": 300,
        "savefig.bbox": "tight",
        "savefig.facecolor": SURFACE,
        "figure.facecolor": SURFACE,
        "axes.facecolor": SURFACE,
    })
```

11.8 build_deck.js

```
// Builds slides/MIRAGE_presentation.pptx from the figures and results.
// Run from the project root: node slides/build_deck.js
const pptxgen = require("pptxgenjs");
const path = require("path");
const fs = require("fs");
const { applyTheme } = require("./apply_theme.js");

const ROOT = path.resolve(__dirname, "..");
const FIG = (n) => path.join(ROOT, "figures", n);
const R = JSON.parse(fs.readFileSync(path.join(ROOT, "results", "results.json")));
const pr = R.performance, mg = R.mandate_gap, cst = R.cost;

const THEME = {
  name: "MIRAGE water",
  headFontFace: "Cambria", bodyFontFace: "Calibri",
  colors: { dk1: "0F2A3F", lt1: "FFFFFF", dk2: "1F4E79", lt2: "EEF3FB", accent1: "2A78D6", accent2: "EB6834", accent3: "6250D6", accent4: "52514E", accent5: "8A8984", accent6: "1BAF7A", hlink: "2A78D6", folHlink: "6250D6" },
};
const pres = new pptxgen();
pres.layout = "LAYOUT_16x9";           // 10 x 5.625 in
pres.theme = { headFontFace: THEME.headFontFace, bodyFontFace: THEME.bodyFontFace };
pres.author = "MIRAGE team, CM617"; pres.title = "MIRAGE: Measuring the Impact and Reality of Rainwater Harvesting Mandates";
const C = pres.SchemeColor;

// ---------- layouts ----------
pres.defineSlideMaster({ title: "DARK", background: { color: "0F2A3F" },
  objects: [{ text: { text: "MIRAGE | CM617 course project", options: { x: 0.5, y: 5.2, w: 5, h: 0.3, fontSize: 9, color: "8FB3D9", isTextBox: true, margin: 0 } } }],
  slideNumber: { x: 9.2, y: 5.2, w: 0.5, h: 0.3, fontSize: 9, color: "8FB3D9" } });
pres.defineSlideMaster({ title: "LIGHT", background: { color: "FFFFFF" },
  objects: [
    { placeholder: { options: { name: "title", type: "title", x: 0.5, y: 0.3, w: 9, h: 0.7, fontSize: 26, bold: true, color: C.text2, margin: 0, valign: "middle" }, text: "" } },
    { text: { text: "MIRAGE | CM617 course project", options: { x: 0.5, y: 5.2, w: 5, h: 0.3, fontSize: 9, color: C.accent5, isTextBox: true, margin: 0 } } }],
  slideNumber: { x: 9.2, y: 5.2, w: 0.5, h: 0.3, fontSize: 9, color: C.accent5 } });

const body = (slide, text, o) => slide.addText(text, Object.assign({ fontSize: 14, color: C.text1, isTextBox: true, margin: 0, valign: "top", paraSpaceAfter: 6 }, o));
const bullets = (arr) => arr.map((t, i) => ({ text: t, options: { bullet: true, breakLine: i < arr.length - 1 } }));
const stat = (slide, x, y, w, big, small, col) => {
  slide.addText(big, { x, y, w, h: 0.8, fontSize: 40, bold: true, color: col || C.accent1, isTextBox: true, margin: 0, align: "left", valign: "bottom" });
  slide.addText(small, { x, y: y + 0.82, w, h: 0.7, fontSize: 12, color: C.accent4, isTextBox: true, margin: 0, valign: "top" });
};
const card = (slide, x, y, w, h, head, text, fill) => {
  slide.addShape(pres.ShapeType.roundRect, { x, y, w, h, fill: { color: fill || "EEF3FB" }, line: { color: fill || "EEF3FB" }, rectRadius: 0.08 });
  slide.addText(head, { x: x + 0.15, y: y + 0.1, w: w - 0.3, h: 0.4, fontSize: 14, bold: true, color: C.text2, isTextBox: true, margin: 0 });
  slide.addText(text, { x: x + 0.15, y: y + 0.5, w: w - 0.3, h: h - 0.6, fontSize: 11.5, color: C.text1, isTextBox: true, margin: 0, valign: "top", paraSpaceAfter: 4 });
};
const pct = (v, d = 0) => (v * 100).toFixed(d) + "%";
const pngSize = (file) => { const b = fs.readFileSync(file); return { width: b.readUInt32BE(16), height: b.readUInt32BE(20) }; };
const ALT = { "fig1_performance_gap.png": "Reported yield against theoretical potential, by evidence tier", "fig2_mandate_gap.png": "Installation shares, functional shares and the Chennai cascade", "fig3_failure_modes.png": "Reported reasons systems fail, by number of sources", "fig4_cost_per_kL.png": "Levelised cost per kilolitre by building scale" };
const img = (slide, name, x, y, w, maxH) => { const d = pngSize(FIG(name)); let h = w * d.height / d.width; if (h > maxH) { h = maxH; w = h * d.width / d.height; } slide.addImage({ path: FIG(name), x, y, w, h, altText: ALT[name] || "figure" }); return h; };

// ---------- 1 title ----------
pres.addSection({ title: "Opening" });
let s = pres.addSlide({ masterName: "DARK", sectionTitle: "Opening" });
s.addText("MIRAGE", { x: 0.6, y: 0.9, w: 8.8, h: 1.0, fontSize: 54, bold: true, color: "FFFFFF", fontFace: "Cambria", isTextBox: true, margin: 0 });
s.addText("Measuring the Impact and Reality of Rainwater Harvesting Mandates", { x: 0.6, y: 1.9, w: 8.8, h: 0.6, fontSize: 24, color: "CADCFC", isTextBox: true, margin: 0 });
s.addText("A meta-analysis of rooftop rainwater harvesting performance in Indian cities", { x: 0.6, y: 2.5, w: 8.8, h: 0.5, fontSize: 16, color: "8FB3D9", isTextBox: true, margin: 0 });
s.addText("Vedant Abhangrao (23B2262)  |  Nilay Shahapure (23B2206)  |  Arjun Jhunjhunwala (23B1245)  |  Yash Krishnani (23B1240)", { x: 0.6, y: 3.6, w: 8.8, h: 0.4, fontSize: 13, color: "FFFFFF", isTextBox: true, margin: 0 });
s.addText("CM617 Sustainability Principles and Practices, IIT Bombay. Pathway 1: Sustainability Research. Sector: Water.", { x: 0.6, y: 4.0, w: 8.8, h: 0.4, fontSize: 12, color: "8FB3D9", isTextBox: true, margin: 0 });
s.addNotes("Title slide. MIRAGE is a meta-analysis of published performance data for rooftop rainwater harvesting in Indian cities.");

// ---------- 2 problem ----------
pres.addSection({ title: "Problem and questions" });
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Problem and questions" });
s.addText("Twenty years of mandates, almost no measurement", { placeholder: "title" });
stat(s, 0.5, 1.2, 2.8, "2003", "Chennai makes RWH compulsory for every building; Bengaluru, Delhi, Mumbai and 20+ others follow");
stat(s, 3.6, 1.2, 2.8, "33", "states and union territories have adopted the Model Building Bye-Laws 2016 RWH provision");
stat(s, 6.7, 1.2, 2.8, "None", "consolidated quantitative syntheses of what installed systems deliver that we could find", C.accent2);
card(s, 0.5, 3.0, 9, 1.9, "The gap in the evidence",
  "Mandates are enforced at plan approval: no completion certificate or water connection without a structure. After that, checks are rare and results are not published. Single-building case studies, agency demonstration reports and newspaper audits exist, but we found no study that combines them. MIRAGE collects the published numbers, puts them
```

```
    ↪ on one scale and asks what the mandate delivers.");
s.addNotes("Set up the problem: compliance is enforced on paper, performance is not measured.");

// ---------- 3 questions and searches ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Problem and questions" });
s.addText("Three questions, four searches", { placeholder: "title" });
card(s, 0.5, 1.15, 2.9, 1.7, "Q1. Performance gap", "How much of the theoretical rooftop potential do documented systems
    ↪ actually harvest or recharge?");
card(s, 3.55, 1.15, 2.9, 1.7, "Q2. Functionality", "What share of installed systems stay functional, and why do they fail?");
card(s, 6.6, 1.15, 2.9, 1.7, "Q3. Cost", "What does harvested water cost per kilolitre compared with piped, desalinated and
    ↪ tanker water?");
const strands = [["89", "performance site records"], ["43", "cost records + 37 price benchmarks"], ["46", "surveys, audits,
    ↪ inspections; 40 sources state failure reasons"], ["24", "mandate instruments"]];
strands.forEach((t, i) => stat(s, 0.5 + i * 2.3, 3.1, 2.2, t[0], t[1]));
body(s, "Sources: CGWB compilations and manual, CSE case studies, Lok Sabha replies, Delhi Jal Board and NGT reports, Arghyam
    ↪ , journal papers, campus and company reports, IMD normals, RBI CPI. Every number logged with its URL and the
    ↪ sentence or values taken from the page.", { x: 0.5, y: 4.55, w: 9, h: 0.6, fontSize: 11, color: C.accent4 });
s.addNotes("Explain the four parallel searches and the logging rule: no number without a URL and a quote.");

// ---------- 4 methods ----------
pres.addSection({ title: "Methods" });
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Methods" });
s.addText("One metric, three evidence tiers", { placeholder: "title" });
s.addShape(pres.ShapeType.roundRect, { x: 0.5, y: 1.15, w: 9, h: 0.75, fill: { color: "EEF3FB" }, line: { color: "EEF3FB" },
    ↪ rectRadius: 0.08 });
s.addText([{ text: "Potential P = Area (m2) x Rainfall (mm) x C / 1000   [kL/yr]      ", options: { bold: true } }, { text: "
    ↪ Performance ratio = reported volume / P.   C = 0.80 (CGWB manual, Table 7.2: concrete 0.7 to GI sheet 0.9);
    ↪ sensitivity 0.70 and 0.90." }],
  { x: 0.7, y: 1.2, w: 8.6, h: 0.65, fontSize: 12.5, color: C.text1, isTextBox: true, margin: 0, valign: "middle" });
card(s, 0.5, 2.1, 2.9, 1.75, "Observed", `Source names a flow meter, a measuring device at the recharge well or a tank-
    ↪ filling record. ${pr.T1.n} observations, ${pr.T1.sites} sites.`, "E3EDFB");
card(s, 3.55, 2.1, 2.9, 1.75, "Agency-reported", `Annual volume stated by CGWB or another agency, no instrument named, or
    ↪ given as a share of estimated runoff. ${pr.T2.n} observations, ${pr.T2.sites} sites.`, "FBE8E0");
card(s, 6.6, 2.1, 2.9, 1.75, "Design calculation", `CSE model projects and one CGWB design note where the 'volume harvested'
    ↪ is a stated share of area x rainfall. ${pr.T3.n} sites.`, "ECEAFA");
body(s, bullets(["Screening: 89 site records found (17 not accessible, 7 added from the cost search and the CGWB compilation)
    ↪ , 79 screened, 47 excluded (20 potential only, 9 no area or volume, 3 modelled, 10 no roof area, 5 mixed or
    ↪ inconsistent), 35 observations at 31 sites included",
  "Costs: deflated to 2024 rupees (CPI-IW); 15-year life and O&M 2% of capital are CGWB norms, 8% discount is our assumption
    ↪ (CGWB example: 15%); divided by design yield, then rescaled to observed yield",
  "Functionality: 13 functional-share estimates from 12 surveys; 90 failure-mode mentions from 40 sources coded into 11
    ↪ categories"]), { x: 0.5, y: 4.0, w: 9, h: 1.2, fontSize: 11 });
s.addNotes("Walk through the formula and why the tiers matter: the method the source used decides what the number means.");

// ---------- 5 result 1 figure ----------
pres.addSection({ title: "Results" });
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Results" });
s.addText("Result 1: in the seven observed cases, yield is about a quarter of potential", { placeholder: "title" });
img(s, "fig1_performance_gap.png", 0.5, 1.1, 9.0, 2.9);
body(s, "Left: reported volume against potential (log scales), 1:1 and 1:4 lines. Right: performance ratio by evidence tier,
    ↪ bars are medians. Observed volumes sit well below agency-reported ones, which cluster at 100% of potential.", { x:
    ↪ 0.5, y: 4.15, w: 9, h: 0.9, fontSize: 11.5, color: C.accent4 });
s.addNotes("The headline figure. Point out the blue observed points below the 1:4 line and the orange agency points at or
    ↪ above 1:1.");

// ---------- 6 result 1 numbers ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Results" });
s.addText("Who measured it decides the answer", { placeholder: "title" });
stat(s, 0.5, 1.15, 2.9, pct(pr.T1.median), `Observed median, n = ${pr.T1.n} at ${pr.T1.sites} sites. 95% CI ${pct(pr.T1.ci_lo
    ↪ )} to ${pct(pr.T1.ci_hi)}; range ${pct(pr.T1.min)} to ${Math.floor(pr.T1.max * 100)}%`);
stat(s, 3.55, 1.15, 2.9, pct(pr.T2.median), `Agency-reported median, n = ${pr.T2.n} at ${pr.T2.sites} sites. ${Math.round(pr.
    ↪ T2.share_ge_1 * pr.T2.n)} of ${pr.T2.n} at or above 99% of potential, the most a roof can give at C = 0.80`, C.
    ↪ accent2);
stat(s, 6.6, 1.15, 2.9, pct(pr.T3.median), `Design-calculated median, n = ${pr.T3.n}. These are the shares CSE assumed, not
    ↪ measurements`, C.accent3);
card(s, 0.5, 3.0, 4.4, 2.0, "Sensitivity checks",
  `Observed median between ${pct(pr.T1.median_c090)} and ${pct(pr.T1.median_c070)} for C from 0.90 to 0.70. Dropping the
    ↪ partial-year Goa 2008 observation: ${pct(pr.T1_excl_partial_year.median)}. Dropping the two storage-limited houses:
    ↪ ${pct(pr.T1_excl_storage_limited.median)}.\nObserved vs agency-reported: Mann-Whitney U = ${pr.T1_vs_T2_mannwhitney.
    ↪ U.toFixed(0)}, p = ${pr.T1_vs_T2_mannwhitney.p.toFixed(3)}.`);
card(s, 5.1, 3.0, 4.4, 2.0, "The one borderline case",
  `Warud (112%) counts as observed by the rule (flow meter named) but CGWB gives its volume as a round "90%" of rain received
    ↪ , so it may be an assumed share. Graded agency-reported instead: observed median ${pct(pr.T1_excl_warud.median)} (n
    ↪ = ${pr.T1_excl_warud.n}), agency-reported median ${pct(pr.T2_incl_warud.median)}, p = ${pr.T2_incl_warud.
    ↪ mannwhitney_p.toFixed(3)}.\nPanjab University's agency figure is stated as "93% of total generated runoff": a design
    ↪  share.`);
s.addNotes("The main inference: agency numbers are design numbers. The few metered sites are the only evidence of real
    ↪ performance.");

// ---------- 7 reconciliation with the mid-term progress report ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Results" });
s.addText("The six midway cases, verified and recomputed", { placeholder: "title" });
const rows = [
  [{ text: "Case", options: { bold: true, color: "FFFFFF", fill: { color: "1F4E79" } } }, { text: "Midway check-in", options:
    ↪  { bold: true, color: "FFFFFF", fill: { color: "1F4E79" } } }, { text: "Final (C = 0.80, stated rainfall)", options:
    ↪  { bold: true, color: "FFFFFF", fill: { color: "1F4E79" } } }, { text: "Status", options: { bold: true, color: "
    ↪ FFFFFF", fill: { color: "1F4E79" } } }],
  ["KITS Ramtek (flow meter)", "about 9%", "11% of potential; 19 of 215.7 m3 received", "Verified in CGWB 2011 p.73; observed
    ↪ "],
  ["Goa University", "61 to 76%", "76% (2010); 28% (2008, partial); 19% (2009)", "Verified in CSE 2014; observed"],
  ["BEL Bengaluru", "63 to 75%", "64% (IMD normal 1,077.6 mm)", "Verified; storage, not recharge; agency-reported"],
  ["Panjab University", "about 79%", "112% (2,100 m2, 1,059 mm)", "Volume is '93% of runoff': design share; agency-reported
    ↪ "],
  ["Warud (flow meter)", "90% (runoff assumption)", "112% of potential; 90% of rain received", "Verified in CGWB 2011 p.73;
    ↪ observed by rule, agency-reported in sensitivity"],
```

```
  ["IIT Delhi Block VI", "not comparable (rainfall NR)", "82% with IMD Safdarjung normal", "Verified in CGWB 2011 p.44; agency-reported"],
];
s.addTable(rows, { x: 0.5, y: 1.1, w: 9, colW: [2.0, 1.8, 2.7, 2.5], fontSize: 9.5, color: C.text1, border: { type: "solid", pt: 0.5, color: "D9D9D9" }, rowH: 0.38, valign: "middle" });
body(s, "All six cases are kept. With one runoff coefficient and a stated rainfall basis, two cases (Panjab University, Warud) move above 100% of potential; a volume above potential is a design share, not a measurement. Umaid Heritage and PAU Ludhiana stay excluded (mixed catchment, no roof area); Sri Aurobindo Ashram is a design calculation.", { x: 0.5, y: 4.45, w: 9, h: 0.7, fontSize: 10.5, color: C.accent4 });
s.addNotes("Show continuity with the midway check-in and explain each revision.");

// ---------- 8 result 2 figure ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Results" });
s.addText("Result 2: installed is not the same as working", { placeholder: "title" });
img(s, "fig2_mandate_gap.png", 0.5, 1.1, 9.0, 2.85);
body(s, `Installation: 60% of 2.72 lakh houses inspected in Chennai (2019), 56% in Hyderabad, 18% of connected properties in Bengaluru, below 1% in Pune and Nagpur. Functionality: ${mg.functionality_n} estimates from ${mg.functionality_surveys} surveys, median ${mg.functionality_median.toFixed(0)}% of installed systems working (0 to 84%). Dwarka 2024: 84% structurally intact, only 32% recharging water free of faecal coliform.`, { x: 0.5, y: 4.0, w: 9, h: 1.0, fontSize: 11.5, color: C.accent4 });
s.addNotes("Two separate gaps: installation and functionality. Definitions differ between surveys; sample sizes in brackets.");

// ---------- 9 cascade native chart ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Results" });
s.addText("Chennai: from mandate to water delivered", { placeholder: "title" });
const casc = mg.chennai;
s.addChart(pres.ChartType.bar, [{ name: "Share of mandated potential", labels: ["Mandated", "With a structure", "Needing no repair", "Delivered at observed ratio"], values: [100, casc.with_structure * 100, casc.functional * 100, casc.delivered * 100].map(v => Math.round(v * 10) / 10) }],
  { x: 0.5, y: 1.1, w: 5.4, h: 3.9, barDir: "col", chartColors: ["2A78D6"], showValue: true, dataLabelPosition: "outEnd", dataLabelFormatCode: "0\"%\"", dataLabelFontSize: 11, dataLabelFontFace: "+mn-lt", dataLabelColor: "0B0B0B",
    catAxisLabelFontSize: 10, catAxisLabelFontFace: "+mn-lt", catAxisLabelColor: "52514E", valAxisLabelFontSize: 9, valAxisLabelFontFace: "+mn-lt", valAxisLabelColor: "52514E", valAxisMaxVal: 110, valAxisMinVal: 0, valGridLine: { color: "E6E5E1", size: 0.5 }, catGridLine: { style: "none" }, showLegend: false, showTitle: false, barGapWidthPct: 60 });
stat(s, 6.3, 1.2, 3.2, pct(casc.with_structure), "of 2.72 lakh inspected houses had a structure (GCC, Aug-Sep 2019 snapshot)");
stat(s, 6.3, 2.5, 3.2, pct(casc.functional), "needing no repair: 77% of the 1.62 lakh structures, same inspection", C.accent2);
stat(s, 6.3, 3.8, 3.2, pct(casc.delivered), `delivered, at the observed yield ratio of ${pct(pr.T1.median)} (${pct(mg.chennai_excl_warud_delivered)} with Warud regraded). Bengaluru by the same chain: about ${pct(mg.bengaluru.delivered)}`, C.text1);
s.addNotes("The cascade multiplies three independent figures. Say clearly that the last step uses the observed-tier median from a small sample.");

// ---------- 10 failure modes ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Results" });
s.addText("Result 3: why systems fail", { placeholder: "title" });
img(s, "fig3_failure_modes.png", 0.5, 1.15, 5.2, 3.2);
const fmTop = R.failure_modes.slice(0, 4);
card(s, 6.0, 1.15, 3.5, 3.9, "Management causes are cited more often than component faults",
  fmTop.map(f => `${f.category}: ${f.distinct_sources} sources`).join("\n") + "\n\nNo maintenance is cited most often; missing incentive or enforcement ties with poor design or construction. Silted pits (7), disconnected downpipes (6) and clogged filters (3) are the faults a routine inspection would catch.");
body(s, `${R.failure_mode_counts.mentions} mentions from ${R.failure_mode_counts.sources} surveys, audits and reports, counted by distinct source, ${R.failure_mode_counts.categories} categories.`, { x: 0.5, y: 4.45, w: 5.2, h: 0.5, fontSize: 11, color: C.accent4 });
s.addNotes("Connect to the course theme: infrastructure as a service that needs an operator, not a one-time artefact.");

// ---------- 11 cost ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Results" });
s.addText("Result 4: cheap at design yield, below tanker prices at observed yield", { placeholder: "title" });
img(s, "fig4_cost_per_kL.png", 0.5, 1.15, 5.6, 2.6);
stat(s, 6.4, 1.1, 3.1, `Rs ${cst.overall.median.toFixed(0)}/kL`, `median levelised cost at design yield, 27 systems (IQR ${cst.overall.q25.toFixed(0)} to ${cst.overall.q75.toFixed(0)}); recharge Rs ${cst.system_recharge.median.toFixed(0)}, storage Rs ${cst.system_storage.median.toFixed(0)}`);
stat(s, 6.4, 2.85, 3.1, `Rs ${cst.overall_delivered_median.toFixed(0)}/kL`, `at observed yield (multiplier ${cst.delivered_yield_multiplier.toFixed(1)}): within piped tariffs (Rs 7 to 48), below desalination (Rs 68) and tankers (Rs 78 to 375)`, C.accent2);
body(s, bullets([`Cost per kL falls steeply with catchment area (Spearman rho = ${cst.spearman_area_cost.rho.toFixed(2)}, p < 0.001); scale groups not separable at these counts (Kruskal-Wallis p = ${cst.kruskal_scale.p.toFixed(2)})`,
  `Twenty rows are CSE Delhi projects of 2000 to 2004 (median Rs ${cst.cse_delhi_only.median.toFixed(0)}); the five post-2010 rows median Rs ${cst.post2010.median.toFixed(0)}, about Rs ${cst.post2010_delivered_median.toFixed(0)} at observed yield, inside the tanker range. 2024 rupees, 15-year life, 8% (Rs ${cst.sensitivity_15pct_median.toFixed(0)} at 15%), O&M 2%`]), { x: 0.5, y: 3.85, w: 5.6, h: 1.3, fontSize: 10.5 });
s.addNotes("Cost does not explain the gap. Even at observed yield harvested water beats the tanker water the same cities buy in dry years.");

// ---------- 12 discussion ----------
pres.addSection({ title: "Discussion" });
s = pres.addSlide({ masterName: "DARK", sectionTitle: "Discussion" });
s.addText("Unmeasured and under-delivering; reported failures point to enforcement", { x: 0.5, y: 0.35, w: 9, h: 0.7, fontSize: 22, bold: true, color: "FFFFFF", fontFace: "Cambria", isTextBox: true, margin: 0 });
s.addText(bullets(["The mandates increased installation. Agencies report what they designed; we found only five metered sites, and those deliver about a quarter of potential.",
  "Structures installed is an output indicator. Water delivered is the outcome. The mandates count structures, not water.",
  "Reported failures point to maintenance and enforcement first; the technical faults listed are ones an inspection would catch.",
  "Cost: within piped tariffs at design yield and below tanker prices even at observed yield for the median system. For an owner, recharge still costs more than piped water while the benefit is shared, so incentives matter."]),
  { x: 0.5, y: 1.15, w: 4.6, h: 3.9, fontSize: 13, color: "FFFFFF", isTextBox: true, margin: 0, valign: "top", paraSpaceAfter: 8 });
const recs = [["1. Measure", "Require a flow meter or recharge-well level logger on systems above a size threshold, so performance can be verified at all."],
```

```
  ["2. Inspect, then pay", "Tie rebates and penalties to a periodic functional inspection, not a one-time completion certificate. Dwarka shows intact structures can recharge contaminated water."],
  ["3. Publish", "Report installed-versus-functional counts every year, as Chennai did in 2019, so the gap is visible."]];
recs.forEach((r, i) => { s.addShape(pres.ShapeType.roundRect, { x: 5.4, y: 1.15 + i * 1.3, w: 4.1, h: 1.15, fill: { color: "1F4E79" }, line: { color: "1F4E79" }, rectRadius: 0.08 });
  s.addText(r[0], { x: 5.55, y: 1.2 + i * 1.3, w: 3.8, h: 0.35, fontSize: 13, bold: true, color: "CADCFC", isTextBox: true, margin: 0 });
  s.addText(r[1], { x: 5.55, y: 1.55 + i * 1.3, w: 3.8, h: 0.75, fontSize: 10.5, color: "FFFFFF", isTextBox: true, margin: 0, valign: "top" }); });
s.addNotes("Close the argument and give the three recommendations.");

// ---------- 13 limitations ----------
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Discussion" });
s.addText("Limitations", { placeholder: "title" });
card(s, 0.5, 1.15, 4.4, 1.85, "Small observed tier", `${pr.T1.n} observations at ${pr.T1.sites} sites, two of them storage-limited houses and one borderline (Warud). The ${pct(pr.T1.median)} figure is indicative, not representative. Publication bias favours showcase systems, so the true gap is probably larger. Ramtek's low ratio is partly an intake limit of the dug well.`);
card(s, 5.1, 1.15, 4.4, 1.85, "Heterogeneous inputs", "Four observations use an IMD city normal and four a CSE city average, not the year's rainfall. Panjab University's area and volume come from different documents. CSE design calculations use roof plus paved area.");
card(s, 0.5, 3.15, 4.4, 1.85, "Surveys differ", "Definitions range from 'needing no repair' to 'dry pit'; two estimates come from one Dwarka report and one is an expert estimate. Several figures reach us only through press reports; originals cited where found.");
card(s, 5.1, 3.15, 4.4, 1.85, "Costs and causality", "Cost rows are dominated by one programme (CSE Delhi, 2000 to 2004) and depend on the discount rate. Associations are not causal; a weak private incentive to maintain recharge pits is a competing explanation. All of these show why metering is needed.");
s.addNotes("Be plain about what the data can and cannot support.");

// ---------- 14 contributions and package ----------
pres.addSection({ title: "Closing" });
s = pres.addSlide({ masterName: "LIGHT", sectionTitle: "Closing" });
s.addText("Contributions and what is submitted", { placeholder: "title" });
const team = [["Vedant Abhangrao", "Search strategy, screening log, mandate search"], ["Nilay Shahapure", "Data harmonisation, analysis code, figures, supplementary document"], ["Arjun Jhunjhunwala", "Cost and benchmark search, deflation, levelised cost"], ["Yash Krishnani", "Functionality and failure-mode search, report, presentation"]];
team.forEach((t, i) => card(s, 0.5, 1.15 + i * 0.9, 4.4, 0.8, t[0], t[1]));
card(s, 5.1, 1.15, 4.4, 3.65, "Submitted",
  "Report (2 pages, IMRaD) and Supplementary Information (15 pages: full tables, screening log, midway reconciliation, 168 source URLs)\n\nData: harmonised dataset for performance, cost, functionality, installation, failure modes and mandates; source extraction tables with a URL and verbatim quote for every number\n\nCode: Python scripts that rebuild the dataset, statistics, figures and tables\n\nEvery number in the report is traceable to a source URL and a verbatim quote.");
body(s, "All four members extracted records and cross-checked each other's rows against the source quotes.", { x: 0.5, y: 4.9, w: 4.4, h: 0.28, fontSize: 9.5, color: C.accent4 });
s.addNotes("Statement of contributions, as required for the final presentation.");

(async () => {
  const out = path.join(__dirname, "MIRAGE_presentation.pptx");
  await pres.writeFile({ fileName: out });
  await applyTheme(out, THEME);
  console.log("written", out);
})();
```